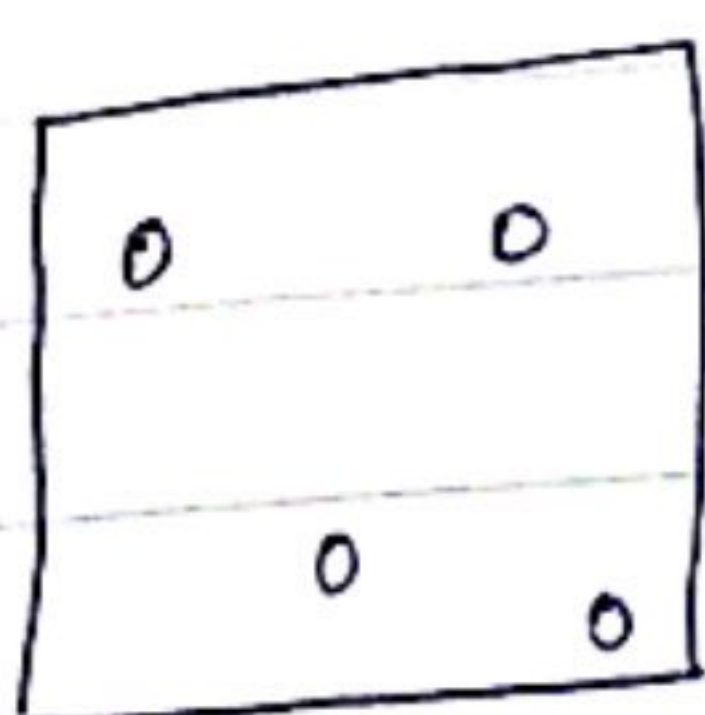
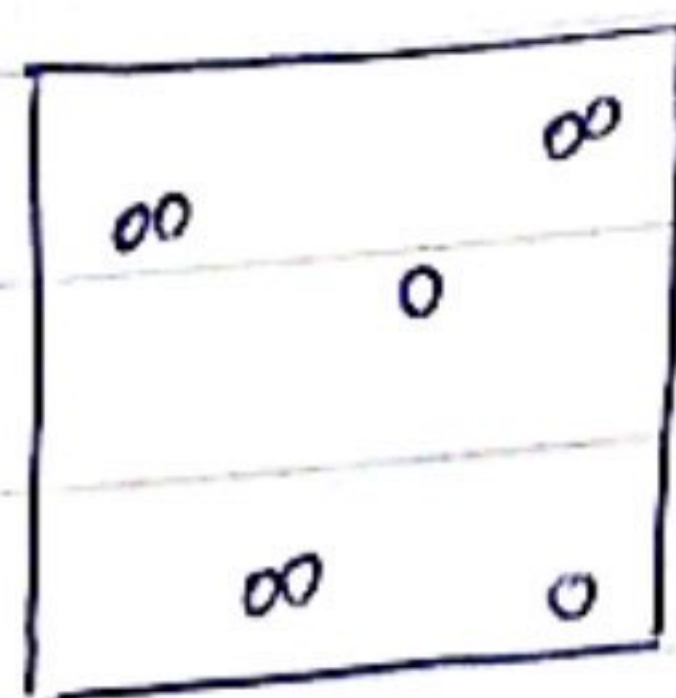


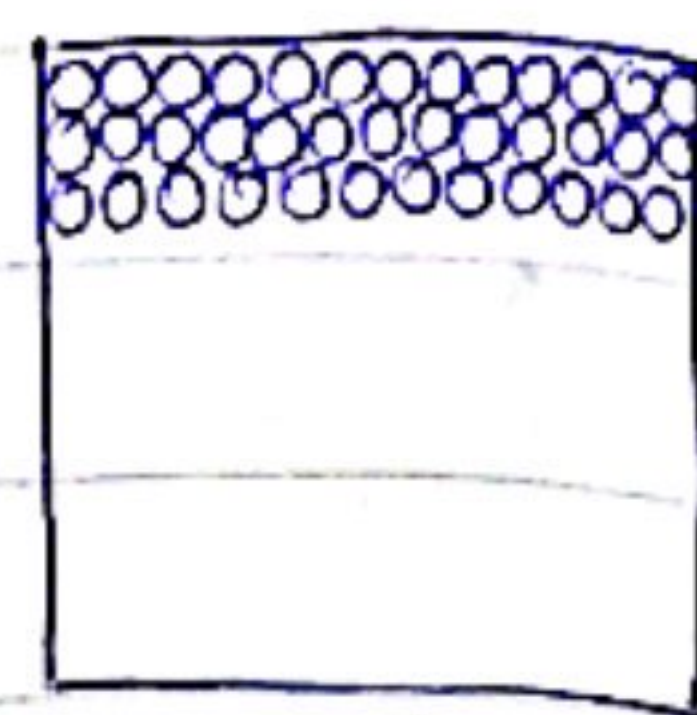
The metal is classified into 3 mainly, gas, liquid and solid



Gas



Liquid



Solid

- | | | |
|--|--|-------------------------------------|
| * Molecules are apart | * Molecules are not closely | * Closely packed. |
| * Less intermolecular attraction between 2 molecules | * Less intermolecular attraction. | * Strong intermolecular attraction. |
| * Molecules have 3 movements | * Have some movements also have more intermolecular space but lower than gas | * Movements are restricted |
| * Have more intermolecular space | * Compressive | * Intermolecular space is minimum |
| * Easy compressiver. | | * Not compressive. |

Phase Transition between the three state of metals.

One state of matter to an another is known as phase transition. If heat energy supplied to a solid, the molecular vibrations enhanced. When the temperature is sufficiently high, the molecules in the solid are able to breakout the well ordered in to the less ordered liquid structure. Energy requirement for this phase transition known as melting and it is referred to as heat of fusion.

If a liquid is heated, and sufficient energy is supplied, then at the temperature known as the boiling point. Molecules in the liquid are able to break out from the state and undergo phase transition or the gaseous state. This process is known as Vaporization. The heat energy requires for this phase transition is known as heat of vaporization.

Molecules of the solid to obtain sufficient thermal energy and transform the solid stage directly into the vapour without

passing through intermediate liquid states and this process is known as sublimation. And the required heat of sublimation of a substance is approximately equal to the sum of heat of fusion and heat of vapourization.

The magnitude of the heat required for the phase transition depends on the strength of inter molecular forces.

Kinetic Molecular Theory of Gases.

Initially we concentrate on the free translational motion and ignore the existence of any interaction between molecules. As such the theoretical model combines the molecular nature of the gaseous state with its continuous, random and chaotic motion. And this is why this theory is named as kinetic molecular theory.

Basic Assumptions :

- 1) A gas is made up of large number of molecules.
- 2) The volume is negligible.
- 3) Gas molecules are continuous, random motion within the container.
- 4) The molecules are hard, spherical and perfectly elastic.
- 5) The molecules do not interact with each other.
- 6) The time spent during the collision is negligible.

The theory stated above and the deduction made on that basis are strictly applicable, under special or ideal conditions.

When the gases are far removed from conditions approaching liquefaction.

These conditions can do a first approximation we summarized as low pressure and high temperature.

The gases which exist under such ideal or perfect conditions referred to as ideal gas or perfect gases.

Mean free path and Velocity of molecules

Mean free path :

The average distance travelled by a molecule between two consecutive collision.

The mean free path increases with an increase of temperature and it decreases with the increase of pressure.

Root Mean Square Velocity :

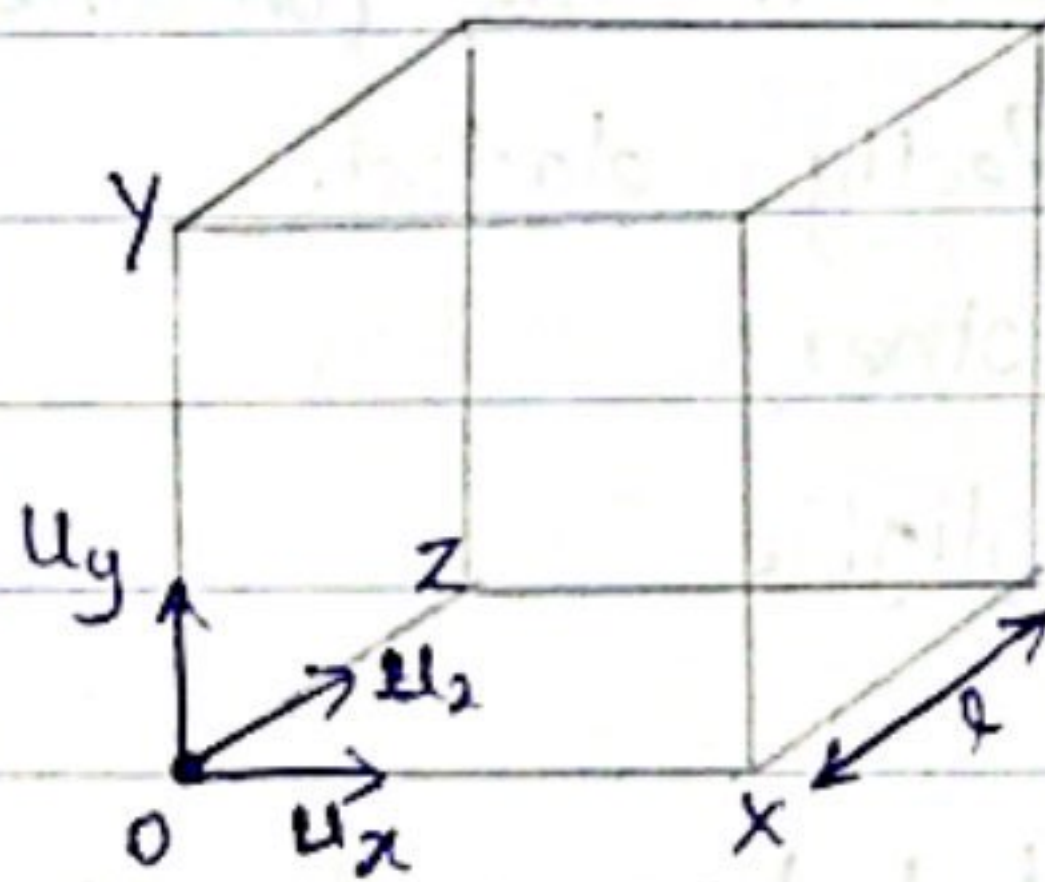
$$\sum u_i^2 = u_1^2 + u_2^2 + \dots + u_n^2 \quad \text{--- (1)}$$

$$\sum n_i = n = n_1 + n_2 + \dots + n_n \quad \text{--- (2)}$$

mean square Velocity $c^2 = \frac{\sum u_i^2}{n} \quad \text{--- (3)}$

$$c = \sqrt{\frac{\sum u_i^2}{n}} \quad \text{--- (4) *}$$

Equations for Pressure of ideal Gas :



A vector can be divided into 3 components

u_x , u_y and u_z

Then,

$$u^2 = u_x^2 + u_y^2 + u_z^2 \quad \text{--- (5)}$$

There are 3 axis which are ^{perpendicular} to the containing a wall as ox , oy and oz .

Let us consider motion of molecules along the ox axis. So momentum can be written as,

$$mu_x - (-mu_x) = 2mu_x \quad \text{--- (6)}$$

Length of the tube along the ox direction is, $ox = l$

$$\text{Change of direction per second} = \frac{u_x}{l} \quad \text{--- (7)}$$

$$\begin{aligned}\text{Change of momentum per second} &= \frac{u_x}{l} \times 2m u_x \\ &= \frac{2m u_x^2}{l} \quad \text{--- (8)}\end{aligned}$$

Similarly ; in the direction of oy ,

$$\text{Change of momentum per second} = \frac{u_y}{l} \times 2m u_y = \frac{2m u_y^2}{l}$$

In the direction of oz ,

$$\text{Change of momentum per second} = \frac{u_z}{l} \times 2m u_z = \frac{2m u_z^2}{l}$$

$$\begin{aligned}\text{Change of total momentum per second} &= \frac{2m}{l} (u_x^2 + u_y^2 + u_z^2) \\ &= \frac{2m \sum u_i^2}{l} \quad \text{--- (9)}\end{aligned}$$

from eqⁿ (3) ;

$$\sum u_i^2 = n c^2$$

$$\therefore \text{Total change of momentum} = \frac{2m n c^2}{l} \quad \text{--- (10)}$$

According to the newton's 2nd law of motion,

$$F = \frac{2m n c^2}{l} \quad \text{--- (11)}$$

$$\text{Total area} = 6l^2$$

$$P = F/A$$

$$P = \frac{F}{A} = \frac{2m n c^2 / l}{6l^2} = \frac{2m n c^2}{6l^3} \quad l^3 = \text{volume}$$

$$P = \frac{1}{3} n m c^2 / V$$

$$\boxed{PV = \frac{1}{3} n m c^2} \quad \leftarrow \text{kinetic molecular theory equation (KMT)}$$

Applications of the kinetic Molecular of Gases.

Reduction of gas laws from the KMT equations.

(01) Boyle's Law,

* { At constant temperature, the pressure of a given sample is inversely proportional to the volume.

$$pV = \text{Constant} \quad \text{--- (1)}$$

From thermodynamics, the energy of an ideal gas depends only on temperature. ($E \propto T$)

$$E = \frac{1}{2} \times (\text{Total mass}) \times (\text{mean square velocity})$$

$$E = \frac{1}{2} \times (nm) \times C^2 \quad \text{--- (2)}$$

from KMT eqⁿ;

$$pV = \frac{1}{3} mnC^2 \quad \text{--- (3)}$$

from (2) and (3),

$$pV = \frac{2}{3} E, \text{ where } E \propto T \quad \text{--- (4)}$$

$$\text{At constant } T, \quad pV = k$$

(02) Charles's law,

* { At constant pressure, the volume of the given gas sample is directly proportional to the temperature.

$$\frac{V}{T} = k \quad \text{--- (1)}$$

$$pV = \frac{2}{3} E \quad \text{--- (2)}$$

p constant,

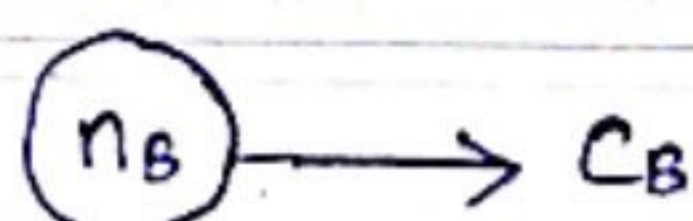
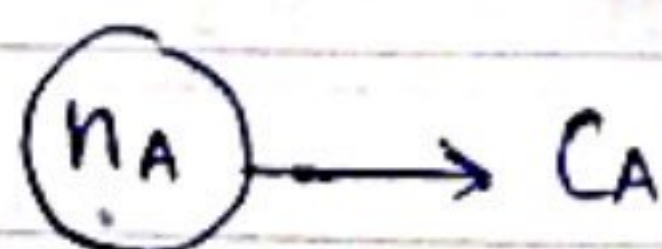
$$pV \propto T \quad \text{--- (3)}$$

$$V = KT$$

$$\frac{V}{T} = k$$

(03) Avagadro's Law,

* { At constant pressure and temperature the molar volumes of all gasses are approximately the same.



$$pV = KT \quad \text{--- (1)}$$

from KMT eqⁿ;

$$\text{for A; } pV = \frac{1}{3} n_A m_A C_A^2 \quad \text{--- (2)}$$

$$\text{for B; } pV = \frac{1}{3} n_B m_B C_B^2 \quad \text{--- (3)}$$

$$\frac{1}{3} n_A m_A C_A^2 = \frac{1}{3} n_B m_B C_B^2 \quad \text{--- (4)}$$

The 2 gases are at the same T .

Hence the mean translational kinetic energy per molecule should be the same.

$$\frac{1}{2} m_A C_A^2 = \frac{1}{2} m_B C_B^2 \quad \text{--- (5)}$$

$$\text{(4)/(5)} \Rightarrow n_A = n_B$$

Ideal gas equation.

$$K.M.T \text{ eq}^n; \quad PV = \frac{1}{3} n m C^2 \quad \text{--- (1)}$$

$$E = \frac{1}{2} m C^2$$

$$E \propto T \quad \text{--- (2)}$$

$$PV = \frac{2}{3} n E \quad \text{or} \quad PV \propto n T \quad \text{--- (3)}$$

$$PV = k n T \quad \text{--- (4)}$$

$k \propto$ Boltzmann constant.

$$\text{Total moles } n' = \frac{n}{A}$$

A : Avagadro constant.

$$PV = n' A k T$$

$$A k = R \quad R : \text{Universal gas constant.}$$

$$PV = n R T$$

28.08.2023

Molar Translational Kinetic Energy

$$m = M/A \quad \text{--- (1)}$$

$$n' = \frac{n}{A} \quad \text{--- (2)}$$

A : Avogadro constant

$$PV = \frac{1}{3} n' m C^2 \quad \text{--- (3)}$$

$$PV = n R T \quad \text{--- (4)}$$

$$n R T = \frac{1}{3} n' m C^2$$

$$C^2 = \frac{3 R T}{M}$$

$$C^2 \propto T$$

$$\sqrt{C^2} = \sqrt{\frac{3 R T}{M}} \quad \text{--- (5)}$$

molar translational KE,

$$E = \frac{1}{2} m C^2 \quad \text{--- (6)}$$

From (5), $MC^2 = 3RT$ — (7)

$E = \frac{3}{2} RT$ — (8)

Per molecule, $E = \frac{3RT}{2A}$

KE $U = \frac{3}{2} KT$ — (9)

Distribution of Molecular Velocity.

$$\frac{dn}{n} = \left(\frac{m}{2\pi KT} \right)^{1/2} \cdot e^{-mv^2/2KT} \cdot dv$$

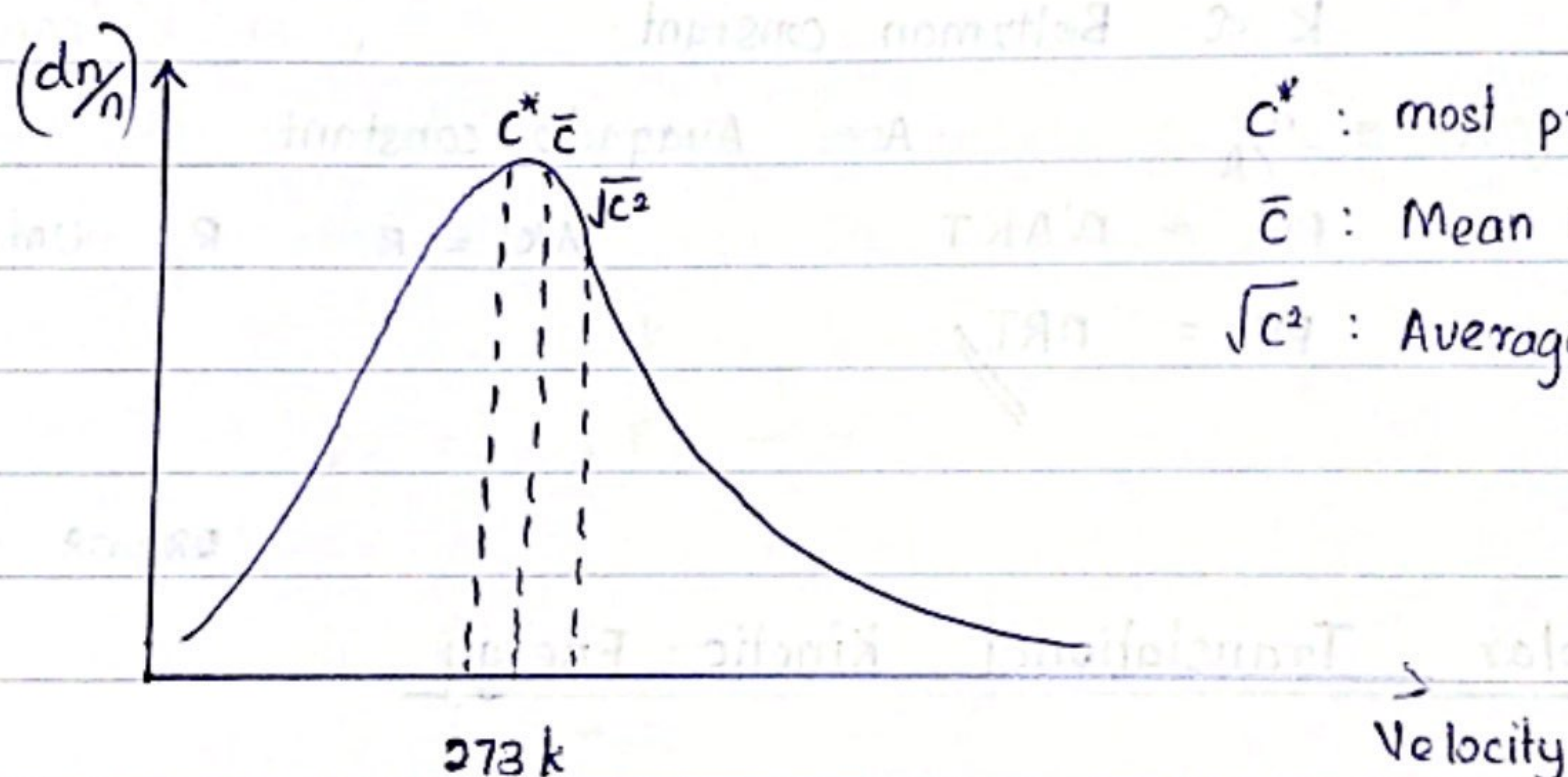
from maxwell - boltzman law,

k : Boltzman constant

T : Absolute temperature (K)

m : mass

dn/n : fraction of molecules moving in the velocity of



c^* : most probable speed.

$\bar{c}$: Mean velocity

$\sqrt{c^2}$: Average root mean velocity.

• The curve is not symmetrical. More molecules at higher velocities than in lower one. The majority of the molecules lie within the fairly narrow range of velocity.

• The maximum curve corresponds to what can be referred to as most probable velocity. (c^*)

• The largest fraction of molecules will possess that velocity as compared to any other speed. $c^* = \sqrt{2RT/M}$

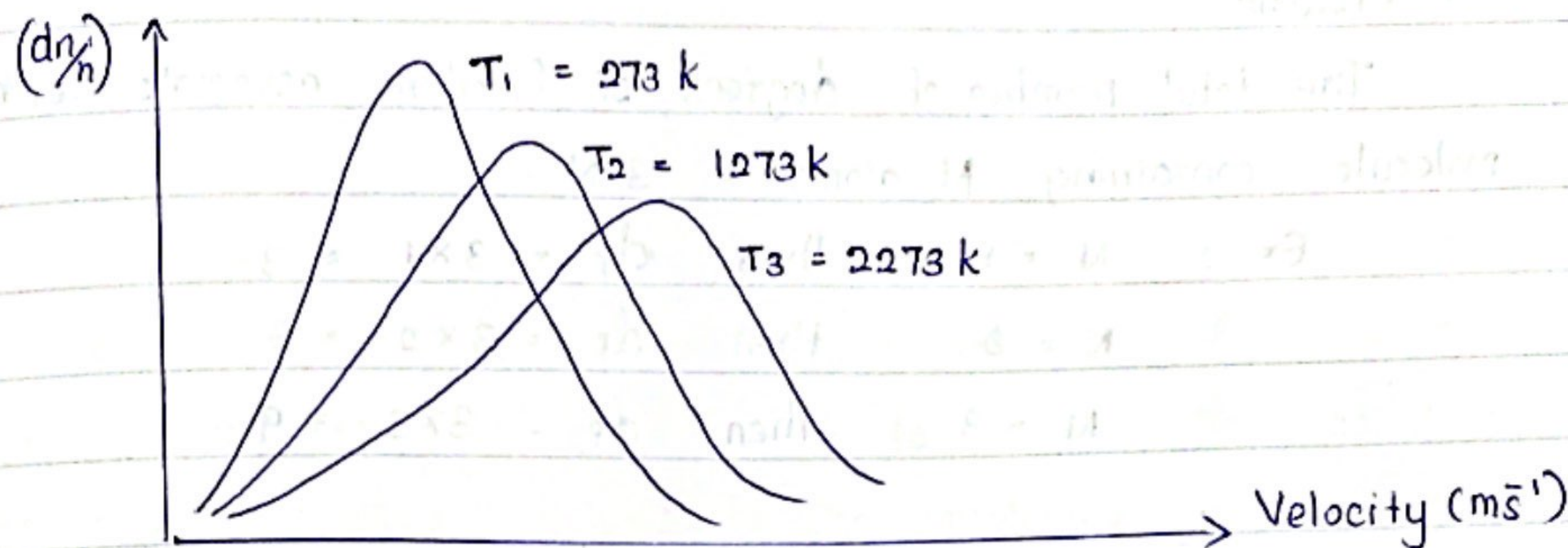
• The average or mean velocity ($\bar{c}$) which means the conventional arithmetical mean can be defined as the sum of the velocities of all molecules divided by the total number of molecules.

$$\bar{c} = \left(\frac{n_1 v_1 + n_2 v_2 + \dots}{n_1 + n_2 + \dots} \right)$$

$$\bar{c} = \sqrt{8RT/\pi M}$$

If the curve is symmetrical $\bar{c} = c^*$

and the ratio between $c^* : \bar{c} : \sqrt{c^2} = 1 : 1.128 : 1.225$



The graph shows the distribution of N_2 at 3 different temperatures with the increase of temperature. There is a greatest spread of molecules with higher velocities but not the proportionate increases. Then the area of the graph shifts to the right side.

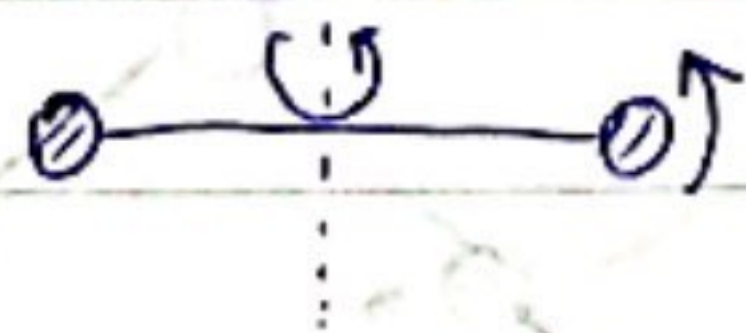
Total Kinetic Energy of Molecules

(i) For a monoatomic gas, this will be the only contribution to the total KE. At such total kinetic energy of a small monoatomic gas molecule taken as $\frac{3}{2} RT = \frac{3}{2} KT$ (when $n = 1$).

(2) In the case of polyatomic molecules the situation is equal, the contribution to the TKE from rotational & vibrational energy.

$$E_t = E_{TKE} + E_v + E_r$$

In addition to translational energy, rotational energy is the energy possessed by a molecule due to the possibility of rotation of the molecules about various axes.



Vibrational energy, the energy possessed by a molecule due to the vibrations of atoms.

(3) Degrees of freedom possess by ideal gas molecules. These are independent components of the motion of the system.

In terms of the types of energy these degrees of freedom are interm described as translational rotational & vibrational degrees of freedom.

The total number of degrees of freedom associate with the molecule containing N atoms = $3N$.

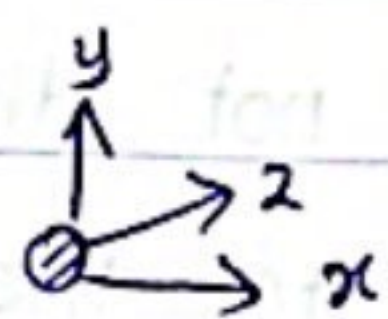
Ex : $N = 1$ then $df = 3 \times 1 = 3$

$N = 2$ then $df = 3 \times 2 = 6$

$N = 3$ then $df = 3 \times 3 = 9$.

→ Translational degrees of freedom

A gas molecule can travel above 3 independent directions in the space. so the molecule possess 3 translational degrees of freedom.



→ Rotational degrees of freedom

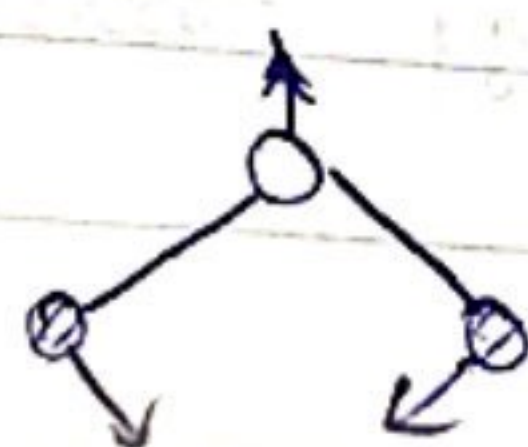
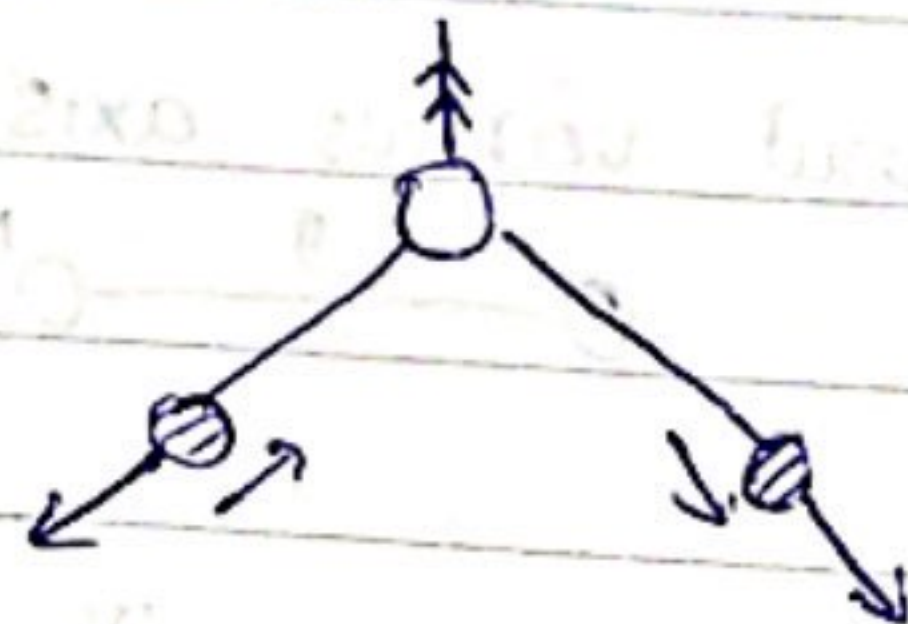
In general 3 perpendicular axis are available for the rotation. In a case of a diatomic molecule or a linear polyatomic molecule. one such axis lies along the intermolecular axis, and rotation about that axis would be negligible.

however non linear molecules possessing 3 or more atom would be able to rotate about all the 3 axis & set to possess 3 rotational degrees of freedom.

→ Vibrational degrees of freedom

This is depend on number of atoms present in the molecule. The vibrational modes commonly defined as

- (1) Stretching
- (2) Bending
- (3) Twisting



Bending

the total number of degrees of freedom associated with molecular and the number of vibrational degree of freedom would be as follows :

* Diatomic & Linear polyatomic $(3N - 5)$

* Non Linear polyatomic $(3N - 6)$

Real gases

The KMT of gases assumes the absence of the intermolecular forces as well as the molecular volume. At low pressure total gas volume is relatively large, to assume that gases molecules possess a negligible volume. The molecule present at low pressure there will be a large average distance of separation between molecules and then assume that intermolecular forces are absent.

At low pressure the real gases behave like ideal gases in accordance with the KMT. When the pressure is higher or T is relatively lower molecules are near to each other. So that it is no longer possible to assume the absence of intermolecular interactions and then volume of the gases will not be relatively negligible.

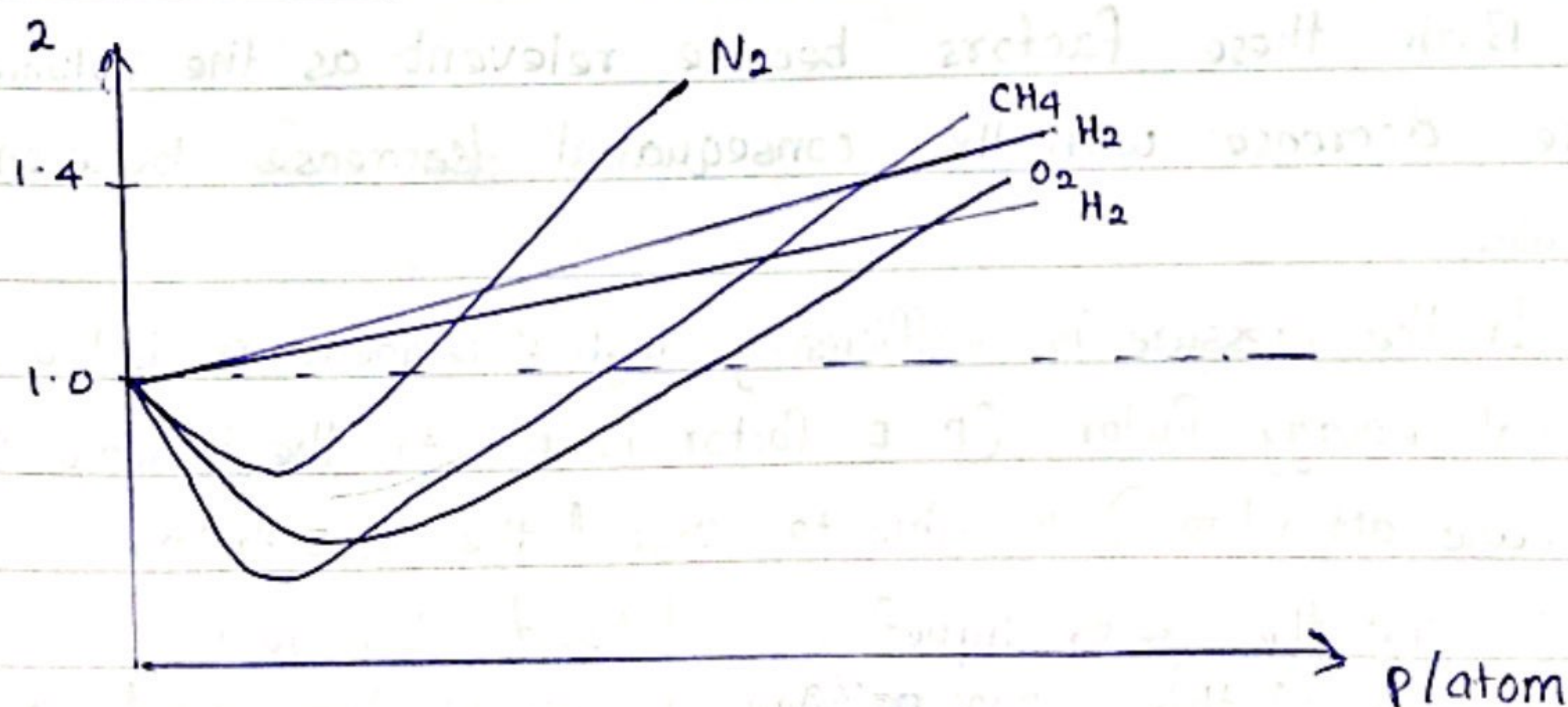
Thus lower P & high T represent most optimum conditions for the behavior of ideal gases. Even the noble gases behave as real gases at high P & low T .

The extent of deviation from ideal gas behaviour & the KMT depends on the nature of the gas.

for ideal gas ; $PV = nRT$ z : Compressibility factor

$$PV = z nRT$$

For an ideal gas $z = 1$ and this measure the nonideal gas,



behave like ideal gas

At low P & of all the gases approaching the value on.

When P increases the Z of the gases greater than one. that means they are more difficult to compress than an ideal gas. Under these conditions the gas molecules are close to each other. The repulsive forces. as a result all the gases are less compressive than an ideal gas.

Van der Waals Equation of State

The existence of intermolecular forces implies molecules cannot come closer than the certain distance of each other.

Instead of the volume of the container volume, the actual volume available for travel by molecules is less than V .

So Van der Waals identified correction term b as corresponding to the total correction factor necessary to be made per mole of gas. $V' = (V - nb)$

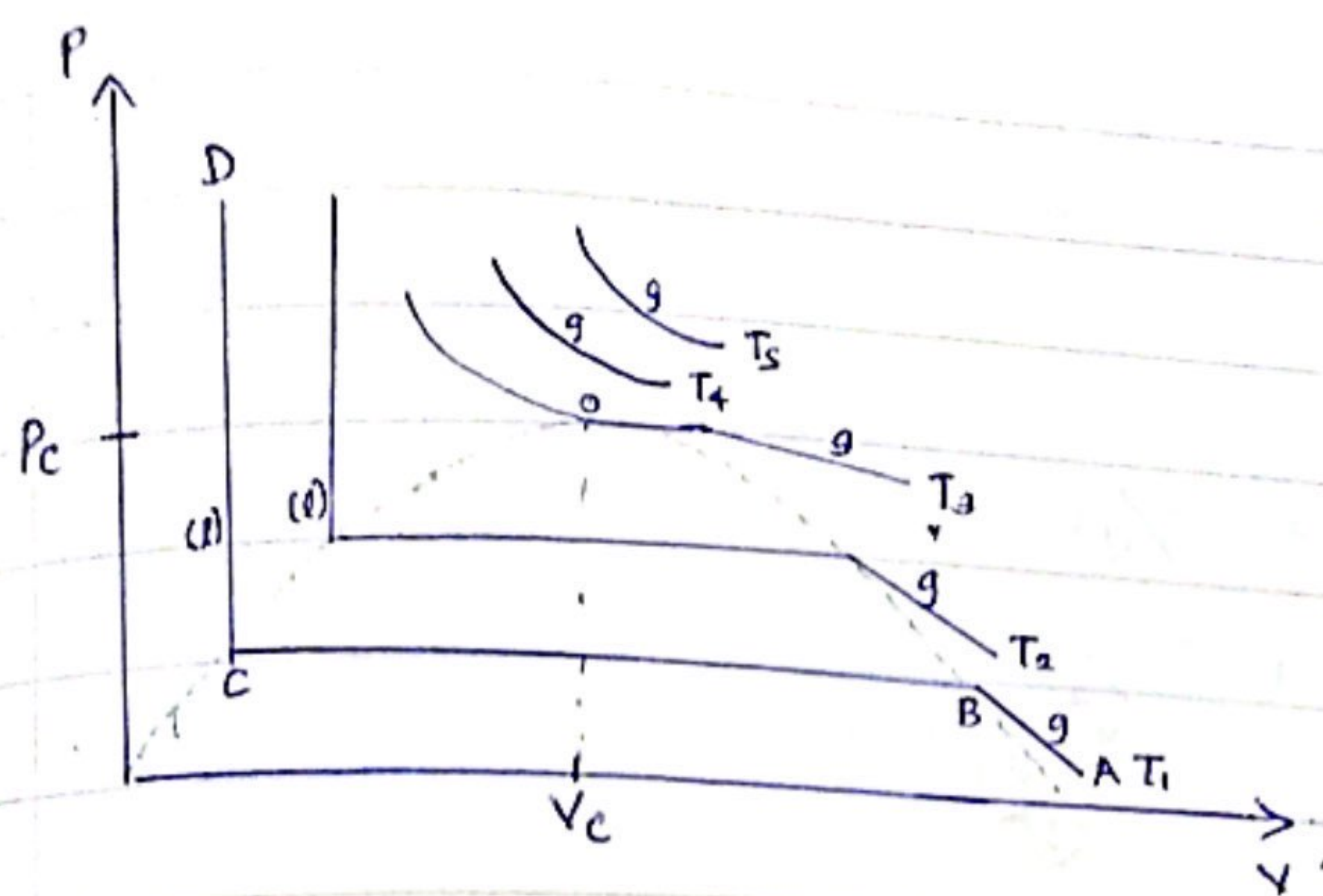
The interacting forces which are essentially of an attractive nature usually tend to be all the gas molecules together. So the force exerted by a molecule on the container wall is reduced due to the attractive forces exerted by the remaining volume. So the actual $P' = P + (n^2 a / V^2)$

'a' represent proportionality constant

$$P + \left(\frac{n^2 a}{V^2} \right) (V - nb) = nRT$$

Liquefaction

- Both these factors become relevant as the volume & temperature are decrease with the consequential decrease between the molecules
- If the pressure is sufficiently high & Temperature is low, the potential energy factor (P.E factor is due to the increase of inter-molecule attractions) is able to exceed the K.E factor.
- So that the gases liquefies and liquid state results.
- Because of this process the density of the liquid increases.



$$T_1 < T_2 < T_3 < T_4 < T_5$$

P_c : Critical pressure

V_c : Critical volume

O : Critical point.

• The condition of liquifaction & critical temperatures & volume of real gas molecule define by Andrew's isotherms and determine in terms of the Vander wal's constant.

• At high Temperature or above the critical volume T_m the isotherms for liquifaction of gases follow ideal gases law (above O, it acts as an ideal gas)

• At $\downarrow T_m$ or below the critical T_m (T_c) the nature of curves have altogether different appearances.

• As the pressure increases the volume of the gas decrease. In curve $A \rightarrow B$ at this point B liquifaction commences & the volume decreases as the gas is converted into a liquid with the much higher density.

• Point C : liquifaction is complete, the CD of the curve is the evidence (because volume is not changing).

• AB presents the gas state.

• BC represents the liquid or vapour or both in the equilibrium.

• CD shows liquid state only.

• Higher T_m at T_2 we get the similar type of curve to ABCD.

• At $T_m = T_c$ horizontal point is reduce to a mere point called the critical point

→ Critical Temperature :

T_m at which the gas can be liquified on the T_m above which the liquid The maximum pressure required for the liquifaction of gases at the T_c . can exist.

→ Critical volume V_c :

The volume occupied by 1 mol^{-1} gas substance at T_c & pressure.

→ Critical pressure P_c :

The maximum pressure required for the liquifaction of gases at the T_c .

** Critical region & critical constant :

The end point of the pressure, T_m curve that designates condition under the which a liquid and its vapour coexist.

T_c of oxygen = 155K.

$$P = \frac{RT}{V-b} - \frac{a}{V^2}$$

$$\left(\frac{dp}{dv}\right)_T = \frac{RT}{(V-b)^2} - \frac{2a}{V^3}$$

$$\left(\frac{d^2p}{dv^2}\right)_T = \frac{2RT}{(V-b)^3} - \frac{6a}{V^4}$$

At critical state

$$\left(\frac{dp}{dv}\right)_T = 0 \quad \& \quad \left(\frac{d^2p}{dv^2}\right)_T = 0$$

$$\frac{RT_c}{(V_c-b)^2} = \frac{2a}{V_c^3} \quad \& \quad \frac{2RT_c}{(V_c-b)^3} = \frac{6a}{V_c^4}$$

From above eqⁿ.

$$V_c = 3b$$

$$T_c = 8a/27Rb$$

$$P_c = a/27b^2$$

Law of Corresponding States.

- Since the critical constants are characteristic properties of gases, a scale can be set using them. For that reduced variables are introduced by dividing the actual by a corresponding critical constant.

Reduce parameters:

$$\begin{aligned} P_R &= P/P_c \rightarrow P = P_R \cdot P_c \\ V_R &= V/V_c \\ T_R &= T/T_c \end{aligned}$$

From Vander Waal's eqⁿ.

$$n = 1$$

$$(P + a/V^2)(V-b) = RT$$

$$\left[(P_R \times P_c) + \frac{a^2}{(V_R \times V_c)^2} \right] ((V_R \times V_c) - b) = R(T_R \times T_c)$$

$$\left[\left(P_R \times \frac{a}{27b^2} \right) + \left(\frac{a}{V_R^2 \times ab^2} \right) \right] (V_R \times 3b - b) = R T_R \times \frac{8a}{27Rb}$$

$$\left(\frac{P_r}{27} + \frac{1}{9V_r^2} \right) (3V_r - 1) = 8 \frac{T_r}{27}$$

$$\left(P_r + \frac{3}{V_r^2} \right) (3V_r - 1) = 8 T_r$$

If any 2 dissimilar gas have same T_r & P_r that means their V_r will be same. This is low corresponding state. $V_r = f(P_r, T_r)$

Calculate the volume occupied by two mol of N_2 , follows Vander Waal's eqⁿ of state at 200K and 10 MPa. $P_r V_r / T_r = 2.2$, $R = 1/2$ l atom mol⁻¹ K⁻¹

$$P_r V_r / T_r = 2.2 \Rightarrow \frac{P/P_c \times V/V_c}{T/T_c} = 2.2$$

$$PV = nRT$$

$$PV/nT = 2.2 \times P_c \times V_c / T_c$$

$$PV = nRT$$

$$\frac{PV}{nT} = R \left(\frac{P \times V}{T} \right) \times \frac{1/P_c \times 1/V_c}{1/T_c}$$

for Vander Waal's gases at critical point,

$$Z_c = 3/8$$

$$PV = ZnRT$$

$$Z = PV/nRT \quad (n=1)$$

$$Z_c = \left(\frac{a}{27b^2} \right) \frac{3b}{R \times (8a/27Rb)}$$

$$PV = ZnRT$$

$$Z = PV/nRT \quad (n=1)$$

$$P_c V_c / RT_c = Z_c = 3/8$$

$$P_c V_c / T_c = 3/8 R$$

$$\frac{PV}{nT} = 2.2 \times 3/8 \times 1/2$$

$$\frac{10 \times 10^6 \times V}{2 \times 200} = 2.2 \times 3/8 \times 1/2$$

$$V = 2.75 \times 10^{-6} \text{ m}^3$$

Physical Chemistry

I - Gases

II - Thermodynamics - Energy & transformation of the system.

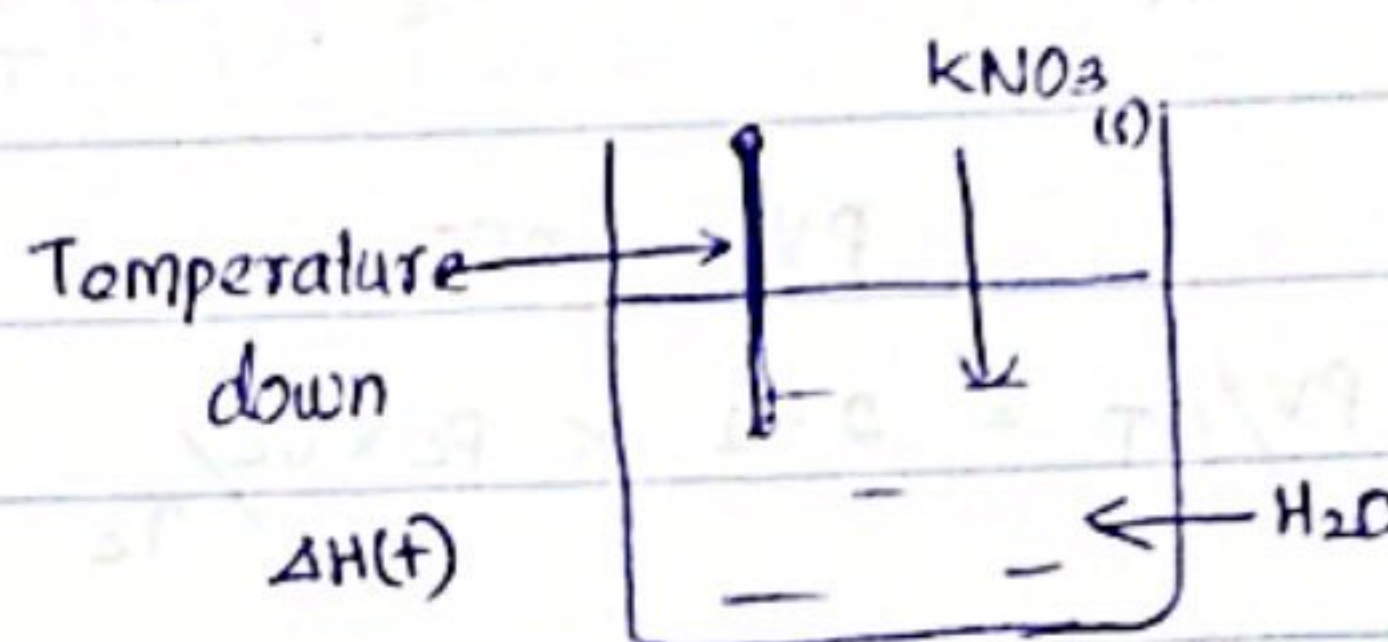
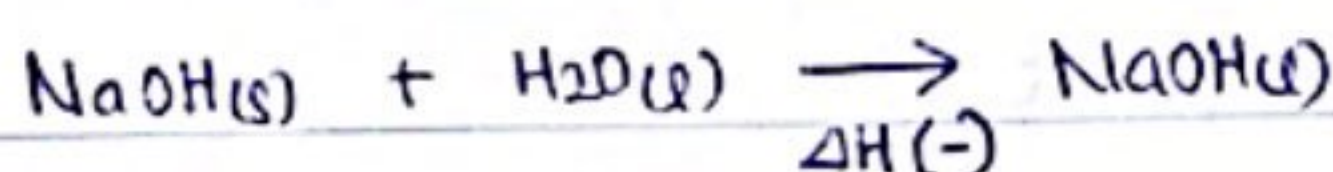
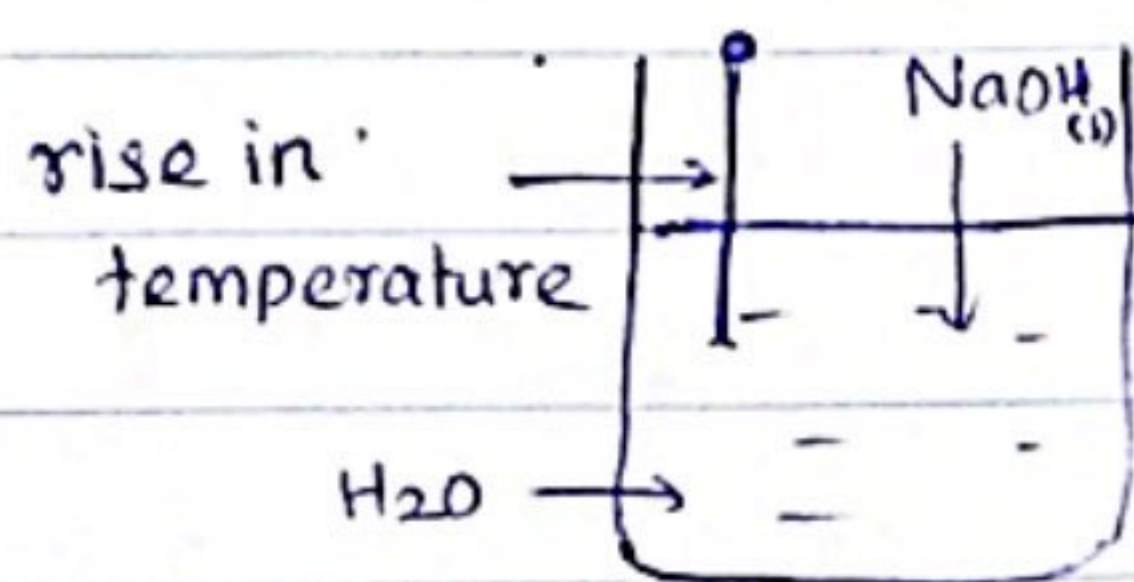
III - Electrochemistry

Thermodynamics



Absorbing of energy
endothermic ($\Delta H +$)

Energy is given out
Exothermic ($\Delta H -$)

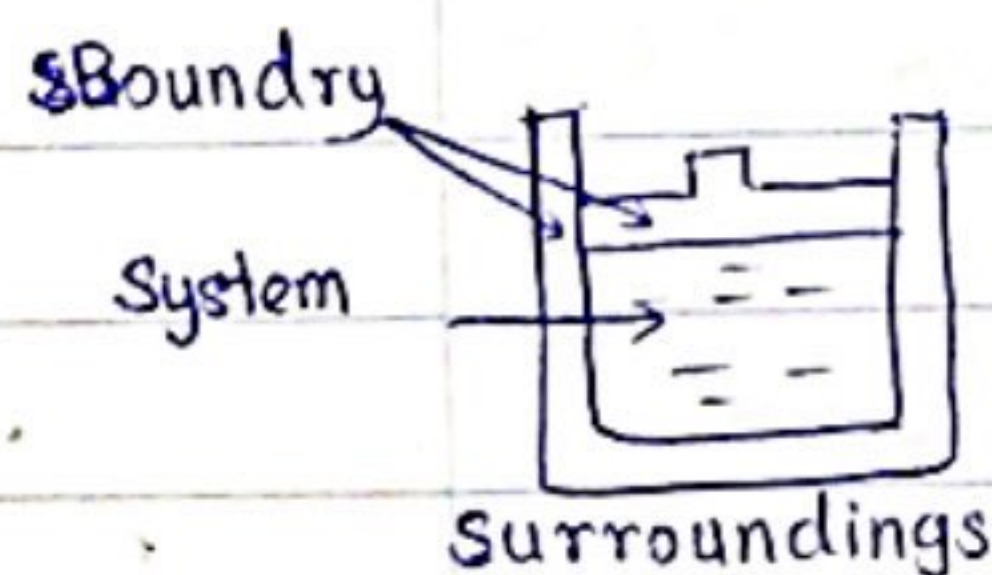


** Terminology used in thermodynamics.

System : Can be big or small (ex: such as biological cell)

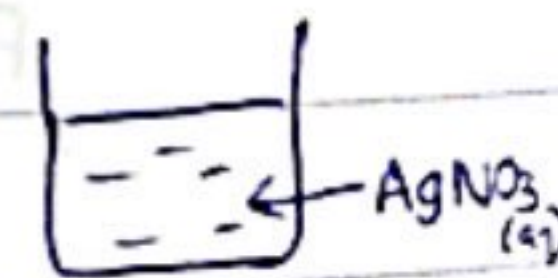
• System is a part of universe which our thermodynamics interest is focused.

• Everything else around the system called as surroundings.



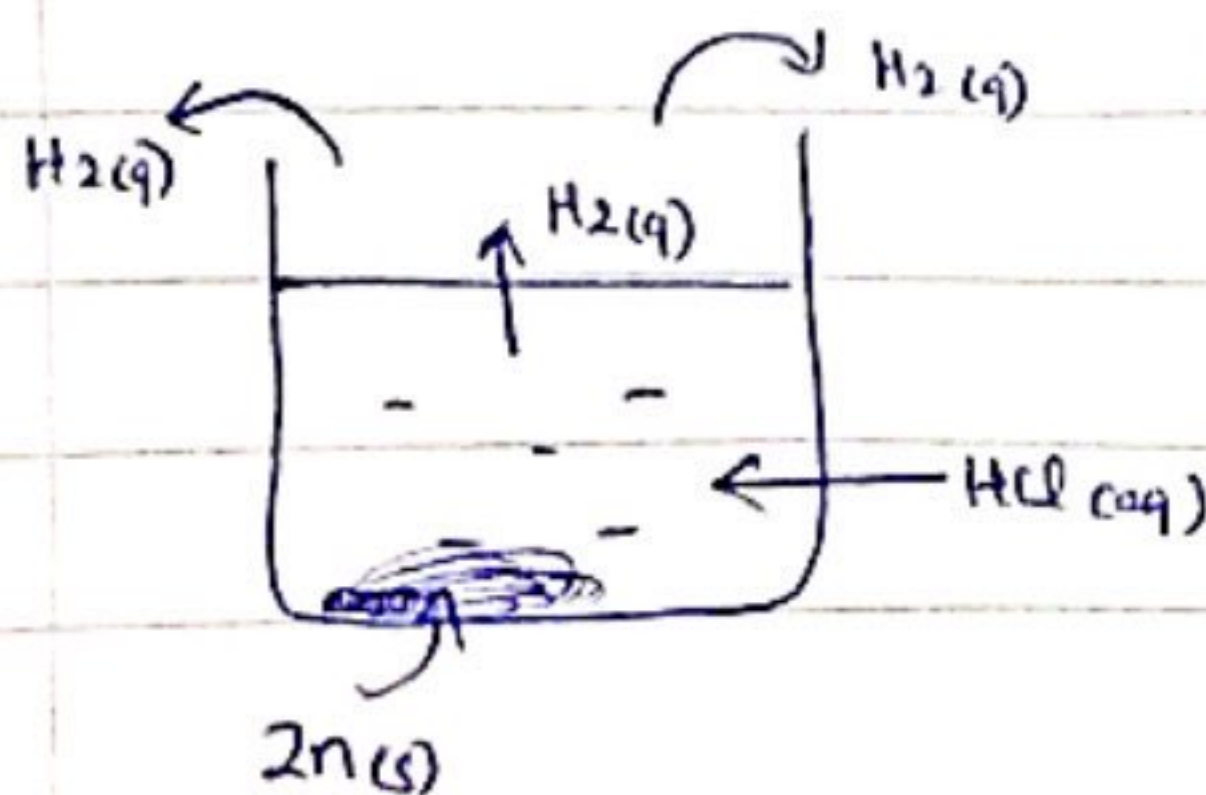
• System + surroundings $\rightarrow$ universe

• There is a boundary between the system & surroundings. It allows exchange heat.



Types of Thermodynamics System

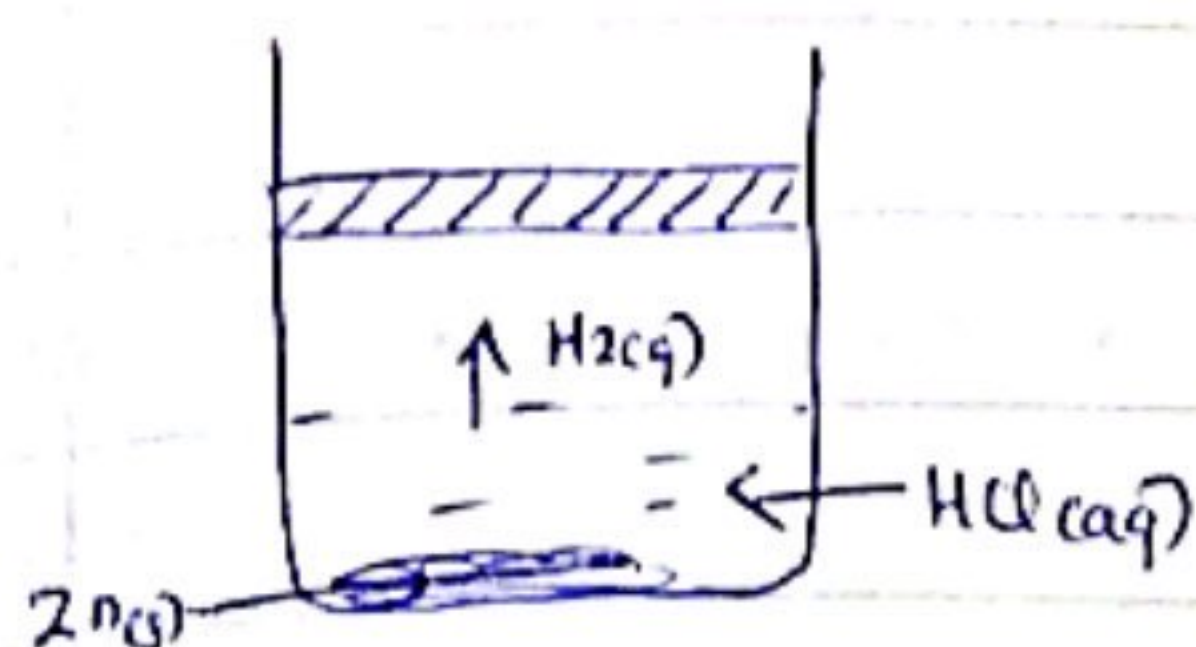
(a) Open system :



Can exchange both energy & matter with its surrounding.

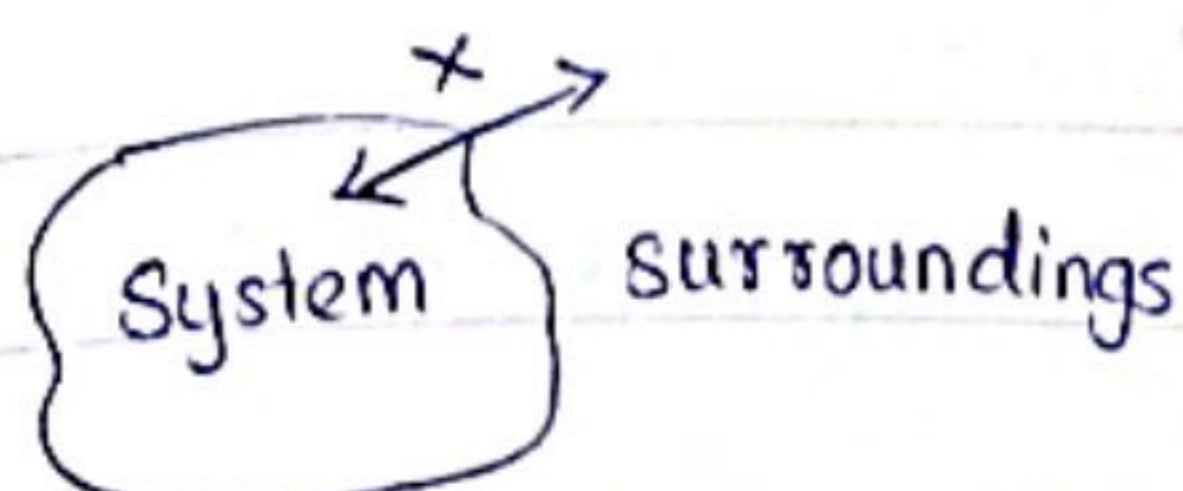
Ex: The boiling tea bottle.

(b2) Closed system :



Energy can exchange with its surrounding
But not matter.

(b3) Isolated system :



This system can exchange neither energy or
matter with its surrounding.

Ex : Universe , Thermoflask .

Adiabatic system : Boundary can be a heat conductor .

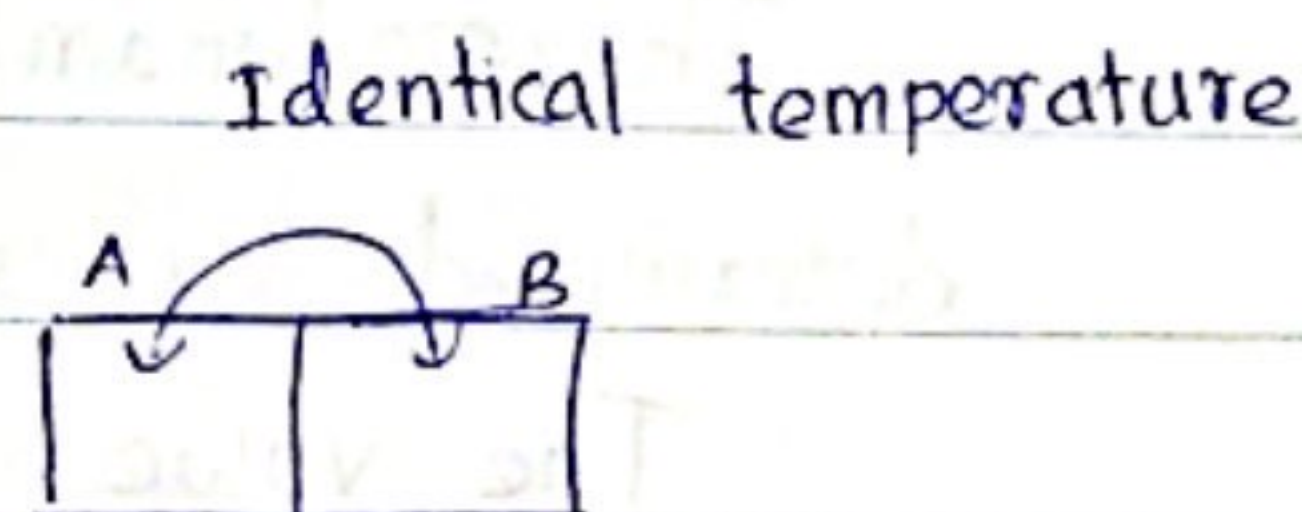
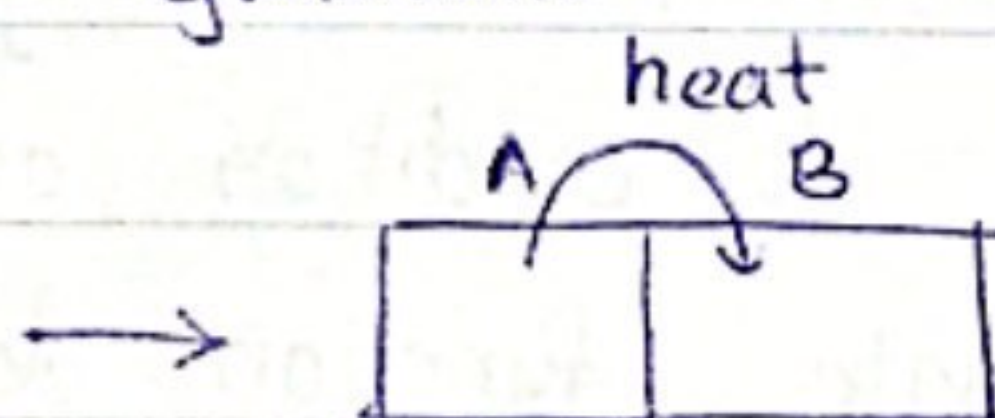
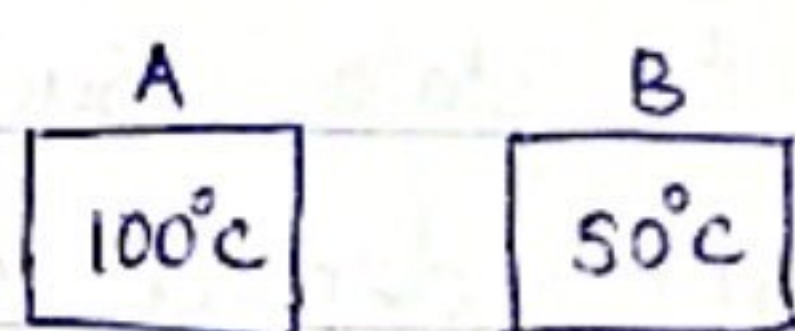
Equilibrium in Thermodynamics System

(1) Mechanical equilibrium :

No unbalanced forces existing between different parts or the system or
between the system & surroundings . Uniformly in P it is essential
zero pressure gradient .

(2) Thermal equilibrium :

Zero temperature gradient .



(3) Chemical equilibrium :

- The composition of the different part of the system is same .
- Zero concentration gradient .

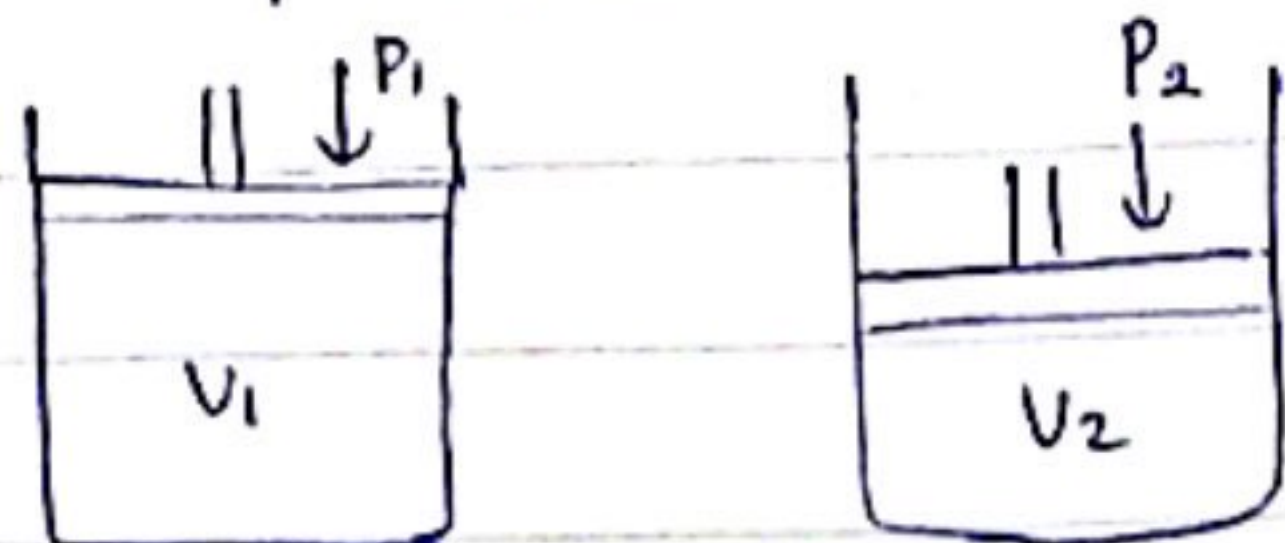
$$c_1 > c_2 \quad c_1 \neq c_2$$

** The system which satisfied all three equilibrium is to be thermodynamic
equilibrium .

(oi) Microscopic Approach :

A system in terms of events occurring at atomic or molecular levels
which cannot be perceived by human senses . Since the number of
constituents particles in the system is significantly higher , a statical
approach is higher .

(02) Macroscopic Approach :



$$\uparrow p \vee \downarrow = nRT$$

Don't care about atomic or molecular structure : care about the bulk of the system.

Classical Thermodynamics : Based on macroscopic behavior in bulk & can be study without ever taking advantages of our knowledge of the structure or behavior of atoms or molecules.

Macroscopic approach bases on the bulk properties of the matter as a whole, such as T, P, V .

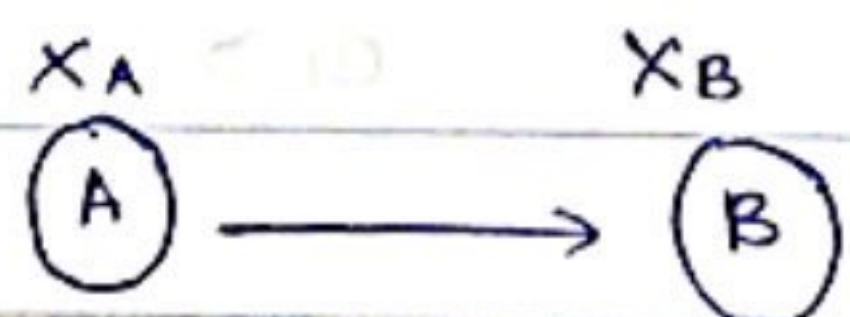
Laws of Thermodynamic

Based on 3 generalized laws of nature. That have been resulted all the years on the basis of natural experience & observations. Therefore these laws are more empirical than theoretical. There are 3 laws.

State Variables / Thermodynamic Functions

Thermodynamic function is a property of the system that is determined by specifying its condition or its state. Such as T, P etc.

The value of a state function doesn't depend on the previous history or mechanism by which it was obtained. It depends only on the initial & final states.

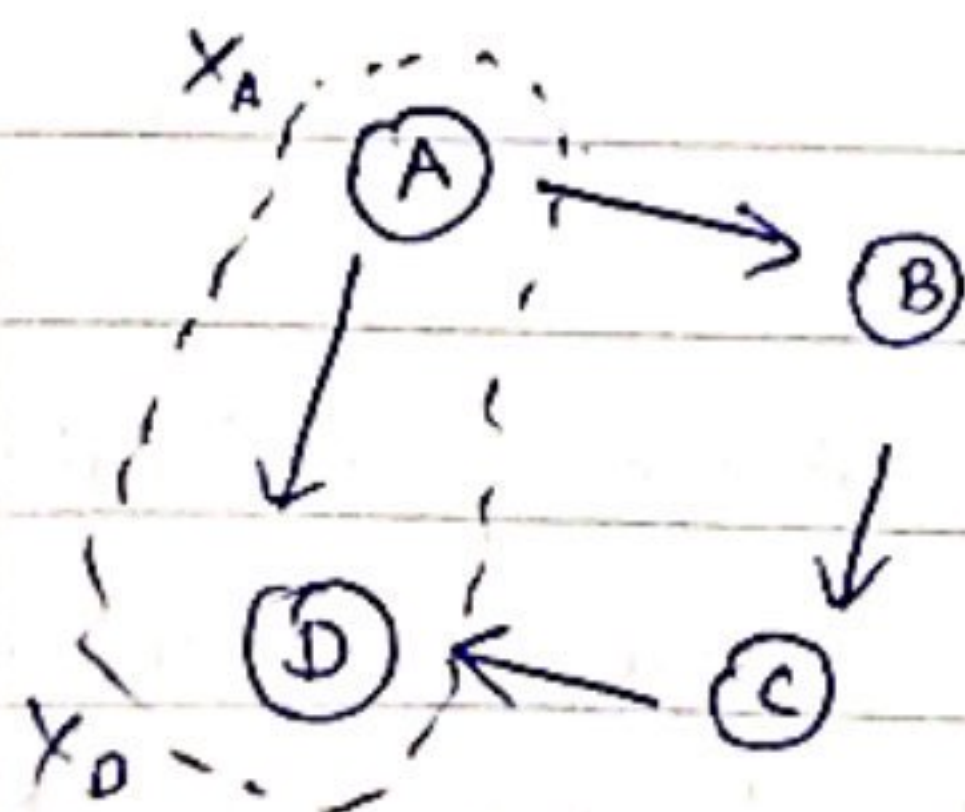


X_A : Initial state

X_B : Final state.

$$\Delta X = X_B - X_A$$

** Thermodynamic functions depends only on initial & final states.

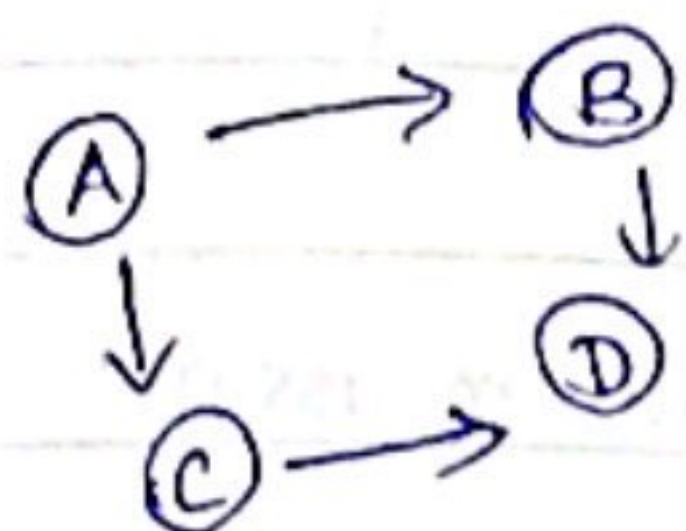


$$\Delta X = X_D - X_A$$

for a small change $dx = X_B - X_A$

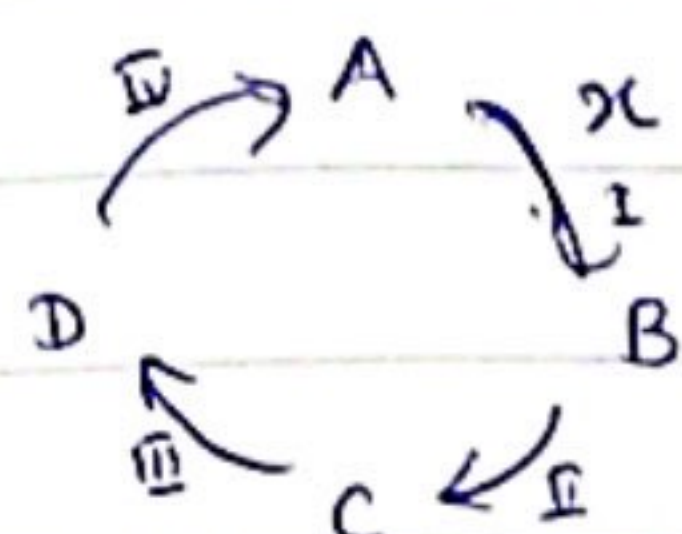
Path Properties

Assistant can exchange energy with its surrounding as the heat (q) or work (w). So the value of q or w depends on the path by which the change occurs.



Depends on initial & final states. not in all the states.

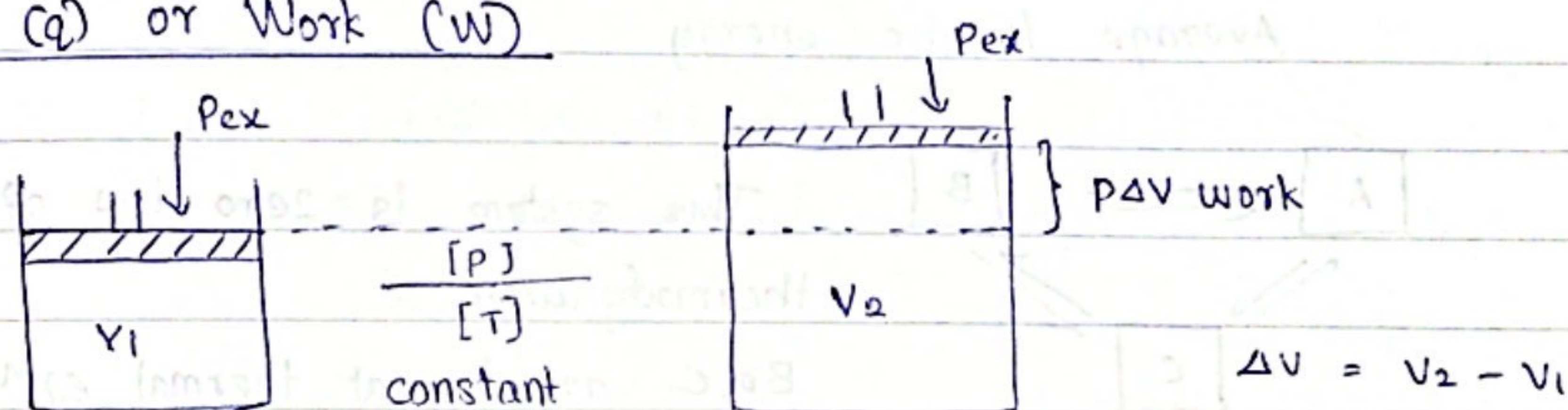
A cycling Process: When a state of a system changes forming new system or new states and finally come back to the original state (A). Then the value of any thermodynamic function (x) is equal to 0.



$$\oint dx = 0$$

Integration has been carried out over the cyclic process

Heat (q) or Work (w)



The change of a state of a system is usually accompanied not only by the performance of work. But also by an exchange of heat between the system & surroundings.

For an infinitesimal volume change dv ,

$$\text{work done, } dw = P_{ex} dv \quad \text{--- (1)}$$

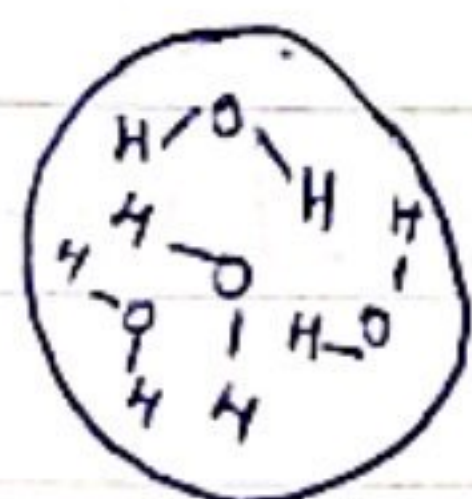
is also very small

For a finite process eq (1) should be integrated,

$$w = \int dw = P_{ex} \int_{V_1}^{V_2} dv$$

$$w = P_{ex} (V_2 - V_1)$$

Internal Energy (U)



Total Energy of the system = KE + PE + Electronic Energy + Nuclear Energy.

$$\Sigma \text{ Translational} + \Sigma \text{ Vibration} + \Sigma \text{ Rotational}$$

movement of the energy
(movement of the 3 dimensional)

Vibrating energy
 $\Rightarrow$

Molecule can rotate around the axis
 $\updownarrow$

** Cannot measure absolute internal Energy. Can measure changing internal energy by monitoring the heat exchange and work done by the process (u).

First Law of Thermo

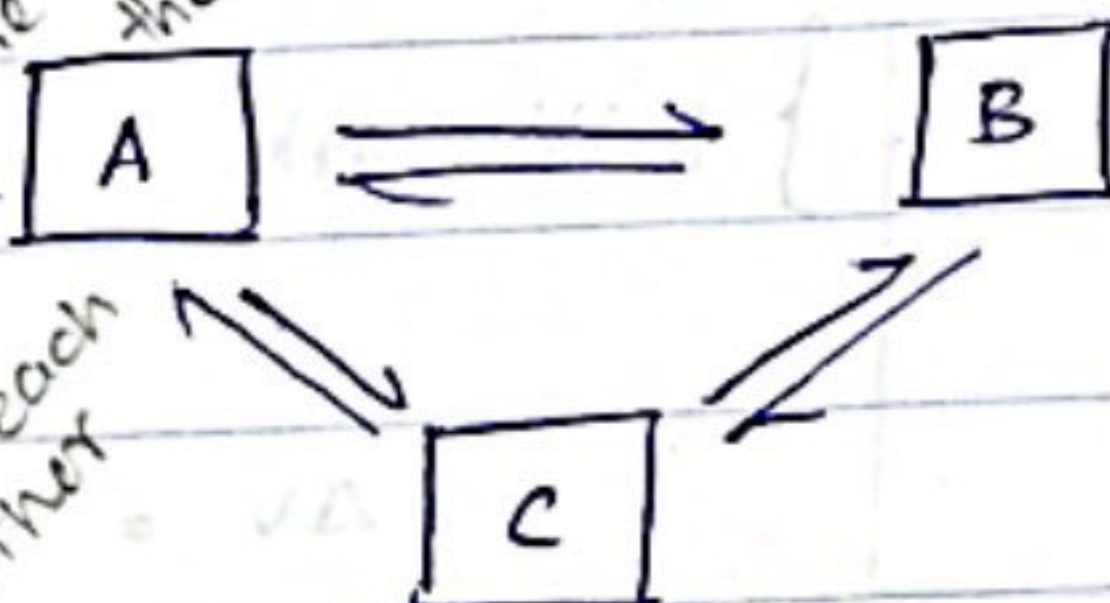
The law of conservation of energy.

18.09.2023

Temperature :

Average kinetic energy

When 2 bds are in thermal equilibrium with 3rd bdy then the 2 bds are also in thermal equilibrium with each other.



This system is zero law of thermodynamic
B & C are also at thermal eq^m.

Temperature is a thermodynamic property. that determines whether or not a system is in thermo eq^m with other system.

Properties :

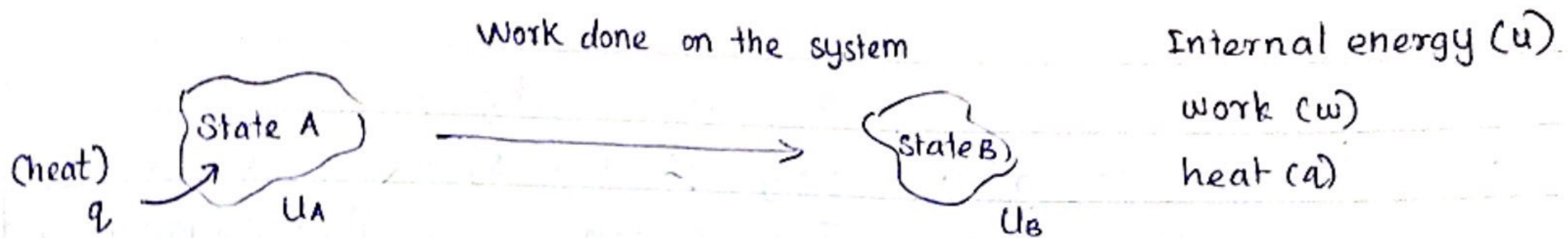
1. Extensive property - depends on the amount of the substance (Ex : mass, volume, heat capacity)
2. Intensive property - Doesn't depend on the amount. (Ex : density, temperature, molar heat capacity)

(2)

First Law of Thermodynamics

The law of conservation of energy.

* $\leftarrow$ Energy of the universe can neither be created nor destroyed.



Total energy of state A = $U_A + q + w$

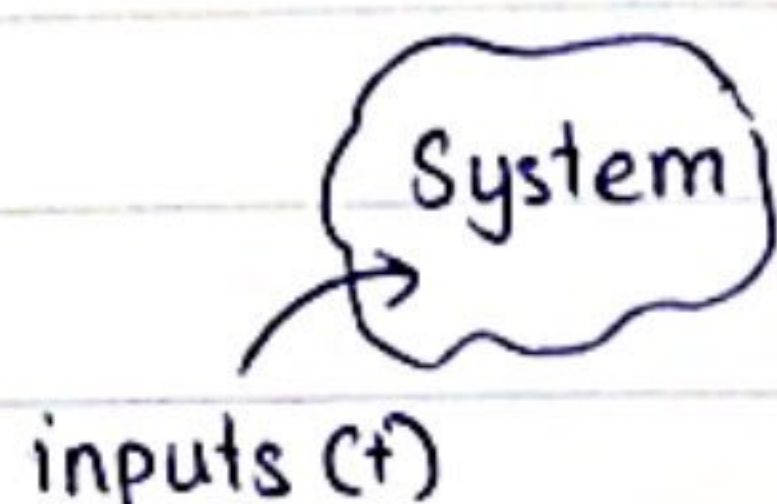
" " " " State B = U_B

$$U_B = U_A + q + w$$

$$\therefore U_B - U_A = q + w$$

Net change in internal energy $\Delta U = q + w$

Sign convention :



The work done on the system is (+) positive.

The work done by the system (-) negative.

Isochoric Process : (constant volume)

$$dv = 0 \quad \therefore \quad p dv (w) = 0$$

$$\therefore \Delta U = q$$

Adiabatic Process ($dq = 0$)

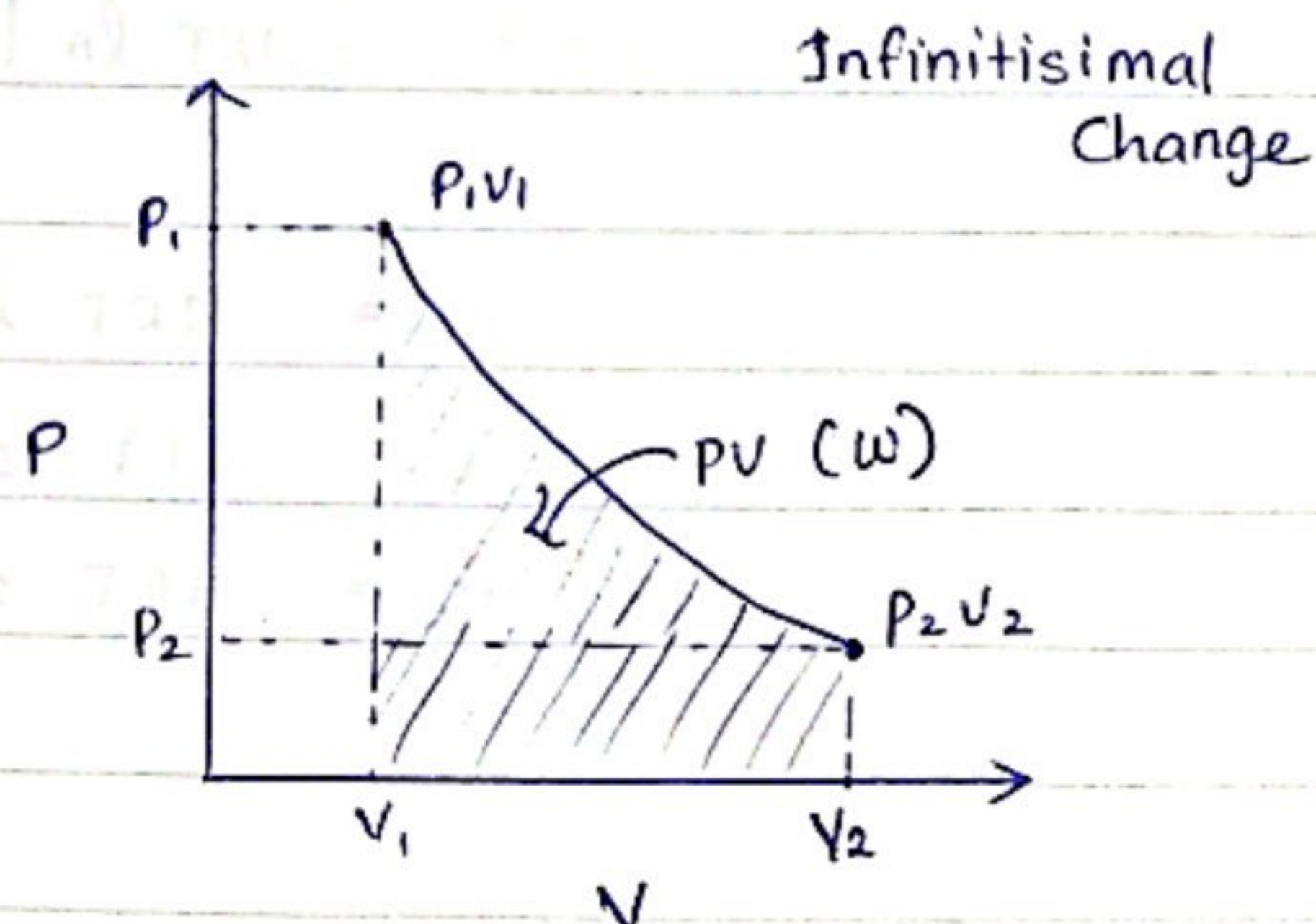
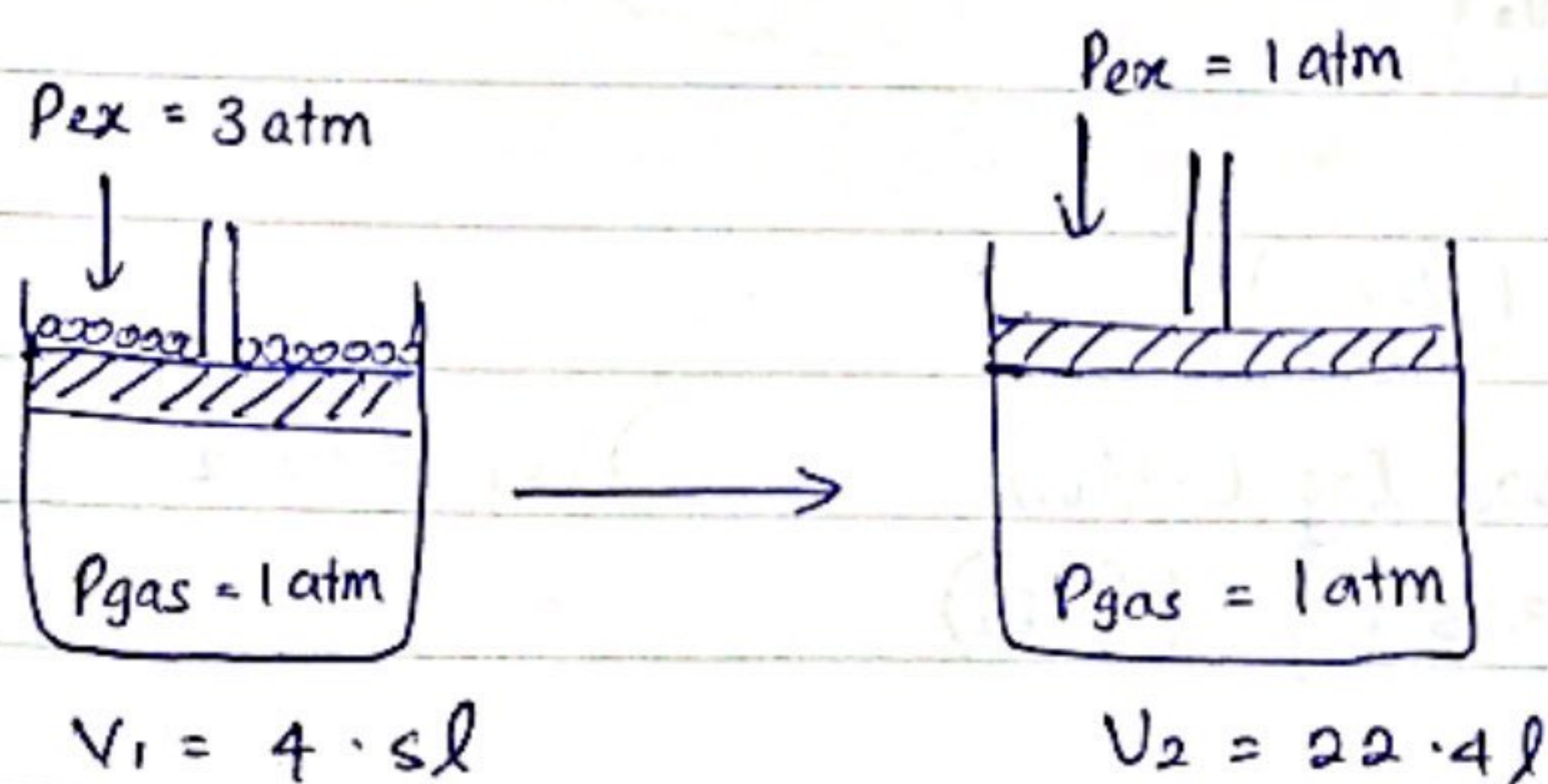
$$\therefore \Delta U = w$$

In a closed system the internal energy is function of v, T .

$$u = f(v, T)$$

$$du = \left(\frac{\partial u}{\partial v} \right)_T dv + \left(\frac{\partial u}{\partial T} \right)_v dT$$

Reversible VS Irreversible process :



A process that can be reversed by inducing infinitesimal change to some property of the system is called a reversible process. This system is in thermodynamic eq^m with its surrounding.

A process that cannot be reversed by inducing infinitesimal change to some property of the system. This system has no thermodynamic eq^m with its surrounding.

Work done by reversible process $\gg$ work done by irreversible process.

for irreversible $W = p (v_f - v_i)$

free expansion :

The stop is open



Vacuum.

the gas expands against the 0 pressure is called free expansion.

$$\begin{aligned} W &= P_{ex} (v_2 - v_1) \\ &= 0 (v_2 - v_1) \\ &= 0 \end{aligned}$$

Reversible expansion :

$$W = P_{ex} dv$$

$$W = P_{sys} dv$$

$$W = nRT \cdot \frac{dv}{v}$$

$$P_{ex} = P_{system}$$

$$Pv = nRT$$

$$P_{sys} = nRT/v$$

(Isothermal process ; $dT = 0$)

$$\begin{aligned} W &= nRT \int_{v_1}^{v_2} \frac{dv}{v} \\ &= nRT \ln v \\ &= nRT \ln [v]_{v_1}^{v_2} \end{aligned}$$

$$\ln x = 2.303 \lg x$$

$$W = nRT \ln (v_2/v_1)$$

$$W = nRT \cdot 2.303 \log (v_2/v_1)$$

$$W = nRT \cdot 2.303 \log (P_1/P_2)$$

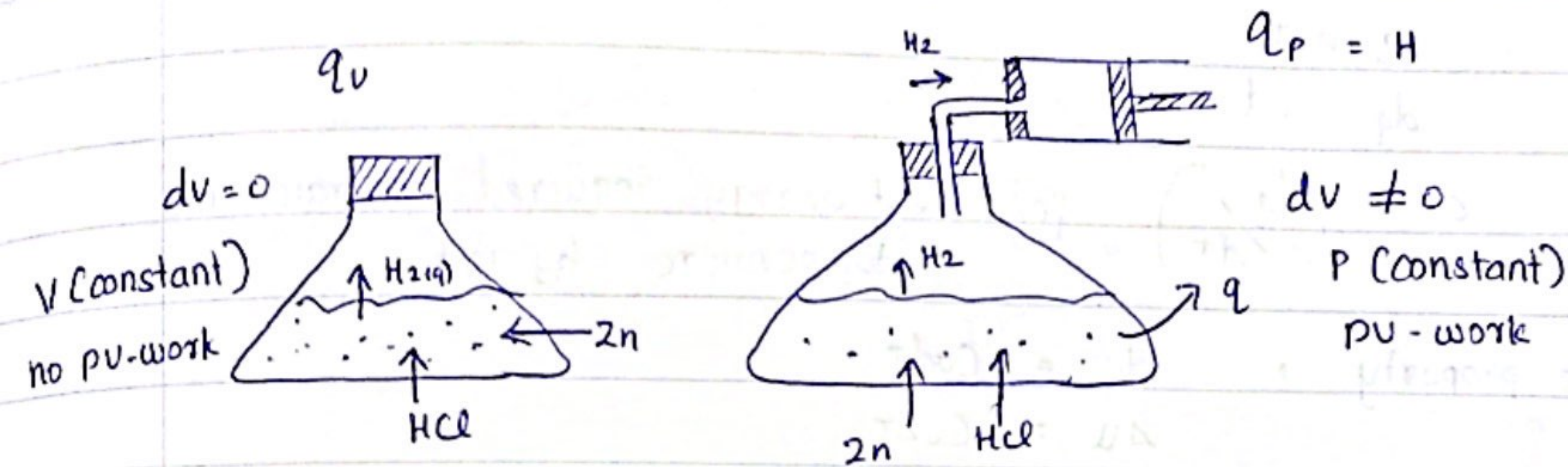
$$P_1 v_1 = P_2 v_2$$

$$q_{\text{system}} = -q_{\text{surrounding}}$$

$$W_{\text{sys}} = -W_{\text{surr}}$$

$$\Delta U_{\text{sys}} = -\Delta U_{\text{surr}}$$

Enthalpy Change (ΔH)



The heat exchange under constant pressure is call as enthalpy

The heat exchange under " volum is call as internal energy

both are not identical

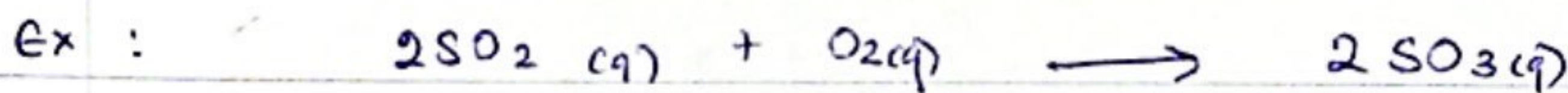
$$q = H + PV$$

$$H = \int (PdT)$$

$$H = \left(\frac{dH}{dP} \right)_T \Delta P + \left(\frac{dH}{dT} \right)_P \Delta T$$

$$\Delta H = \Delta U + P\Delta V$$

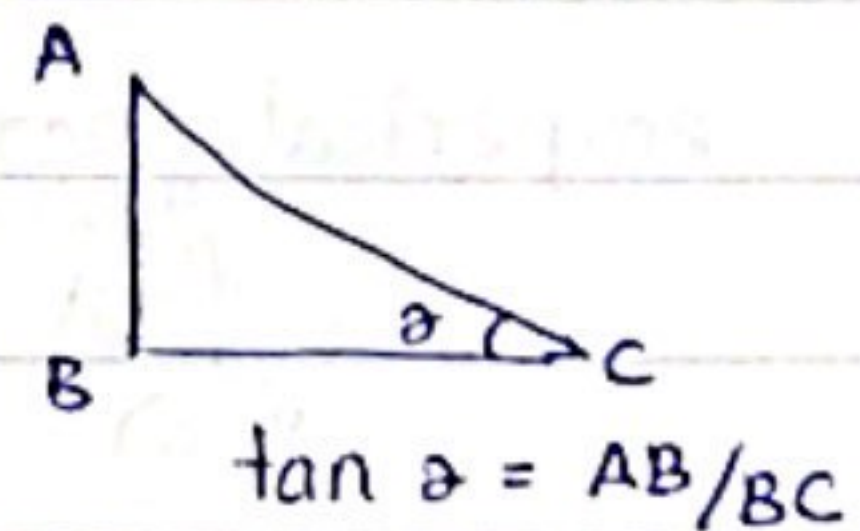
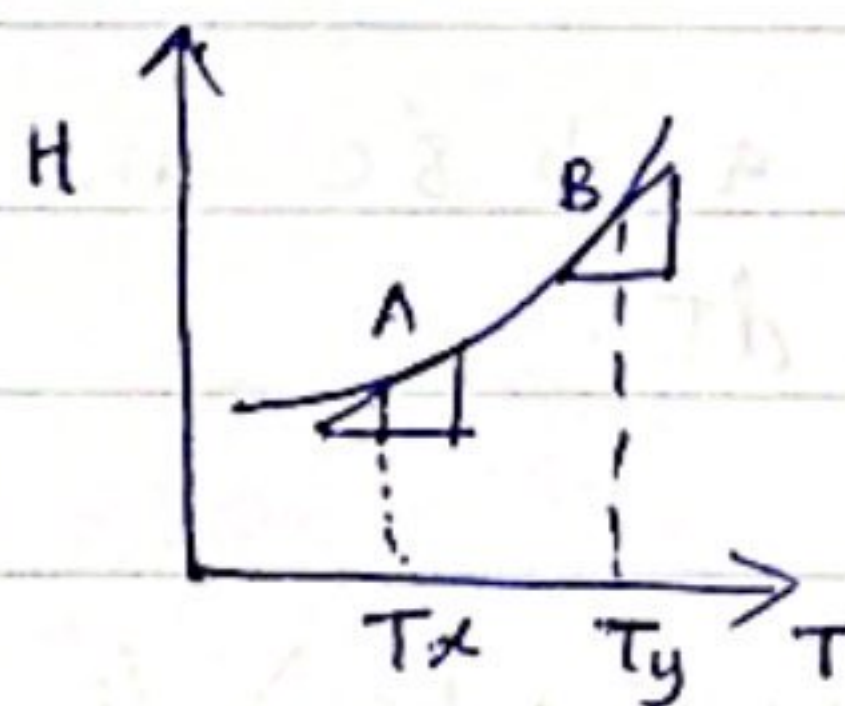
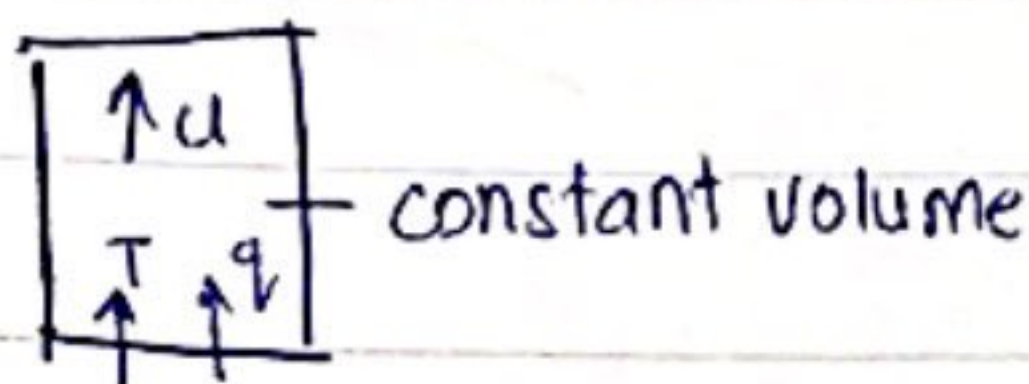
$$\Delta H = \Delta U + \Delta nRT \quad \text{number of moles when gas changes}$$



$$\Delta n = 2 - 3 = -1$$

25.09.2023

Heat Capacity



Slope $\frac{dy}{dx}$ = heat capacity

at constant volume $C_v = \frac{du}{dT}$

at constant pressure $C_p = \frac{dH}{dT}$

$T_m \uparrow$ 1 unit

$$H = U + PV$$

$$\Delta H = \Delta U + P\Delta V$$

$$\Delta n = n_p - n_r$$

$$\Delta H = \Delta U + \Delta nRT$$

$$q_p = dH$$

$$q_v = du$$

$$dq \propto T$$

$$dq = c dT$$

$$c = \left(\frac{dq}{dT} \right)$$

heat energy required to raise the temperature by unit.

extensive property

$$du = C_v dT$$

$$\Delta U = C_v \Delta T$$

$$q_v = n C_v dT$$

$$q_p = n C_p dT$$

$$q_v = n C_v dT$$

$$\int q_p = n C_p \int_{T_1}^{T_2} dT$$

$$\Delta H = n C_p (T_2 - T_1)$$

$$\Delta U = n C_v (T_2 - T_1) \text{ range is very small}$$

monoatomic ideal gas (He)

Variation of heat capacity with Temperature.

$$C_{pm} = a + bT + c/T^2 \dots$$

empirical parameters a , b & c are independent on T_m

$$\int_{H(T_1)}^{H(T_2)} dH = \int_{T_1}^{T_2} C_{pm} dT$$

$$\int_{T_1}^{T_2} dH = \int_{T_1}^{T_2} \left(a + bT + \frac{c}{T^2} \right) dT$$

$$H(T_2) - H(T_1) = a \int dT + b \int T dT + \int \frac{c}{T^2} dT$$

$$= a(T_2 - T_1) + \frac{1}{2}b(T_2^2 - T_1^2) - c\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$C_p = \left(\frac{\partial H}{\partial T} \right)_p \quad \text{--- (1)}$$

$$C_v = \left(\frac{\partial U}{\partial T} \right)_v \quad \text{--- (2)}$$

$$C_p - C_v = \left(\frac{\partial H}{\partial T} \right)_p - \left(\frac{\partial U}{\partial T} \right)_v \quad \text{--- (3)}$$

$H = U + pV$ differentiate with respect to T at constant p .

$$\left(\frac{\partial H}{\partial T} \right)_p = \left(\frac{\partial U}{\partial T} \right)_p + p \left(\frac{\partial V}{\partial T} \right)_p \quad \text{--- (4)}$$

$$U = f(V, T)$$

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT \quad \text{--- (5)}$$

eqⁿ (5) / dT at constant p .

$$\left(\frac{\partial U}{\partial T} \right)_p = \left(\frac{\partial U}{\partial T} \right)_v + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_p \quad \text{--- (6)}$$

Substitute eqⁿ (6) in (4):

$$\left(\frac{\partial H}{\partial T} \right)_p = \left(\frac{\partial U}{\partial T} \right)_v + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_p + p \left(\frac{\partial V}{\partial T} \right)_p \quad \text{--- (7)}$$

$$C_p = C_v + \left(\frac{\partial V}{\partial T} \right)_p \left[p + \left(\frac{\partial U}{\partial V} \right)_T \right]$$

$$C_p - C_v = \left(\frac{\partial V}{\partial T} \right)_p \left[p + \left(\frac{\partial U}{\partial V} \right)_T \right]$$

$$C_p - C_v = \left(\frac{\partial V}{\partial T} \right)_p p + \left(\frac{\partial V}{\partial T} \right)_p \left(\frac{\partial U}{\partial V} \right)_T$$

$$p = nRT/V$$

$$V/T = nR/p$$

$$\left(\frac{\partial V}{\partial T} \right)_p = nR$$

$$C_p - C_v = nR + \left(\frac{\partial V}{\partial T} \right)_p \left(\frac{\partial U}{\partial V} \right)_T$$

For an ideal gas, expansion will not result change in U & H .

$$\left(\frac{\partial U}{\partial V} \right)_T = \left(\frac{\partial H}{\partial p} \right)_T = 0$$

$$\therefore C_p - C_v = nR$$

$$dq = 0$$

Adiabatic equation of state.

$$\Delta U = q + w$$

$$w = -pdv$$

$$\Delta U = w$$

$$dU = -pdv$$

$$p = nRT/V$$

$$C_v dT = -pdv$$

$$C_v dT = -nRT \frac{dv}{v}$$

$$C_v \int_{T_1}^{T_2} \frac{dT}{T} = nR \int_{v_1}^{v_2} \frac{dv}{v}$$

$$dU = C_v dT$$

$$\left(\frac{\partial U}{\partial T} \right)_v = C_v$$

$$\left(\frac{\partial H}{\partial T} \right)_p = C_p$$

$$C_p - C_v = R$$

$$C_p/C_v - 1 = R/C_v \quad \gamma = C_p/C_v$$

$$\gamma - 1 = R/C_v$$

$$C_v = \ln T_2/T_1 = -R \ln v_2/v_1$$

$$\ln T_2/T_1 = \ln (v_1/v_2)^{R/C_v}$$

$$T_2/T_1 = (V_1/V_2)^{\gamma-1}$$

$$(T_2 V_2)^{\gamma-1} = (T_1 V_1)^{\gamma-1}$$

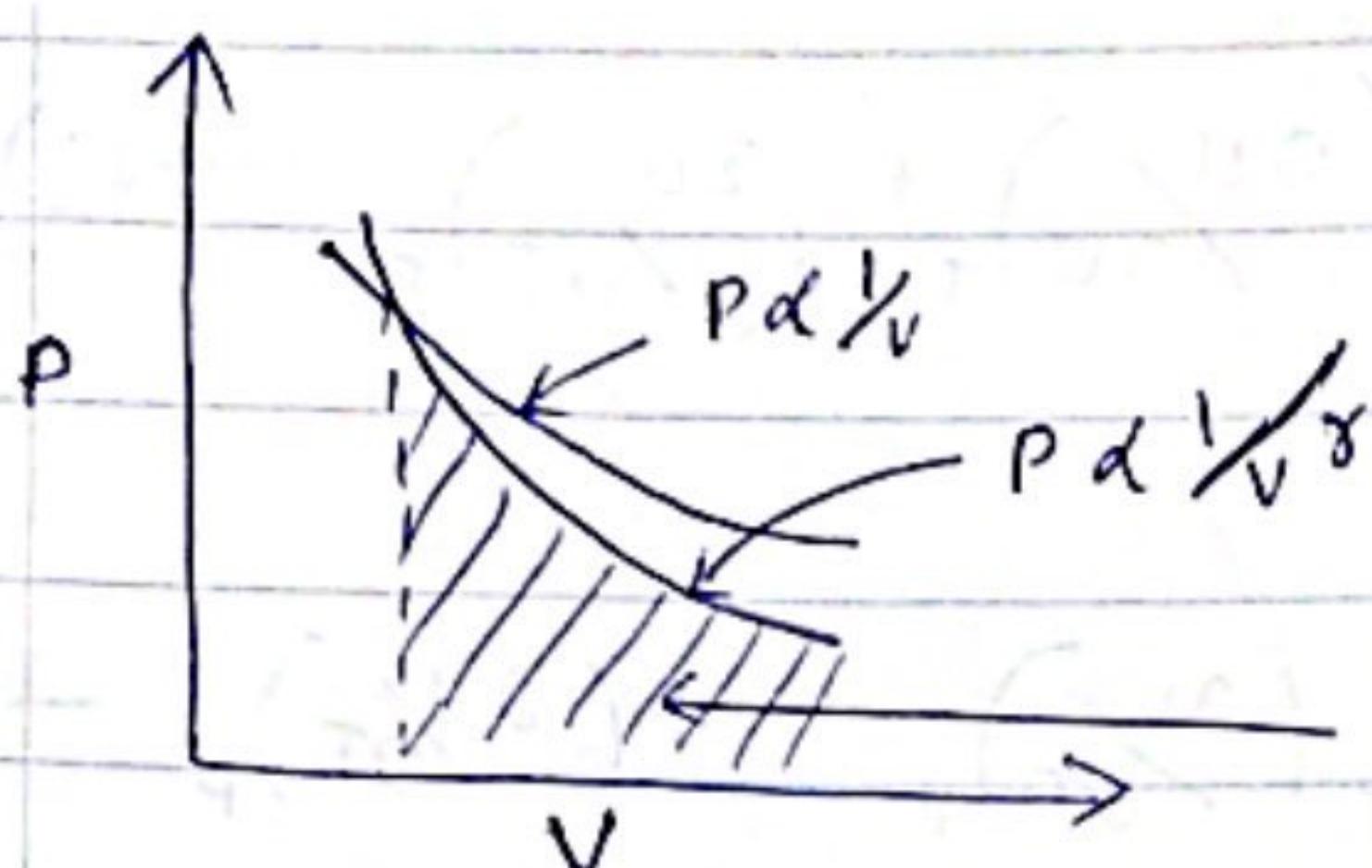
$$P_1 V_1 / T_1 = P_2 V_2 / T_2$$

$$T_2/T_1 = P_2 V_2 / P_1 V_1 = (V_1/V_2)^{\gamma-1}$$

$$P_2 V_2^{\gamma} = P_1 V_1^{\gamma}$$

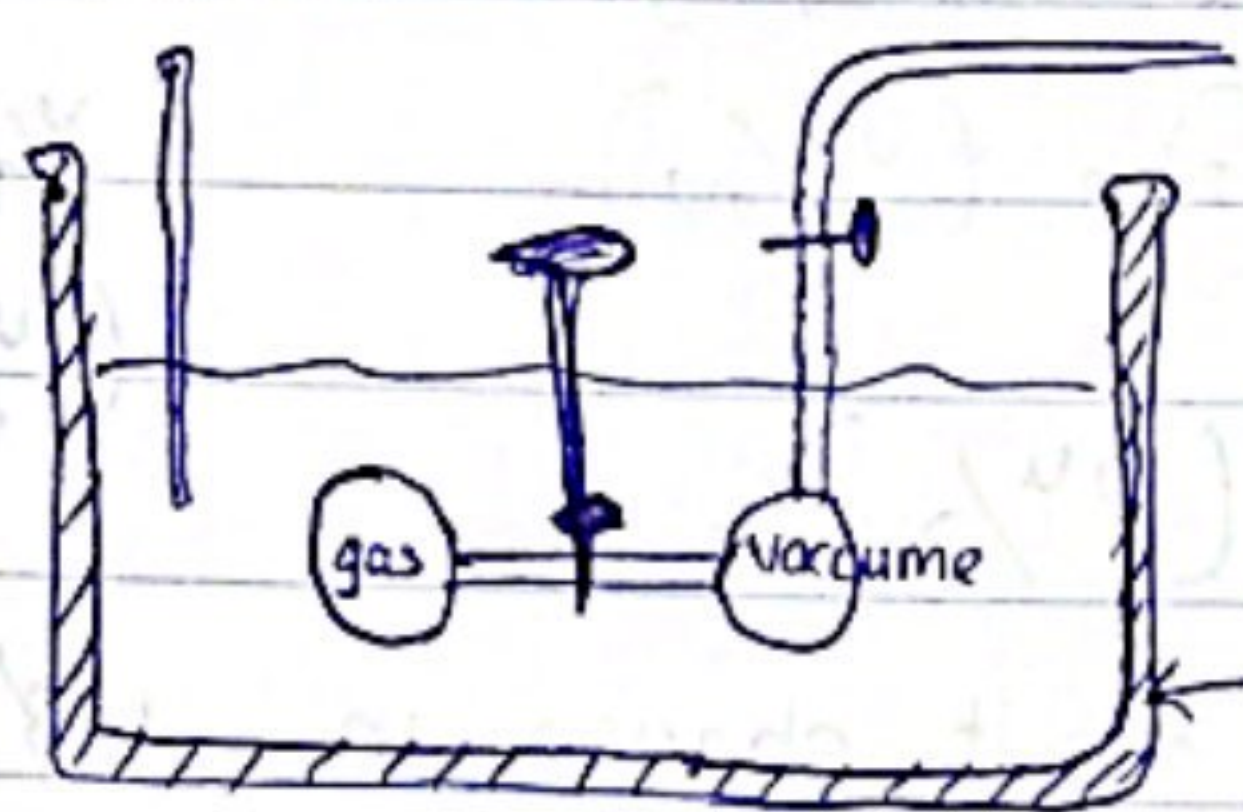
$$PV^{\gamma} = \text{constant}$$

$$C_p > C_v \quad C_p / C_v = \gamma \quad \text{greater than unity}$$



Adiabatic reversible expansion.

Joule - Thomson Experiment.



expansion of a gas to a vacuum ($P_{\text{ext}} = 0$)
with no change in temperature, $dT = 0$

$$w = -P_{\text{ext}} dv = 0$$

$$dq = 0 \quad (\text{adiabatic system})$$

$$\Delta u = q + w$$

$$\Delta u = 0$$

$$\left(\frac{\partial u}{\partial v} \right)_T = 0 : \text{ideal gas}$$

$$\left(\frac{\partial u}{\partial v} \right)_T \neq 0 : \text{real gas}$$

When a gas is allowed to expand or else rigid porous plug from a higher pressure to lower pressure under adiabatic condition fall in temperature of gas was observed. This is called Joule - Thomson effect.

$$C_v, C_p$$

$$dH = n C_p dT \quad \text{for small heat changing enthalpy.}$$

$$du = n C_v dT$$

$$n C_v dT = d \left(n \frac{3}{2} RT \right)$$

$$nC_v dT = 3nR/2 dT$$

$$C_v = 3/2 R \quad \leftarrow \text{monoatomic gas}$$

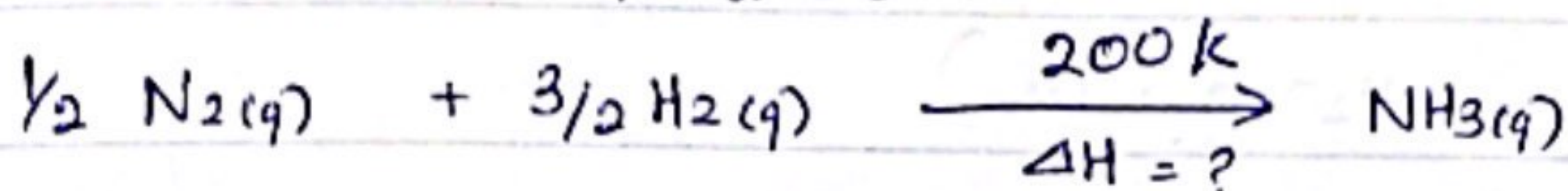
$$nC_p dT = d(n^{3/2}RT) + d(nRT)$$

$$C_p = 3R/2 + R = 5R/2$$

$$C_p = 5R/2$$

for diatomic gas, $C_v = 5R/2$, $C_p = 7R/2$

* Reactants $\longrightarrow$ Products.



Changing

Heat capacity in constant pressure for reaction ΔC_p ,

$$\Delta C_p = \sum C_{p(p)} - \sum C_{p(r)}$$

$$= C_p(NH_3) - \frac{1}{2} C_p(N_2) + \frac{3}{2} C_p(H_2)$$

$$= 33 \text{ kJ mol}^{-1} - (\frac{1}{2} \times 29 + \frac{3}{2} \times 28.8)$$

$$= -22.76 \text{ J mol}^{-1}$$

$$\Delta H = \Delta C_p dT$$

$$\int_{H_1}^{H_2} \Delta H = \Delta C_p \int_{T_1}^{T_2} dT$$

$$\Delta H_2 - \Delta H_1 = \Delta C_p (T_2 - T_1)$$

$$\Delta H_2 = (-22.76 (400 - 298) + 46.11) \text{ kJ mol}^{-1}$$

$$= -48.4 \text{ kJ mol}^{-1}$$

Positive Joule Temperature Effect

The gas expand to high P to low P - Temperature drops.

Negative Joule Temperature Effect

For same gases even the pressure drops temperature increases.

- $H_2(g)$

$$\left(\frac{\partial H}{\partial P}\right)_T = 0, \quad \left(\frac{\partial U}{\partial V}\right)_T = 0$$

$$\left(\frac{\partial P}{\partial T}\right)_H \rightarrow \text{Joule Thomson coefficient } (\mu)$$

$$\Delta T = \frac{-\left(\frac{\partial H}{\partial P}\right)_T \Delta P}{C_p}$$

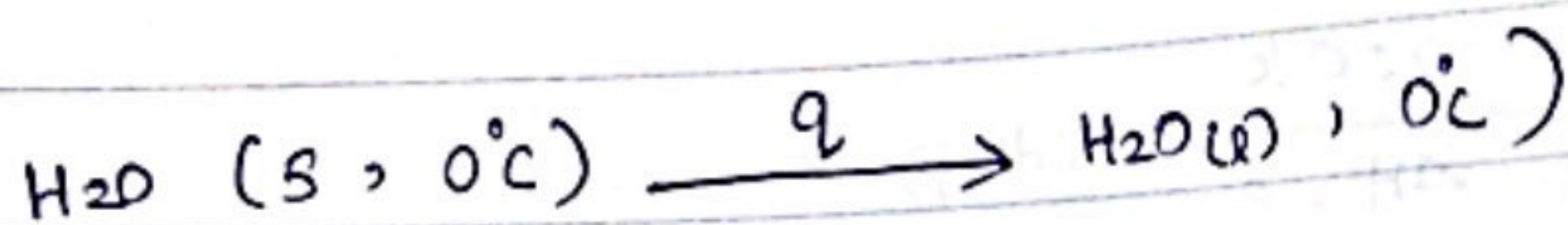
For ideal gas $H = U + PV$

$$H = \left[\frac{\partial (U + PV)}{\partial P} \right]_T / C_P$$

$$= -\frac{1}{C_P} \left[\left(\frac{\partial U}{\partial V} \cdot \frac{\partial V}{\partial P} \right)_T + \left(\frac{\partial (PV)}{\partial P} \right)_T \right]$$

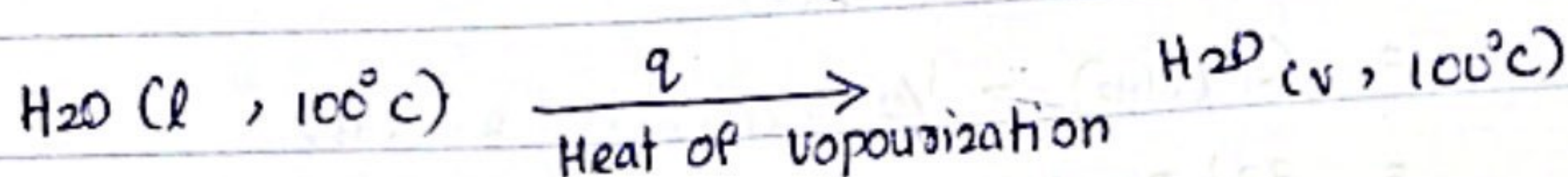
Phase Change

Solid $\rightarrow$ liquid $\rightarrow$ Vapour



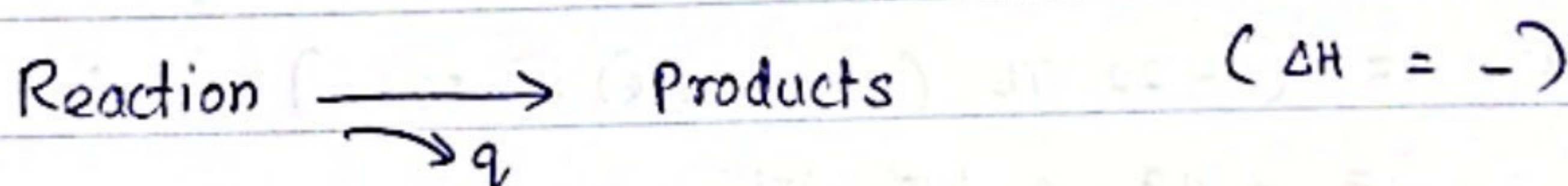
fusion, $q \text{ d.m}$

$$q = m \Delta_{fu} H \text{ at constant } P \quad q_P = \Delta H$$



$$\therefore q = m \Delta_{vap} H$$

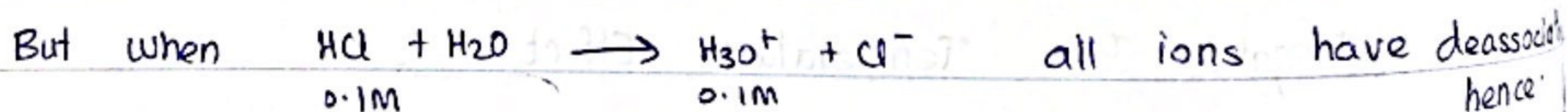
Spontaneous Process of reaction.



If heat given out in the reactions, Products become more stable than the reactants.



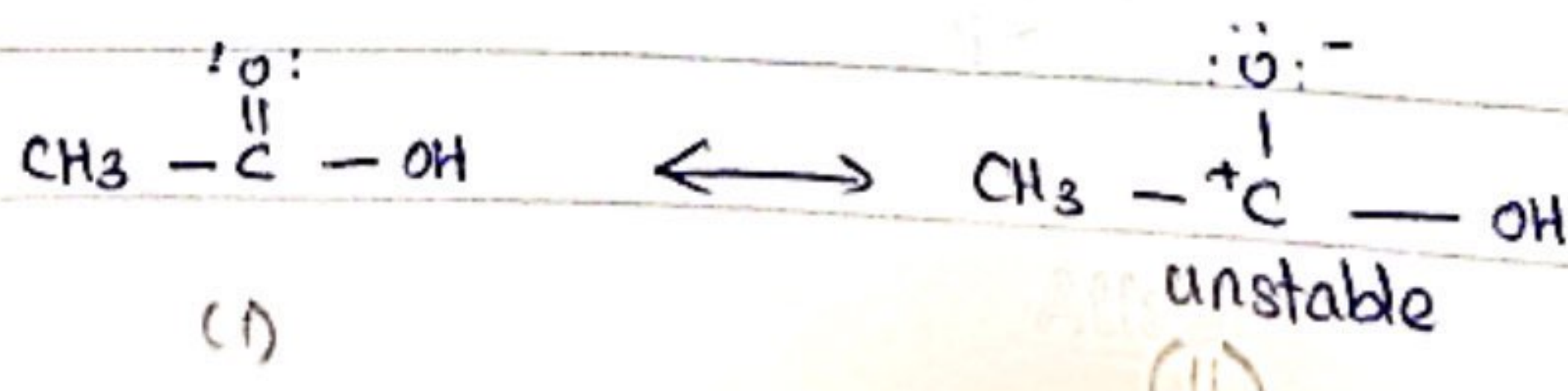
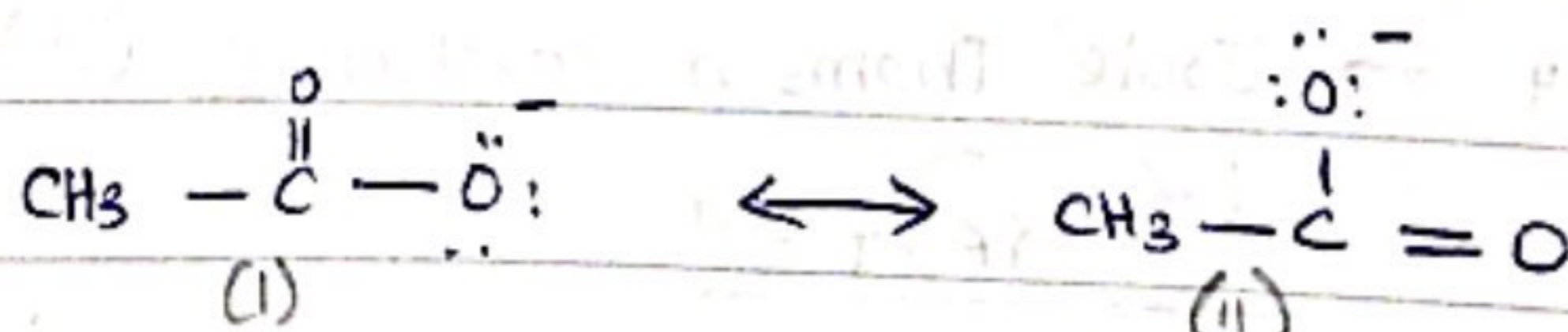
ion dissociation is not complete. $H_3O^+ \ll 1M$

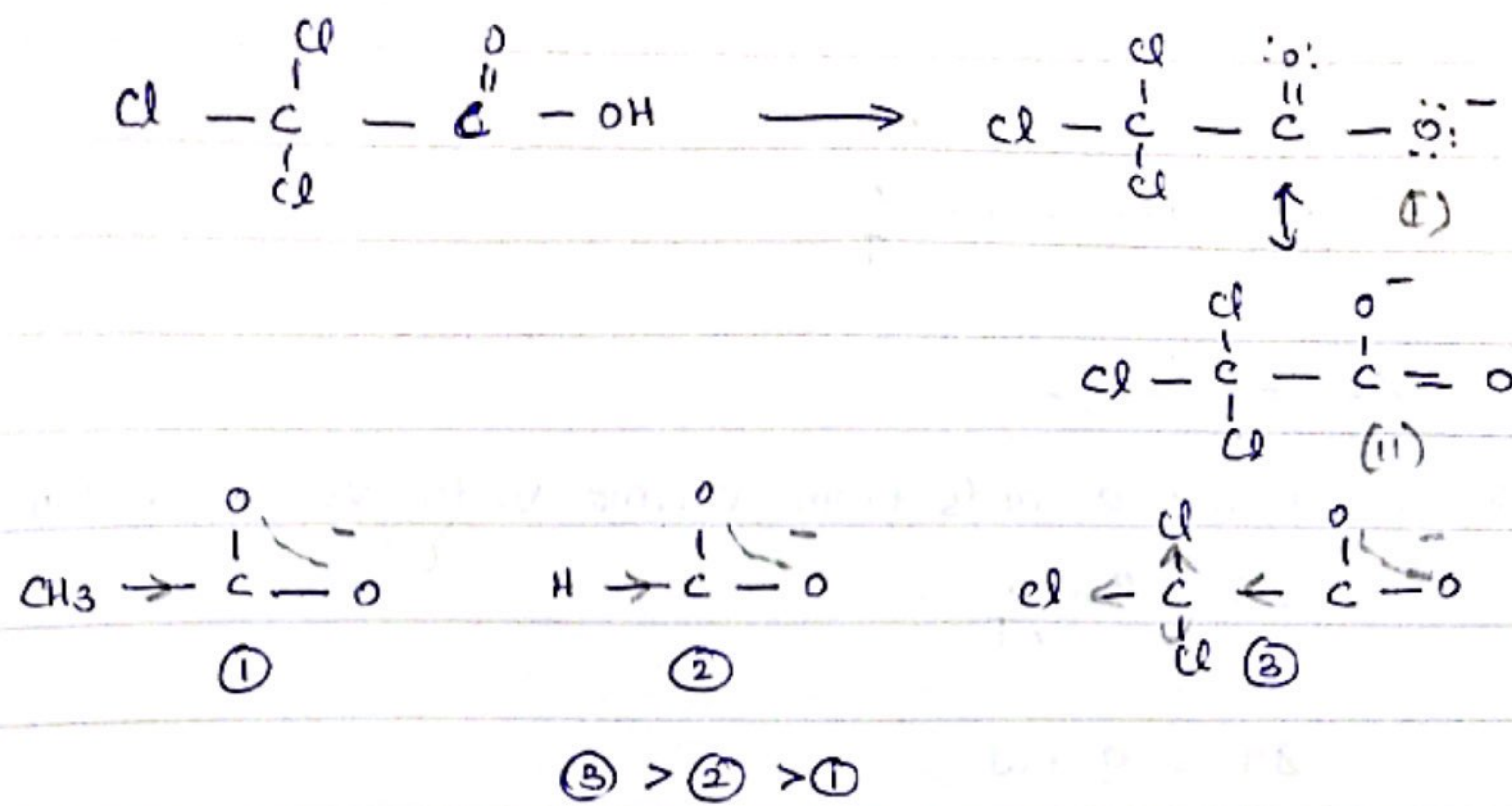


$$pH = -\log_{10} (H^+_{(aq)})$$

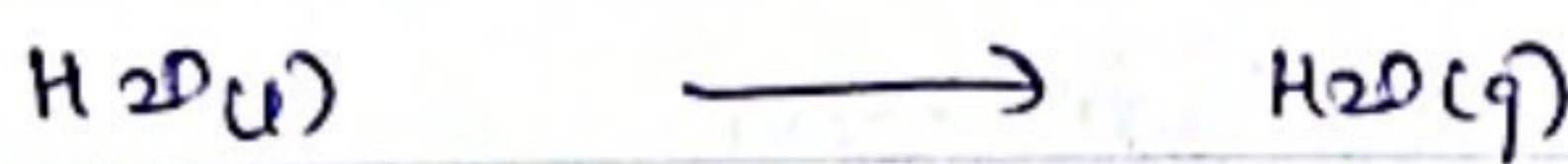
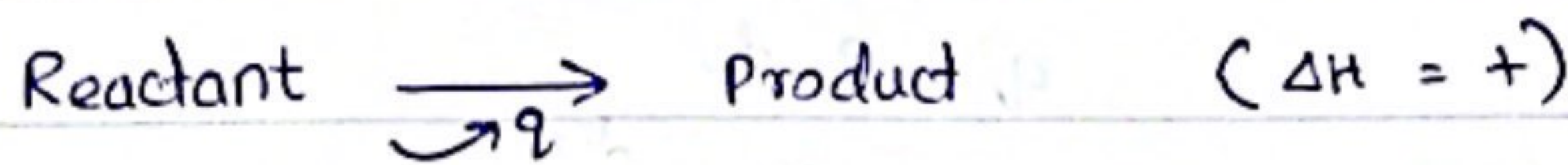
$$= -\log_{10} (1 \times 10^{-1}) = 1$$

Why CH_3COO^- stable,





If heat is absorbed,



Entropy (s)

All the natural processes take place bcs of $\uparrow$ or $\downarrow$ the entropy
2nd law of thermodynamic, $\Delta S \geq 0$

$$\Delta S = q_{\text{rev}} / T$$

16.10.2023

A spontaneous process

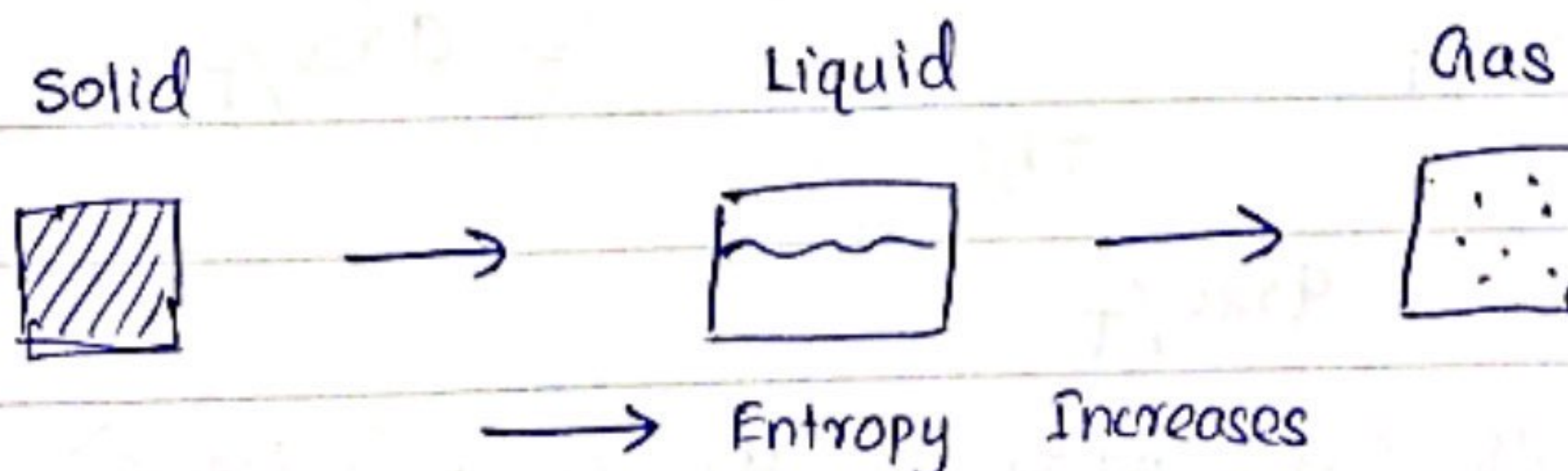
flow of heat from the region of higher to region of lower.

flow of water from a higher level to a lower level.

A spontaneous process is a change that continues without outside intervention
it proceeds on its own.

A process at constant energy can take place continuously only if the entropy increases. $\therefore \Delta S > 0$

Ex :



$$ds = dq / T$$

$$dq = ds \cdot T$$

$$dq_{\text{rev}} = T \cdot \Delta S$$

For an isothermal process,

(i) Phase change (ice $\rightarrow$ water)
$$\Delta S = q_{\text{reversible}} / T$$

$$\Delta S = \Delta H / T$$

1 mol of an ideal gas expands from volume V_1 to V_2 reversibly & isothermally

$$\Delta S = q_{\text{rev}} / T$$

$$\Delta U = q + w$$

$$\Delta U = 0$$

$$\therefore q = -w \leftarrow dw = -P \cdot dv$$

$$q = P \cdot dv$$

$$= nRT \int_{V_1}^{V_2} dv/v$$

$$= nRT \ln V_2/V_1$$

$$= nRT \ln (V_2/V_1)$$

$$\Delta S = nR \ln (V_2/V_1)$$

$$\Delta S = 2.303 nR \ln (V_2/V_1)$$

Constant Pressure Process

$$dq = C_p dT$$

$$\Delta S = C_p \int_{T_1}^{T_2} dT/T$$

$$\Delta S = C_p \ln T_2/T_1$$

$$\int dx = x$$
$$\int dx/x = \ln x$$

Constant Volume Process

$$dq = C_v dT$$

$$\Delta S = C_v \int_{T_1}^{T_2} dT/T$$

$$\Delta S = C_v \ln T_2/T_1$$

Reversible Process

$$\Delta S_{\text{system}} = q_{\text{rev}} / T$$

$$\Delta S = nR \ln (V_2/V_1)$$
$$= q_{\text{rev}} / T$$

$$\Delta S_{\text{surrounding}} = -q_{\text{rev}} / T$$

$$\Delta S_{\text{system}} + \Delta S_{\text{surrounding}} = q_{\text{rev}} / T + (-q_{\text{rev}} / T) = 0$$

Irreversible Process (Spontaneous Process)

$$ds > dq/T$$

$$\Delta S_{\text{system}} + \Delta S_{\text{surrounding}} > 0$$

All spontaneous processes increase the total entropy of the universe.

The total energy of universe is constant. But the entropy of the universe is almost hence to maximum.

Gibbs Free Energy Change (ΔG)

(Driving force of a Reaction)

Composite function of H & T

$$G = H - TS \quad \leftarrow \text{at constant } P \text{ \& } T$$

$$\Delta G = \Delta H - T \Delta S$$

$\Delta G (+)$ $\rightarrow$ not spontaneous

$\Delta G (-)$ $\rightarrow$ spontaneous

$\Delta G = 0$ $\rightarrow$ Equilibrium

$$G = H - TS$$

$$dG = dH - Tds - sdT \quad \text{--- (1)}$$

$$H = U + PV$$

$$dH = dU + PdV + VdP \quad \text{--- (2)}$$

$$q_w = Tds \quad \text{--- (3)}$$

$$dG = VdP - SdT$$

At constant Temperature

$$dG = VdP - \frac{SdT}{0}$$

$$dG = VdP$$

$$\therefore \left(\frac{dG}{dP} \right)_T = V$$

At constant Pressure

$$dG = \frac{VdP}{0} - SdT$$

$$dG = -SdT$$

$$\left(\frac{dG}{dT} \right)_P = -S$$

$$G = f(p, T)$$

with pressure

with temperature

$$dG = (dG/dp)_T dp + (dG/dT)_T dT$$

$$dG = V \cdot dp + (-S) dT$$

$$dG = V \cdot dp - S dT$$

$$\Delta S = \sum S_{\text{product}} - \sum S_{\text{reactant}}$$



$$\Delta G = \sum G_{\text{product}} - \sum G_{\text{reactant}}$$

Method to find ΔG .

- (i) $\Delta G = \Delta H - T \cdot \Delta S$
- (ii) $\Delta G = -RT \ln k$
- (iii) $\Delta G = \sum G_{\text{product}} - \sum G_{\text{reactant}}$
- (iv) $\Delta G = -nEF$

Variation of Equilibrium Constant with Temperature

$$\Delta G = -\Delta S dT$$

$$\left(d\Delta G/dT \right)_P = -\Delta S$$

$$\Delta G = \Delta H - T\Delta S \quad \text{at constant } T$$

$$\Delta G = \Delta H - T \left(-d\Delta G/dT \right)_P$$

$$\boxed{\Delta G = \Delta H + T \left(d\Delta G/dT \right)_P} \quad \text{Gibbs Helmholtz equation.}$$

$$\Delta G^\circ = -RT \ln K_p$$

$$RT \ln K = \Delta H^\circ + T \left(d(RT \ln K_p)/dT \right)_P$$

$$\boxed{d \ln K / dT = \Delta H^\circ / RT^2}$$



$$da = v dp - s dT$$

$$da_A = da_B$$

$$da_A = da_B$$

$$\Delta a = 0$$

$$(v_A dp - s_A dT) = (v_B dp - s_B dT)$$

$$(v_A - v_B) dp = (s_A - s_B) dT$$

$$\Delta v dp = \Delta s dT$$

$$dp/dT = \Delta s / \Delta v$$



$$\Delta s = q_{rev} / T$$

$$dp/dT = \frac{\Delta H / T}{\Delta v}$$

At constant $p, q = H$

$$\Delta s = \Delta H / T$$

$$\boxed{dp/dT = \Delta H / T \Delta v}$$

Clapeyron Equation

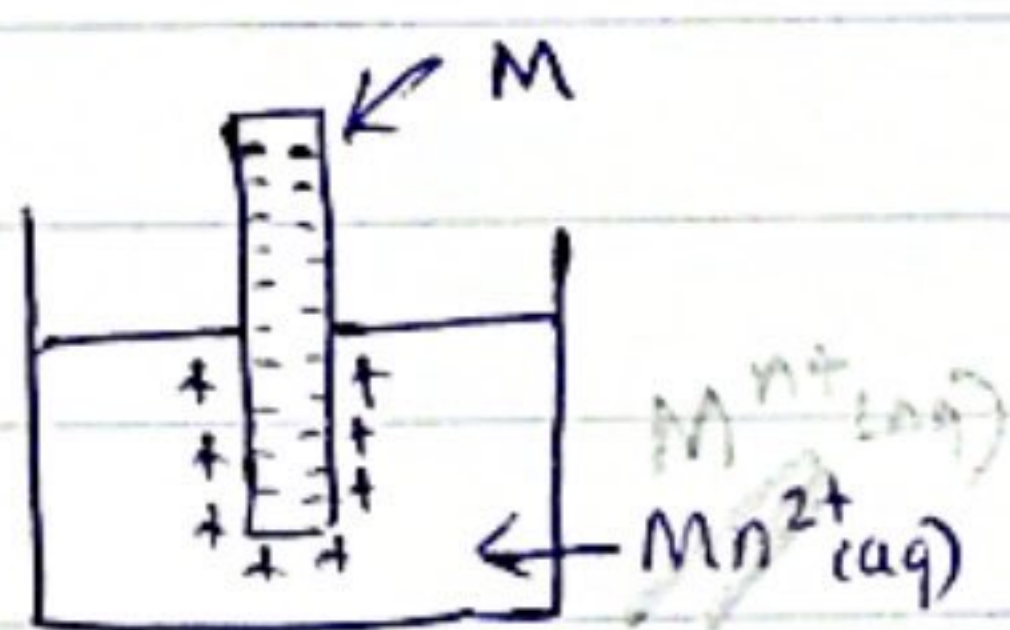
Electrochemistry

Chemical & Electronic Energy.

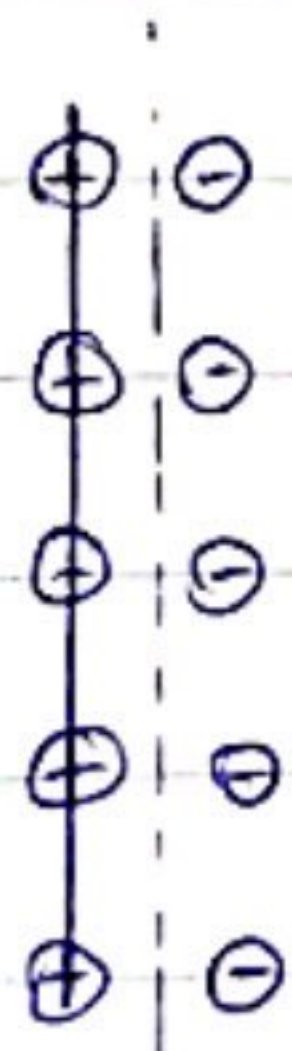
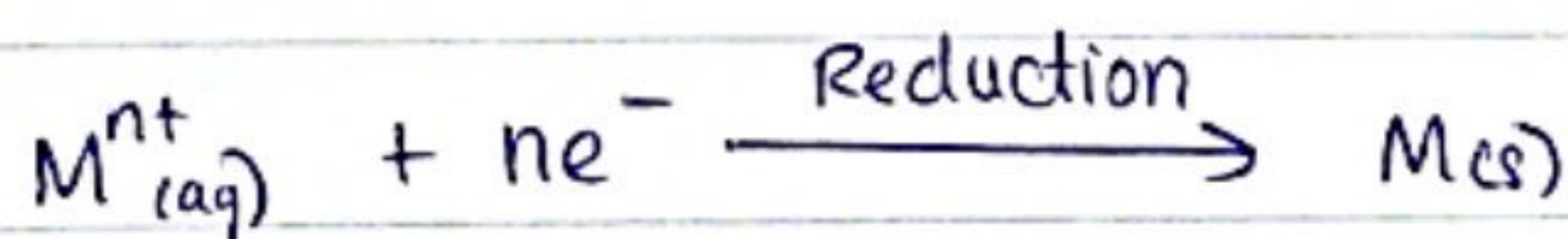
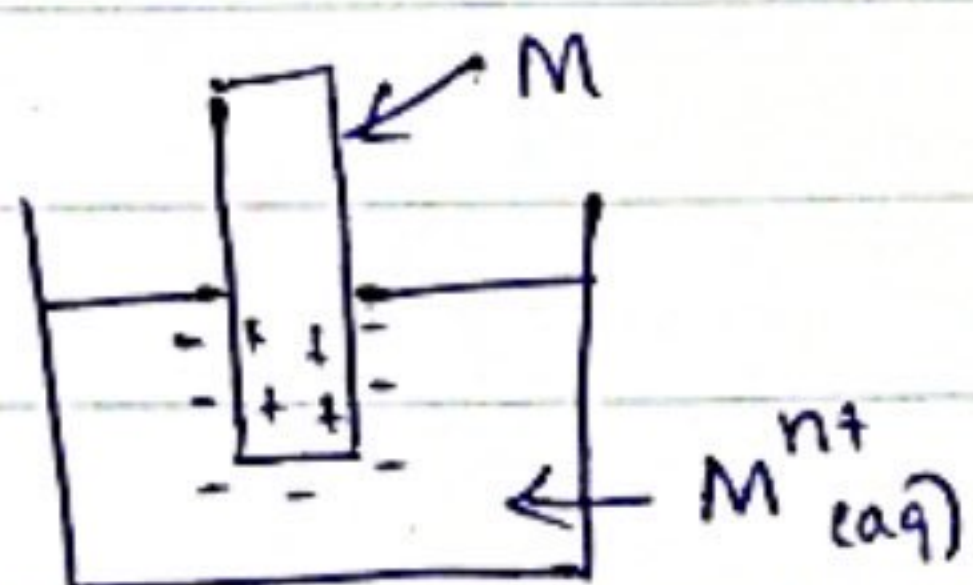
* Galvani cell : Chemical energy $\longrightarrow$ Electrical energy

* Electroly cell : Electrical energy $\longrightarrow$ Chemical energy.

There are different cells to convert energies to chemical energy or Chemical energy to Electrical energy.



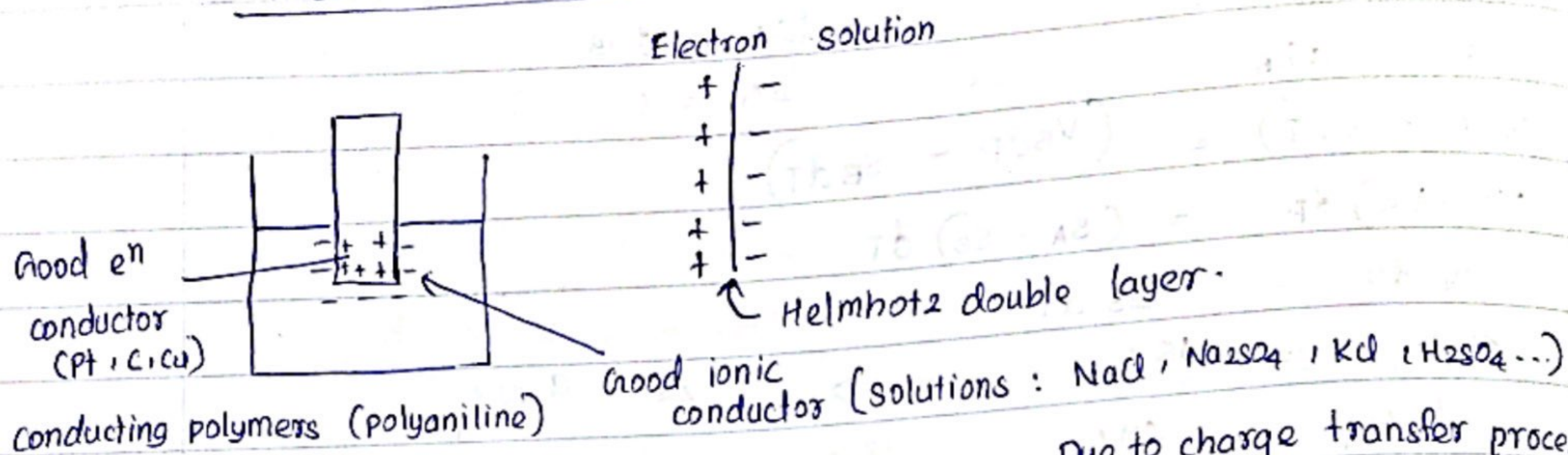
e^- remain with the metal. Metal (-) charge



Helmholtz double layer

- A double layer is formed at the electrode - solution interphase
- Charge separation developed a potential at the electrode solution interphase, called : "Electron Potential".

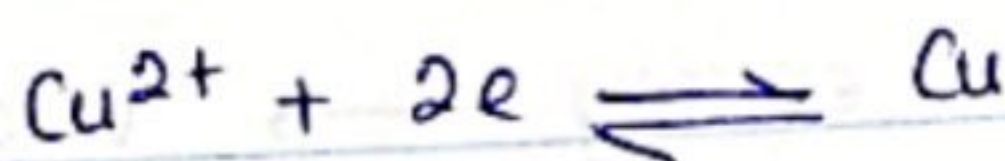
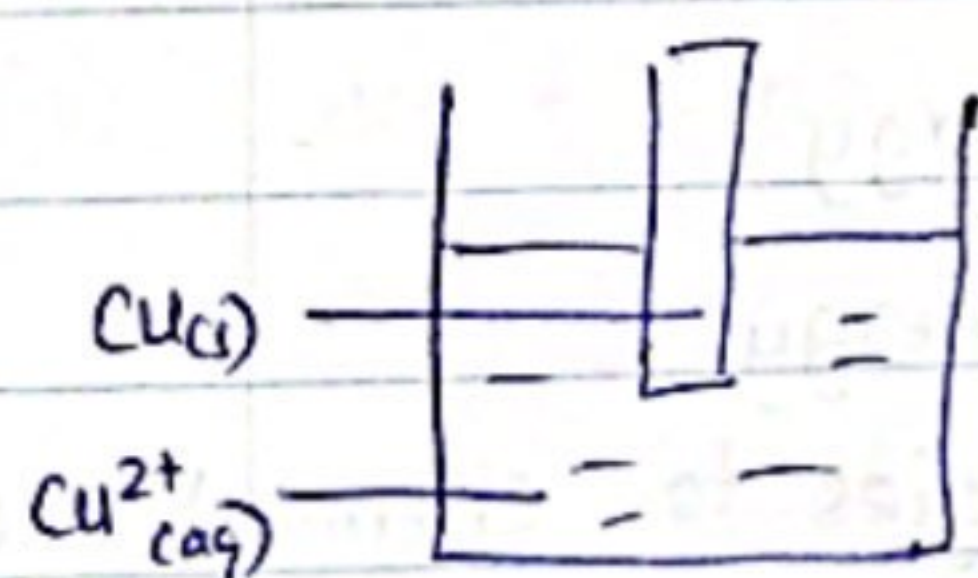
Electron Potential.



- Electro reactions are Heterogeneous. - Due to charge transfer process through e^- - solution interface.
- Chemical reactions are homogeneous

Types of Electrodes

(i) $Mn^{+}(aq) / M(s)$ electrode.



* All reactions should be written as reduction.

Nernst eqn.

$$E = E^{\circ} - \frac{RT}{nF} \ln \frac{[activity\ reaction]}{[activity\ of\ oxidation]}$$

$$E = E^{\circ} - \frac{RT}{nF} \ln \left[\frac{a_{red}}{a_{ox}} \right]$$

in here,

$$a = \gamma \cdot C$$

γ - activity constant, C - concentration.

a - activity / Effective concentration.

exact amount of used Cu^{2+} .

$$\ln x = 2.303 \log x$$

$$\text{in } 25^{\circ}C \quad E = E^{\circ} - \frac{RT}{nF} \ln \left[\frac{a_{red}}{a_{ox}} \right]$$

$$= \frac{8.314 \times 298}{n \times 96480} \times 2.303 \log \left[\frac{a_{red}}{a_{ox}} \right]$$

$$= 0.059$$

$$E = E^{\circ} - \frac{0.059}{n} \log \left[\frac{a_{red}}{a_{ox}} \right] \quad \text{in the electrode reaction are at unit activity}$$

→ e^- potential depend on,

- Temperature.
- concentration of the solution

E° - Standard electrode potential.

Ex:

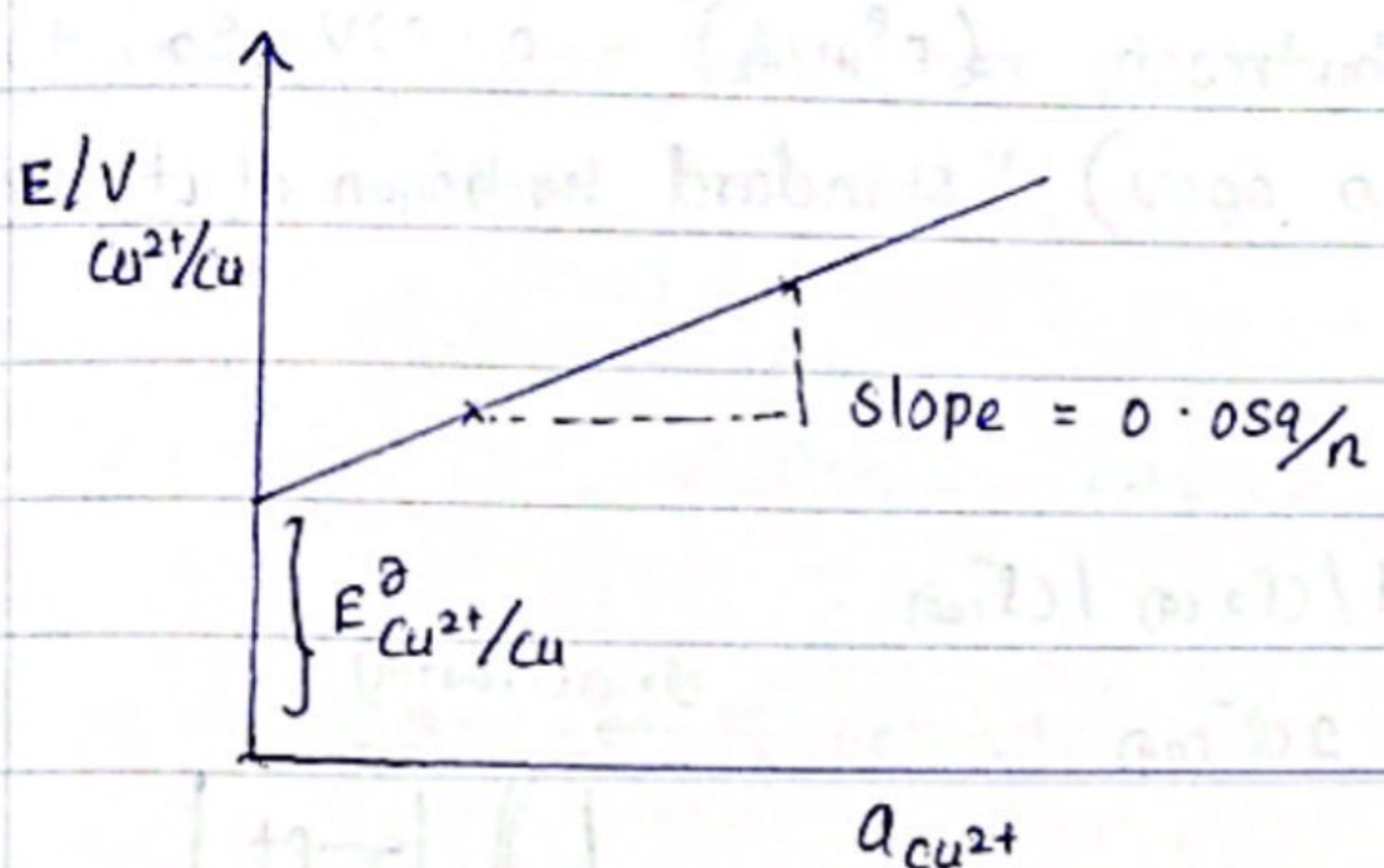
$$E_{\text{Cu}^{2+}/\text{Cu}} = E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} - 0.059/2 \log \frac{a_{\text{Cu}}}{a_{\text{Cu}^{2+}}} \quad * \text{ Cu is pure solid}$$

→ Activity of pure solid or liquid = 1

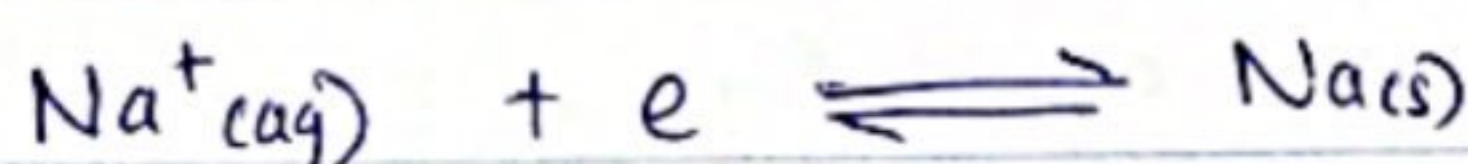
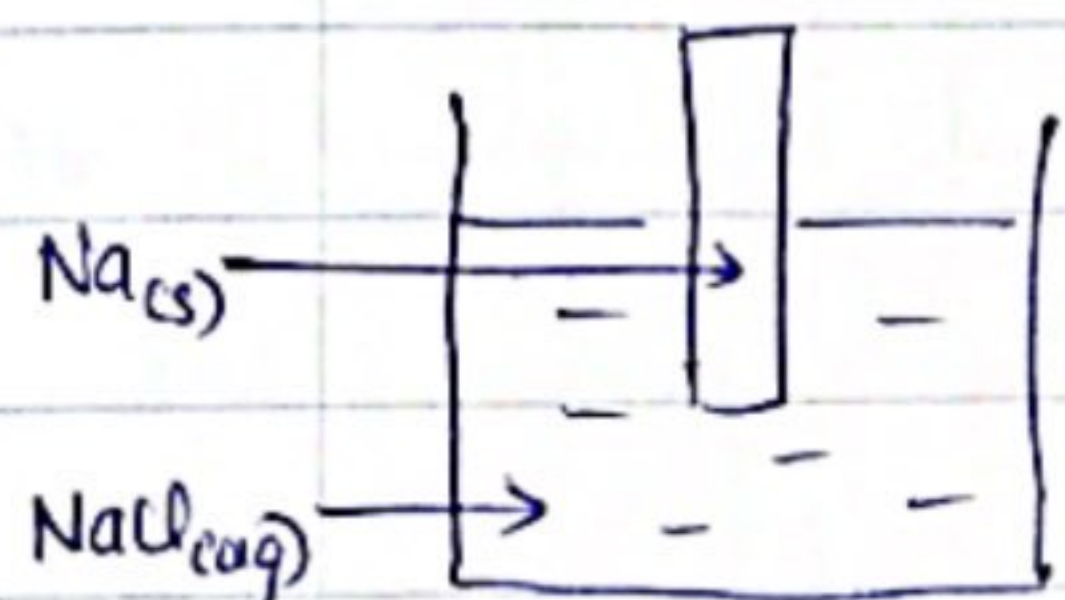
So,

$$E_{\text{Cu}^{2+}/\text{Cu}} = E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} + 0.059/2 \log a_{\text{Cu}^{2+}}$$

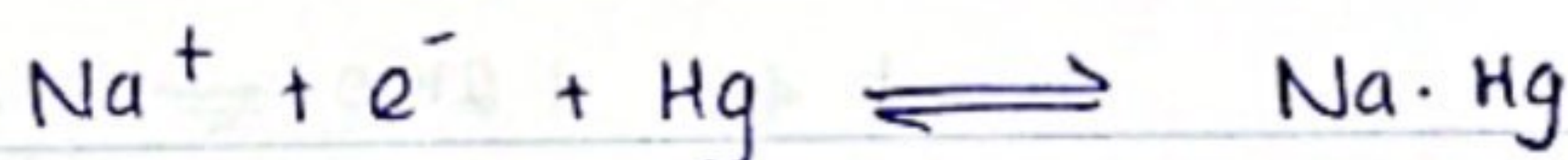
$$y = c + m x$$



(02) $\text{Na}^+ / \text{Na(s)}$ electrode.



Na is highly reactive with water. So have to decrease the reactivity of Na. to that - amalgam (Na, Hg) is used.



$$E = E^{\circ} - 0.059/n \log \frac{a_{\text{Na} \cdot \text{Hg}}}{a_{\text{Na}^+} \cdot a_{\text{Hg}}}$$

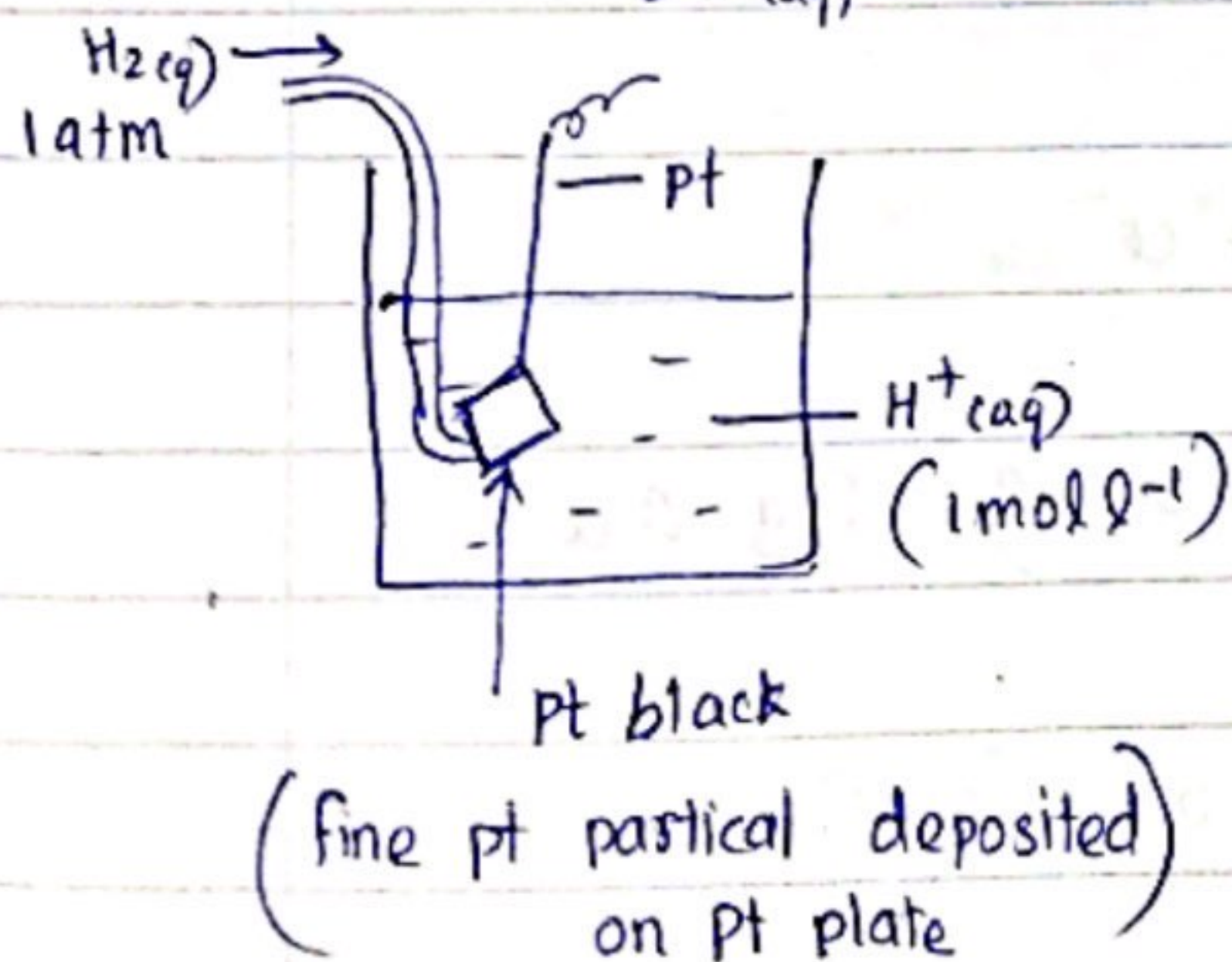
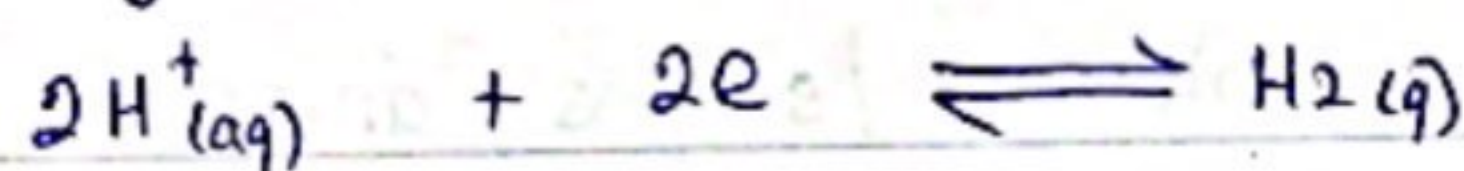
pure liquid $\therefore a = 1$

$$E = E^{\circ} - 0.059/n \log a_{\text{Na} \cdot \text{Hg}} / a_{\text{Na}^+}$$

(03) Gas Electrode

If we put a sponge in water it is completely fill with water - absorbed.

(03) Hydrogen gas electrode



Gas molecules get absorbed on Pt surface - absorption.

absorption $\rightarrow$ Chemisorption very strong
 $\rightarrow$ Physisorption weak.

Electron notation in here $\rightarrow$ Pt / $H_2(1.0 \text{ atm})$ / $H^+(aq)$
 at $25^\circ C$ $E = E^\circ_{H^+/H_2} - 0.059/n \log a_{H_2}/a_{H^+}^2$ shows the different
 bcc 2 H^+ participating

$$E = E^\circ_{H^+/H_2} - 0.059/2 \log 1/a_{H^+}^2$$

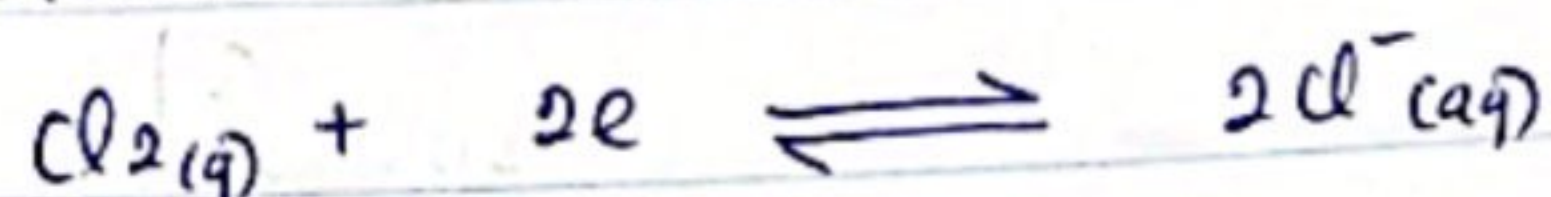
$$E = E^\circ + 0.059/2 \cdot \cancel{2} \log a_{H^+}$$

$$E = E^\circ_{H^+/H_2} + 0.059 \log a_{H^+}$$

Standard electrode potential for hydrogen $(E^\circ_{H^+/H_2}) = 0.00V$ So, this electrode is called $(E^\circ_{SHE} = 0.000V)$ "standard hydrogen electrode".

(04) Cl_2 gas electrode.

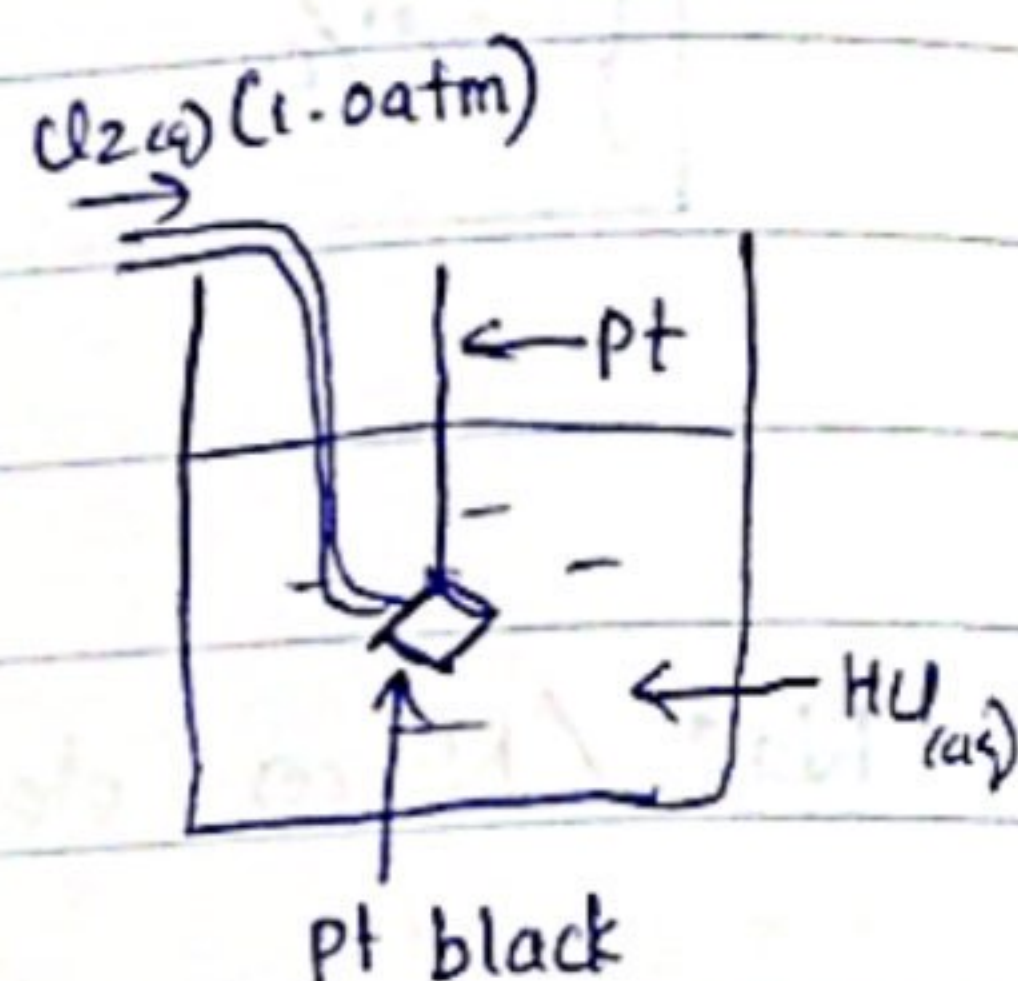
Electrode notation = Pt / $Cl_2(g)$ / $Cl^-(aq)$.



$$E = E^\circ - 0.059/2 \log a_{Cl^-}^2/a_{Cl_2} \leftarrow \text{pure} = 1$$

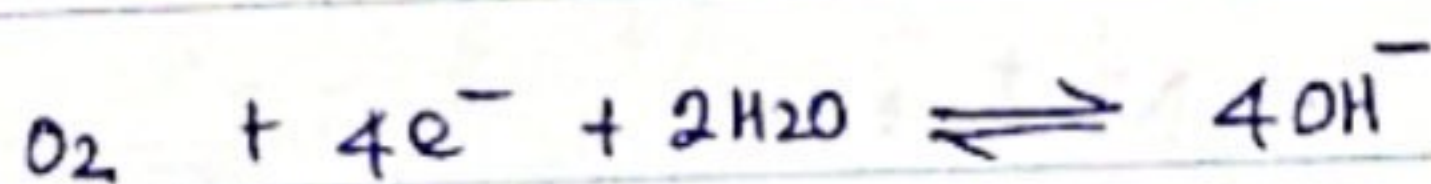
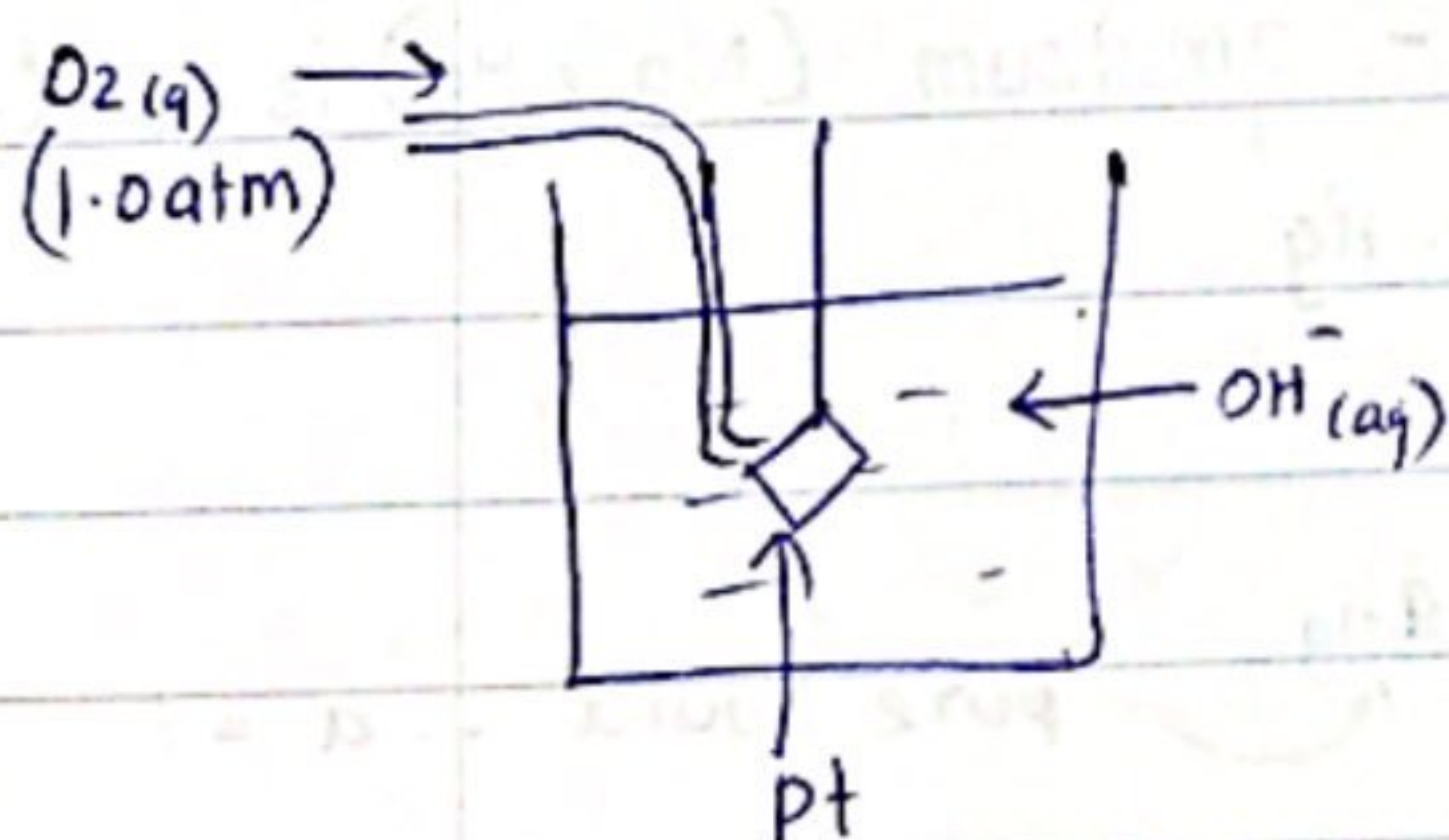
$$E = E^\circ - 0.059/2 \times 2 \log a_{Cl^-}$$

$$E = E^\circ - 0.059 \log a_{Cl^-}$$



(05) O_2 gas electrode

Pt / $O_2(1.0 \text{ atm})$ / $OH^-(aq)$



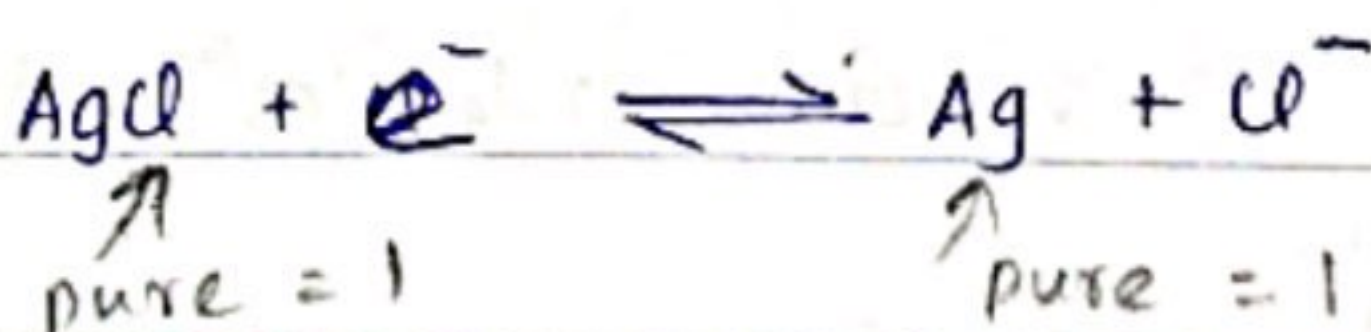
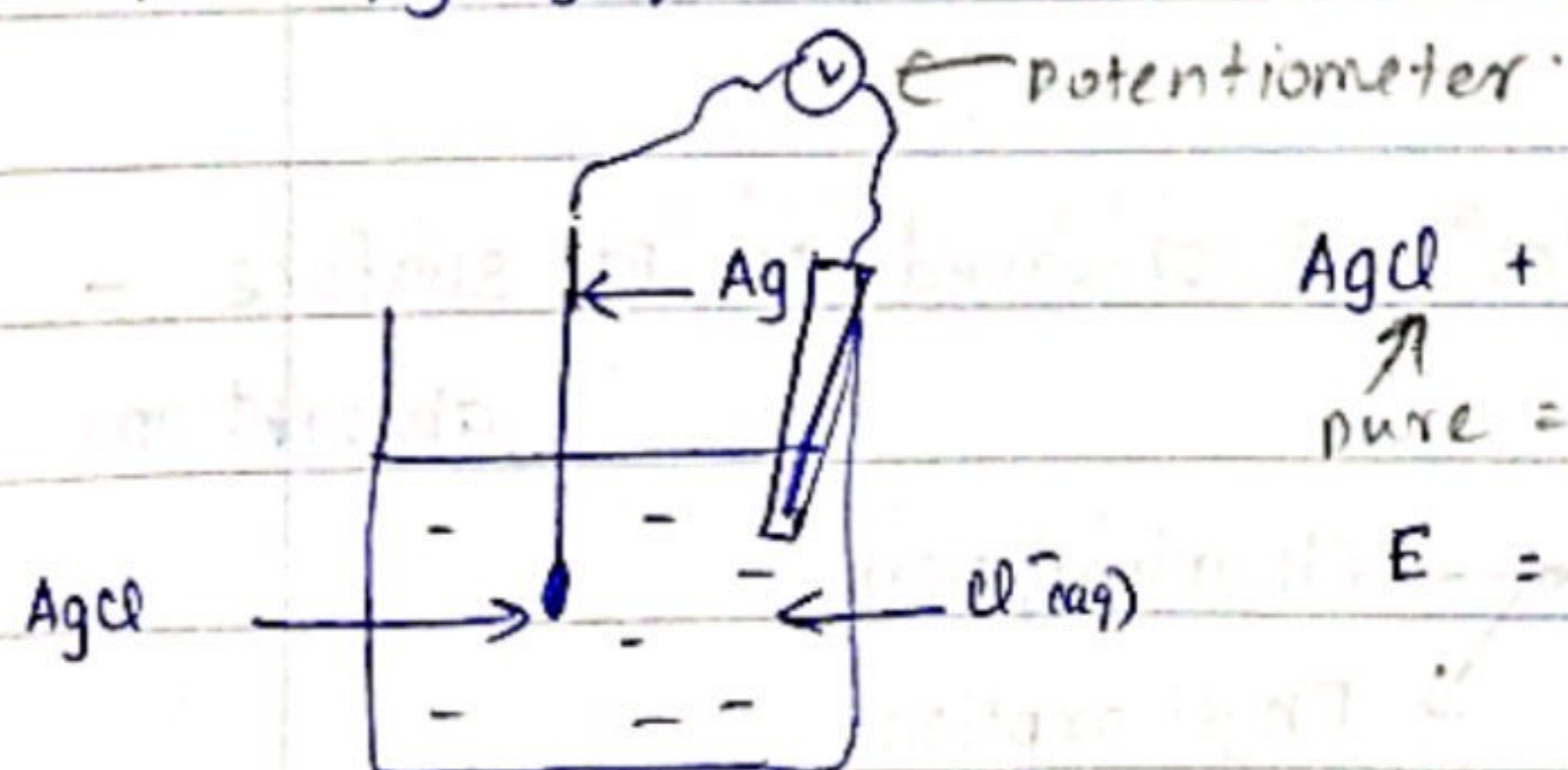
$$E = E^\circ - 0.059/4 \log a_{OH^-}^4$$

$$E = E^\circ - 0.059/4 \cdot 4 \log a_{OH^-}$$

$$E = E^\circ - 0.059 \log a_{OH^-}$$

(06) M / Mx / $X^-(aq)$

(06) $Ag/AgCl/Cl^-$: metal / sparingly soluble salt / s.s.s. anion



$$E = E^\circ_{Ag/AgCl/Cl^-} - 0.059/1 \log a_{Cl^-}$$

$$E = E^\circ_{Ag/AgCl/Cl^-} - 0.059 \log a_{Cl^-}$$

at 25° a electrode gives a constant, reproducible
 $(E = 0.198 \text{ V})$

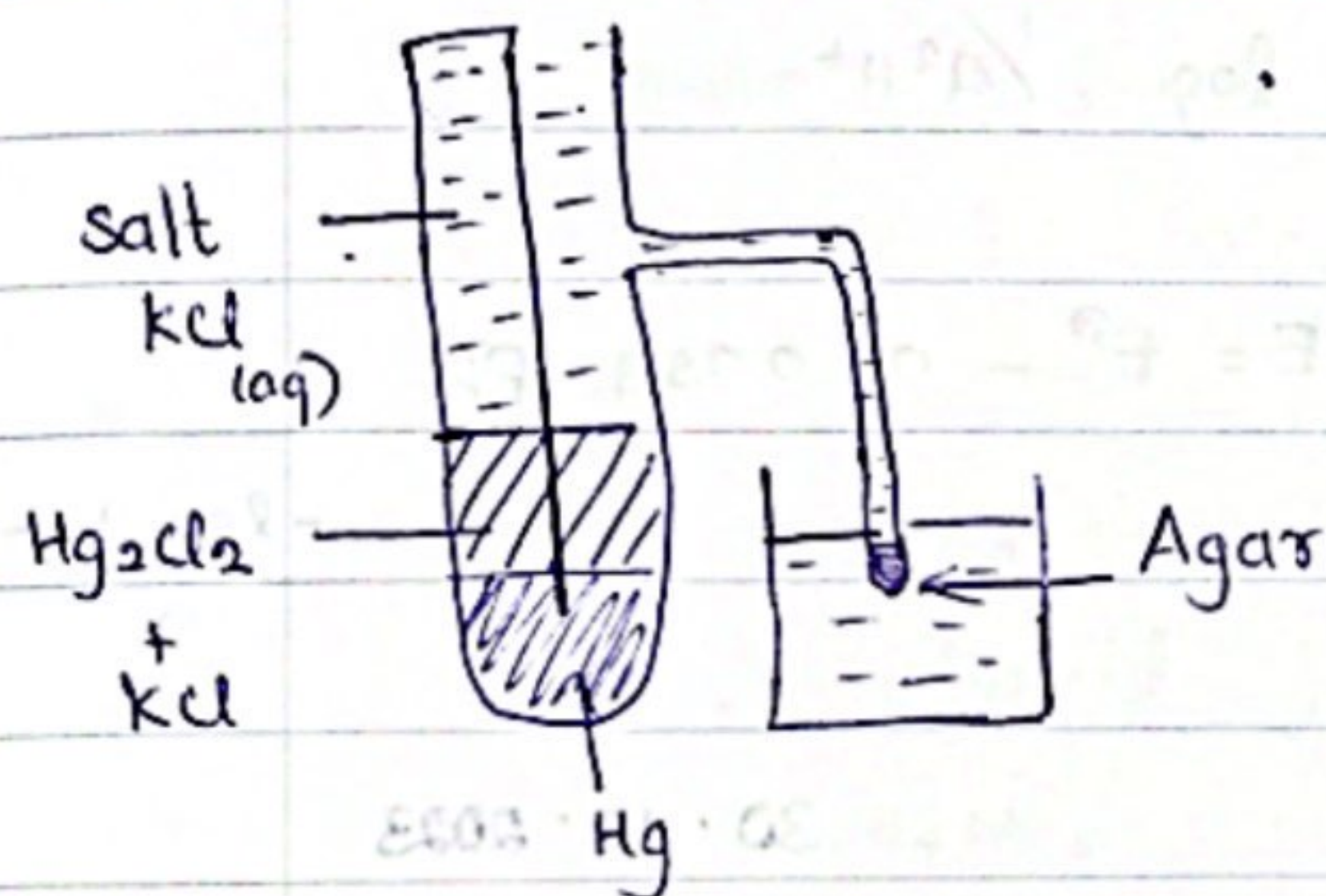
- * Absolute Electron potential of single e^- is cannot be measured.
 (Need more than 1 e^- to measure). (To measure the e^- potential we use - Potentiometer.) But can be measured with respect to another e^- .

SHE = 0.00V is use as a Reference electron to measure the potential of other electron.

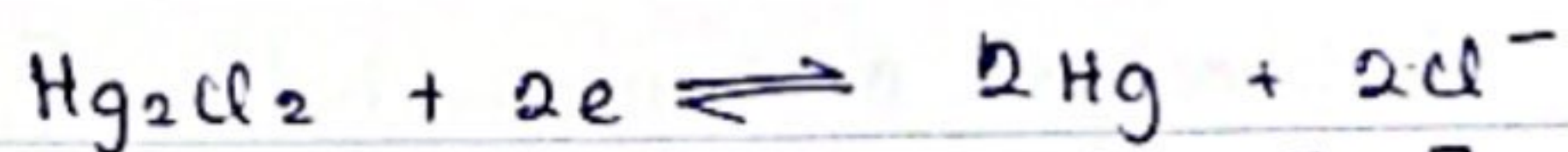
$\left(\frac{\partial E}{\partial T}\right)$ temperature doesn't change significantly with the temperature.

- * Ag / AgCl / Cl⁻ electrode has very low temperature coefficient.

Caramel Electrode.



Saturated electrode



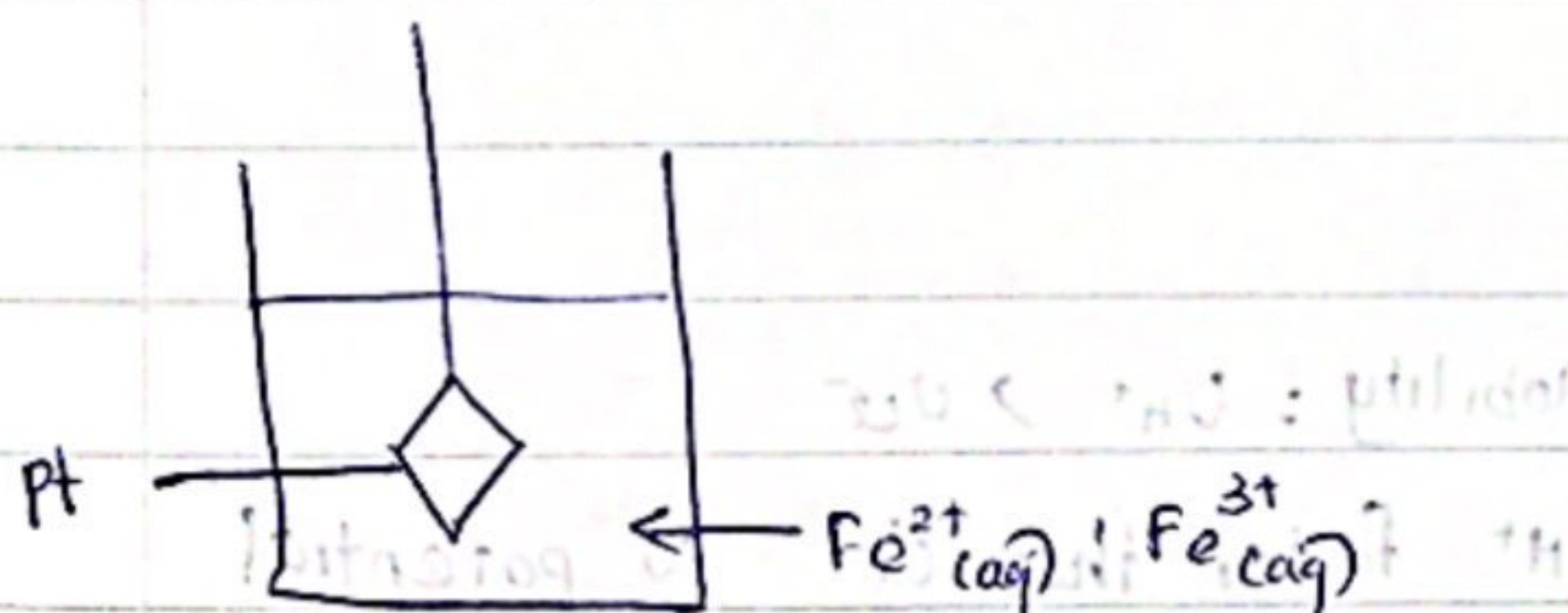
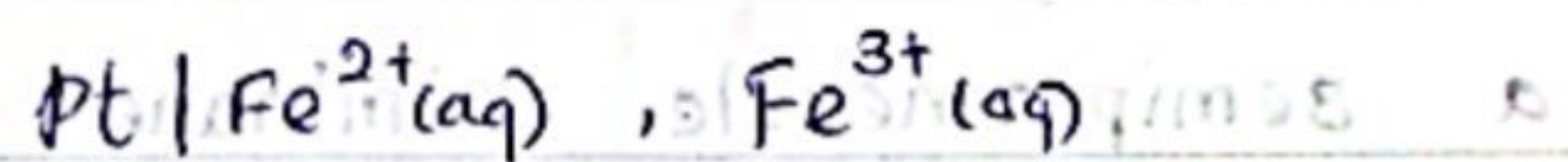
$$E = E^\circ - \frac{RT}{nF} \ln \frac{[a_r]}{[a_{ox}]}$$

$$E = E^\circ - \frac{0.0059}{2} 2 \log a_{\text{Cl}^-}$$

$$E = E^\circ - 0.0059 \log a_{\text{Cl}^-}$$

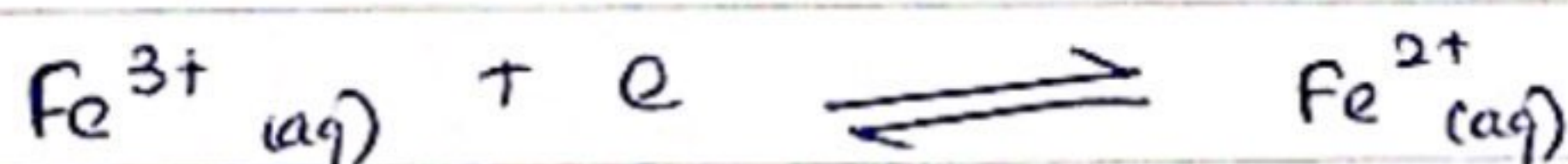
Ex: Pt / Hg / Hg₂Cl₂ / Cl⁻

Redox Electrode Have 2 different solution



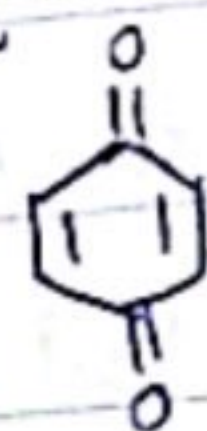
* In same species they separated by commas.

* most reduce species are kept closer to the electrode.

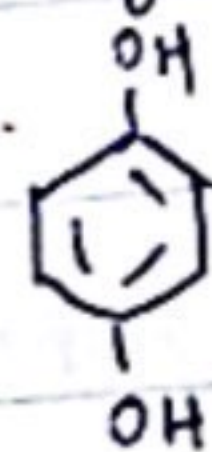


Quinhydrone Electrode

Equal mole mixture of quinone & hydroquinone

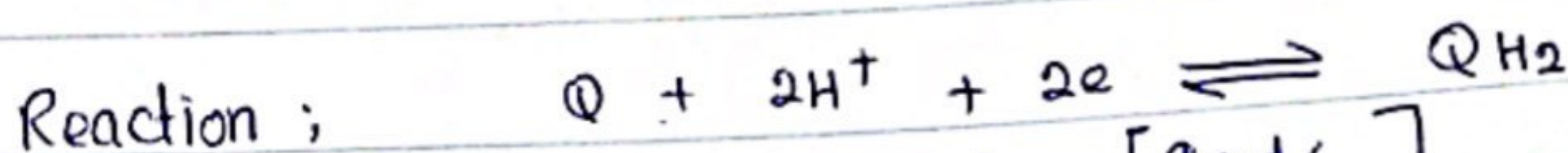
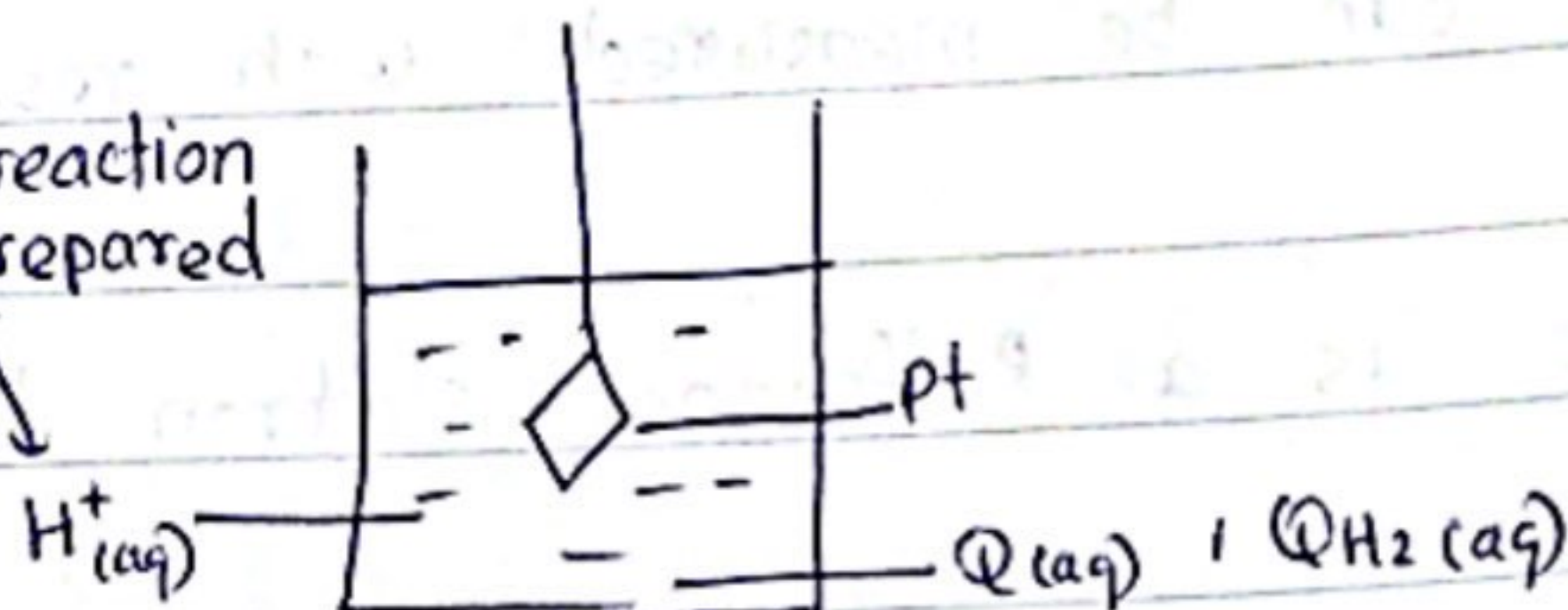


→ Q



→ QH₂

In the reaction
H⁺ prepared



$$E = E^\ominus - \frac{RT}{nF} \ln \left[\frac{a_{red}}{a_{ox}} \right]$$

$$= E^\ominus - 0.0059/2 \log \frac{a_{H_2}}{a_{H^+}^2 \cdot a_Q}$$

Same concentration

$$= E^\ominus - 0.0059/2 \log \frac{1}{a_{H^+}^2}$$

$$= E^\ominus - 0.0059/2 \cdot 2 \log \frac{1}{a_{H^+}^2}$$

$$E = E^\ominus + 0.0059 \log a_{H^+} \Rightarrow E = E^\ominus - 0.0059 \text{ pH}$$

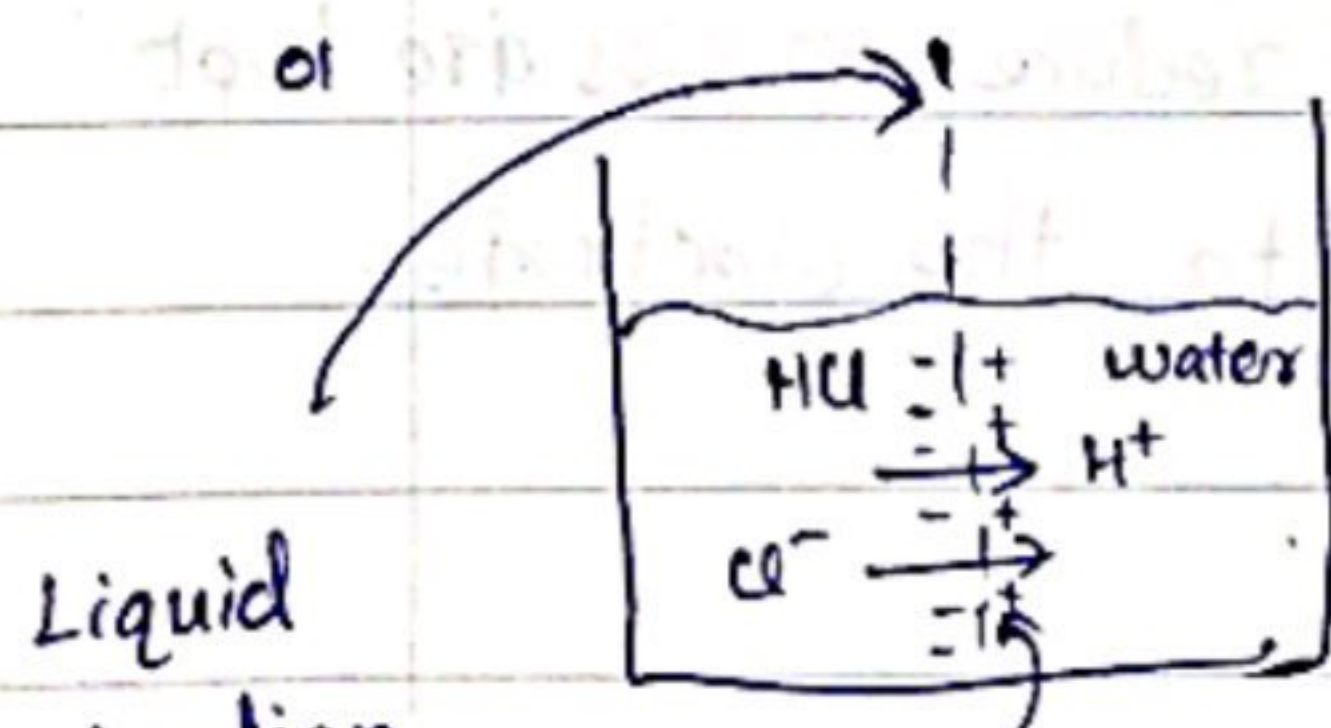
$$\text{pH} = -\log a_{H^+}$$

30.10.2023

Electrochemical Cells

There are 2 ways of combining electrodes.

- (1) Bringing the 2 solutions of the 2 electrodes with the contact through a semipermeable membrane.
- (2) Using a salt bridge.



Liquid
junction
potential

semipermeable membrane - pores comparable to the size of bare ions.

Mobility : $U_{H^+} > U_{Cl^-}$

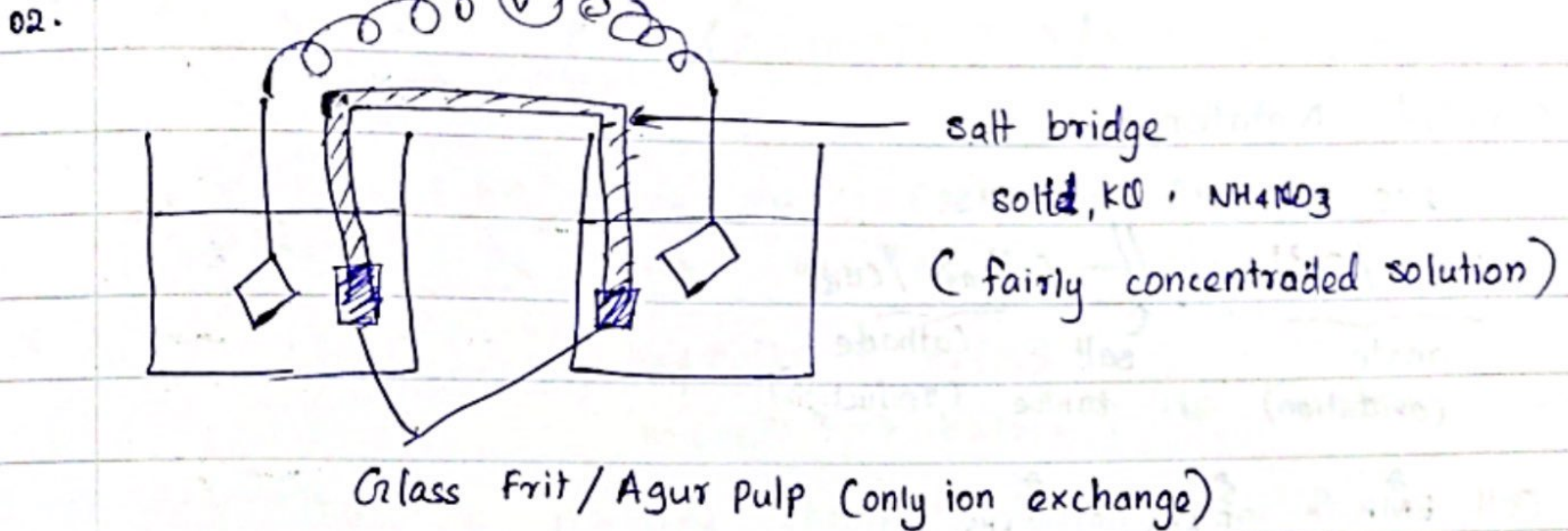
H⁺ faster than Cl⁻ so potential

ea' build. so one side get + charge & other get -

potential difference establish between the membrane is called Liquid junction potential.

This liquid junction potential always ~~at~~ acts against the EMF of the cell. And the lower the obtainable emf of the cell. Now the cells with ~~trans~~ liquid junction potential is called "cells with transference".

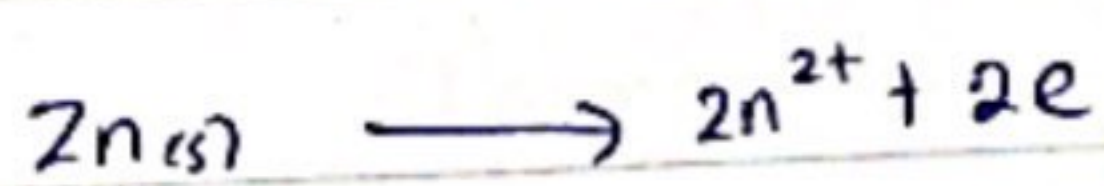
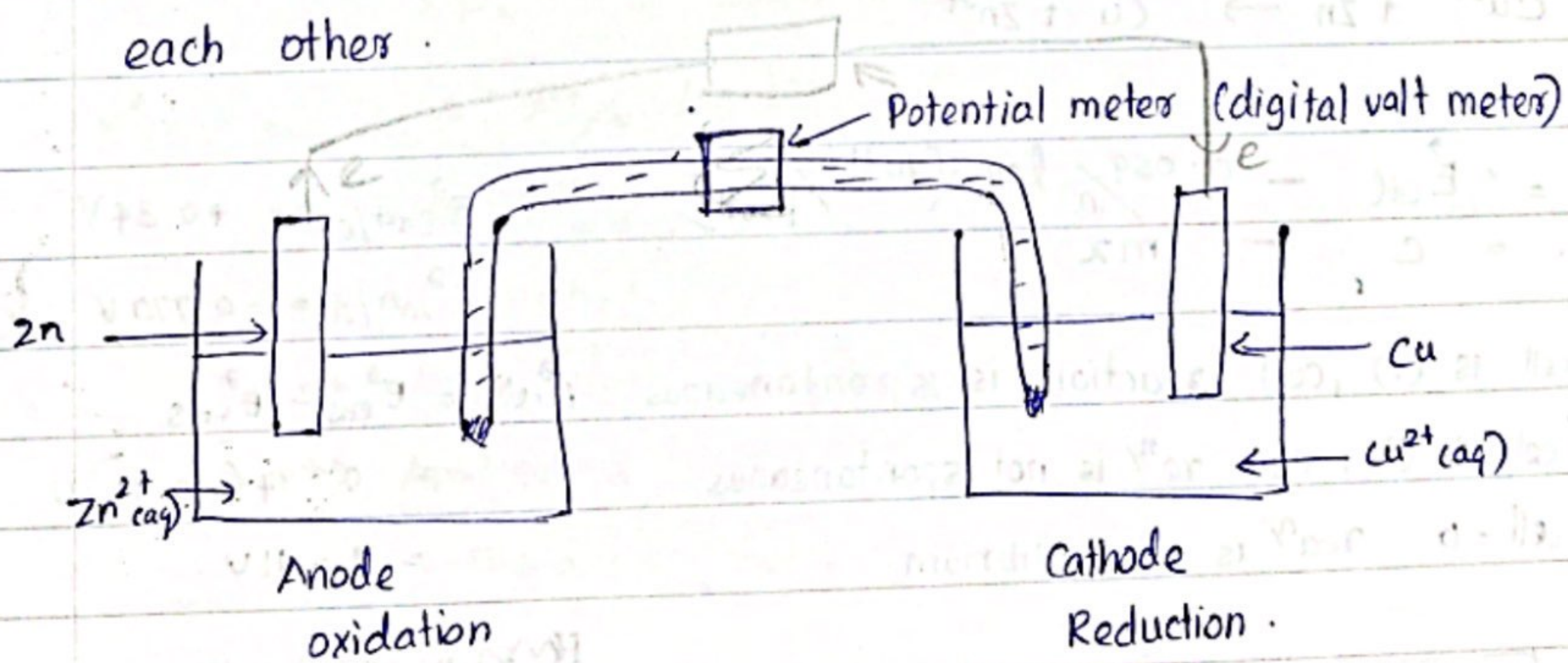
Symbol =



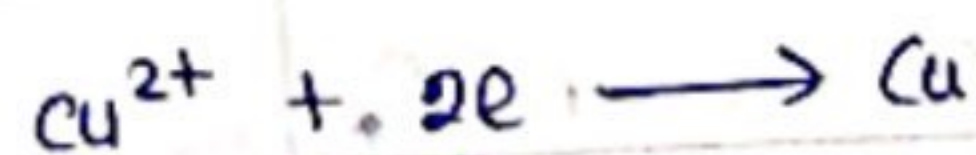
In there, 2 liquid junction potentials are formed at the 2 interfaces. (2 ends of the glass frit) but equal in magnitude but opposite in sign.

liquid junction potential is eliminated.

Without liquid junction potential called "cells without transference". KCl , NH_4NO_3 mobilities are more equal to each other. But $NaCl$ cannot be used. Because their Na^+ & Cl^- mobilities are not equal to each other.

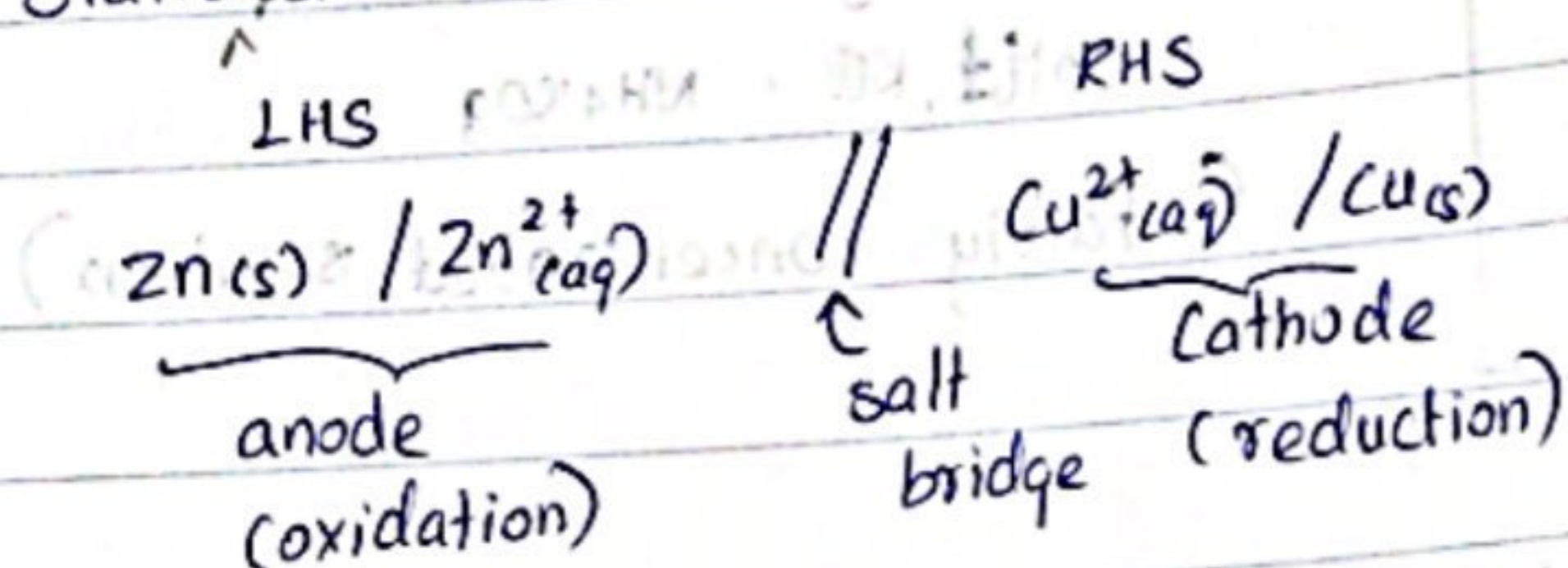


e^- release. (Bcs e^- move anode to cathode)



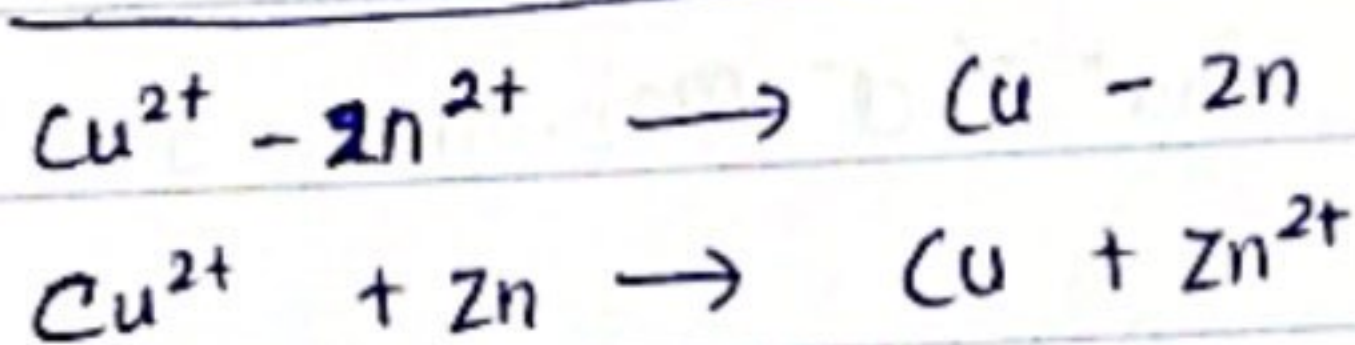
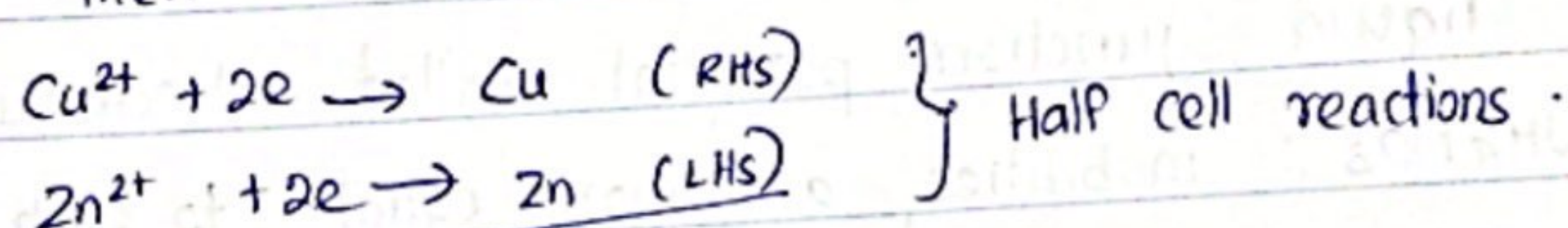
→ Electromotive Force (EMF)
 The potential difference between the 2 electrodes when there is no current pass into the cell is called as EMF.
 Voltmeter have very high internal resistance. So current $i = 0$. Then it's EMF.

→ Standard Notation



$$\text{Cell EMF} = E_{\text{RHS}}^{\circ} - E_{\text{LHS}}^{\circ}$$

While writing the reaction both reaction should be written as reduction reaction. But 1st we should write cathode reaction and then at 2nd should write anode reaction.



$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$

$y = c \cdot x^m$

$E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = +0.34\text{V}$

$E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.770\text{V}$

** If E_{cell}° is (+), cell reaction is spontaneous. $E_{\text{cell}}^{\circ} = E_{\text{RHS}}^{\circ} - E_{\text{LHS}}^{\circ}$

$= 0.34 - (-0.77)\text{V}$

$= 1.11\text{V}$

← If E_{cell}° is (-), cell reaction is not spontaneous.

oxidation anode ← If $E_{\text{cell}}^{\circ} = 0$ reaction is at equilibrium.

If E_{cell}° is (+) cell

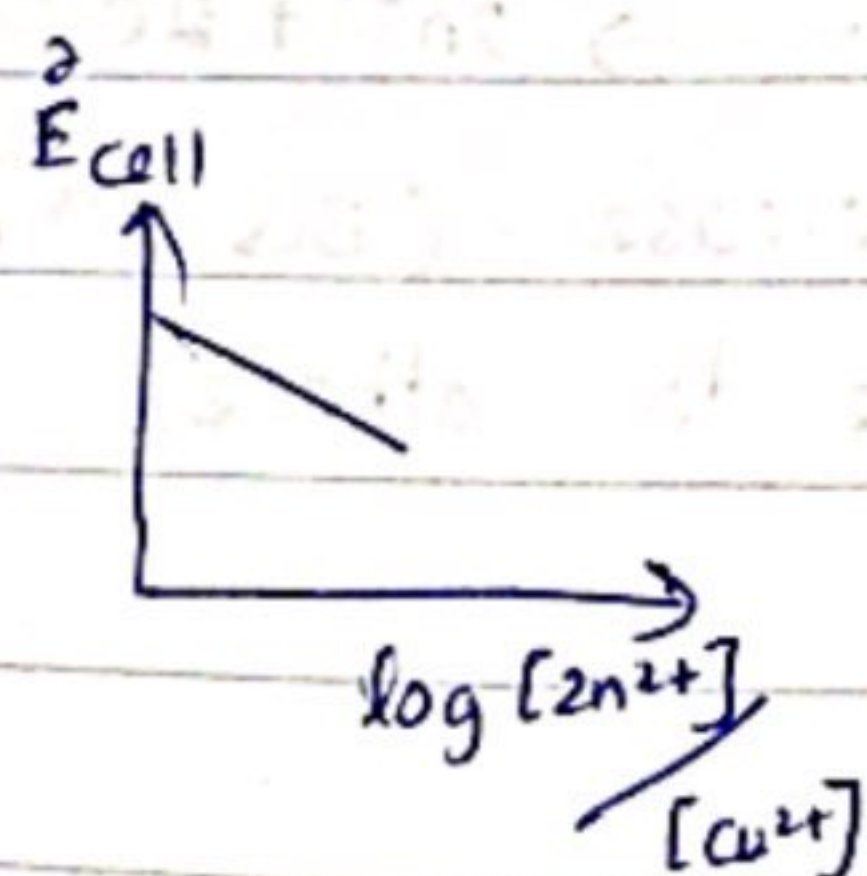
$\Delta G = -nEF_{\text{cell}}$

96480 C mol⁻¹

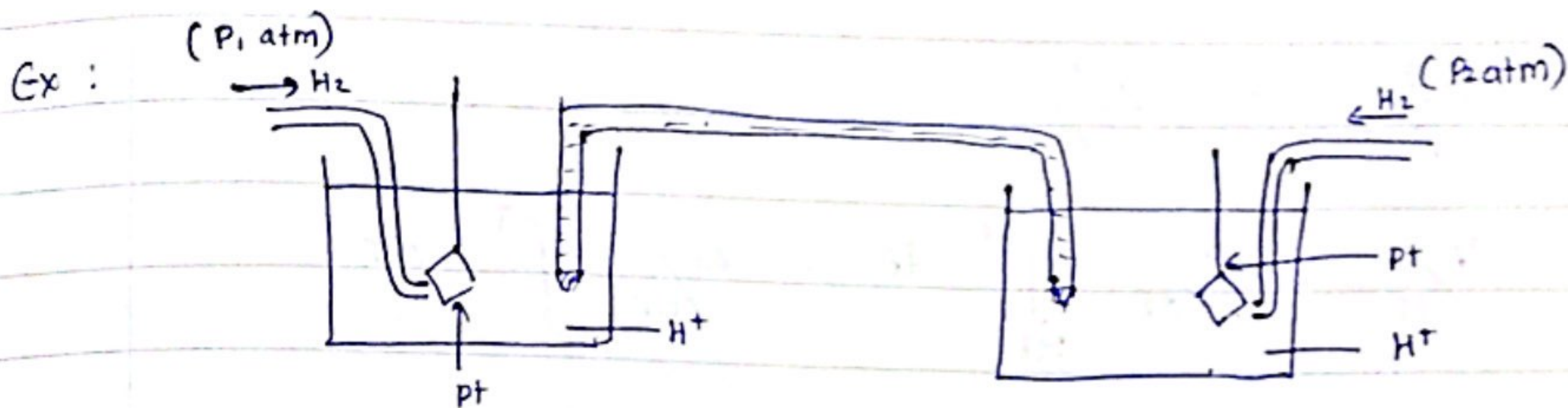
number of E_{cell}

If E_{cell} (+); ΔG (-) → spontaneous.

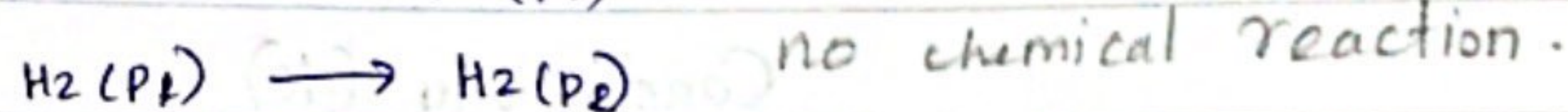
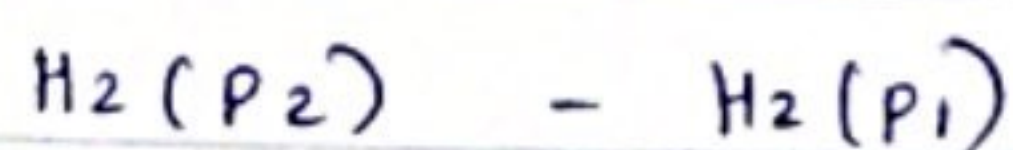
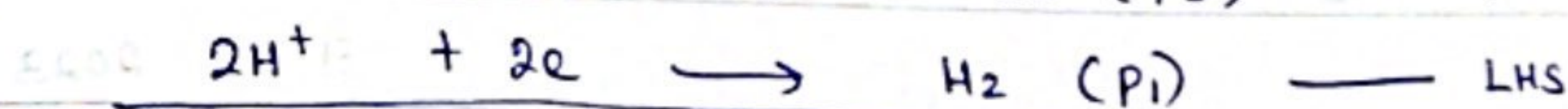
E_{cell} (-); ΔG (+)



$$P_1 > P_2$$



Cell Notation : $\text{Pt} / \text{H}_2 (P_1 \text{ atm}) / \text{H}^+_{\text{aq}} / \text{H}_2 (P_2 \text{ atm}) / \text{Pt}$



There is no any chemical reaction from the reagent of higher pressure to lower pressure.

Natural tendency of flow matter from the reagent of higher concentration to the reagent of lower concentration. Result in decrease in free energy.

Decrease in free energy can be exported generate electrical energy.

$$E^\circ_{\text{cell}} = E^\circ_{\text{cell}} - 0.059/n \log [a_{\text{red}}] / [a_{\text{oxid}}]$$

$$E^\circ_{\text{RHS}} - E^\circ_{\text{LHS}} \longrightarrow \text{H}_2(\text{aq}) = 0$$

$$E^\circ = E^\circ_{\text{cell}} = 0.059/n \log [\text{H}_2(P_2)] / [\text{H}_2(P_1)]$$

$$= 0 - 0.059/n \log [\text{H}_2(P_2)] / [\text{H}_2(P_1)]$$

→ Conductivity

→ Molar conductance

→ Applications

→ Cathode kauraslee law.

Thermodynamic Equations.

$$\begin{aligned} -\Delta S^\circ &= \left(\partial \Delta G^\circ / \partial T \right)_P \\ \Delta G^\circ &= -nE^\circ F \end{aligned}$$

$$d\Delta G^\circ / dT = -nF (dE^\circ / dT)$$

$$\frac{dE}{dT} = \frac{\Delta S}{nF}$$

$$\frac{dE}{dT} = \frac{\Delta S}{nF} \int_{298}^{E_T} dE = \frac{\Delta S}{nF} \int_{298}^T dT$$

$$E_T - E_{298} = \frac{\Delta S}{nF} (T - 298)$$

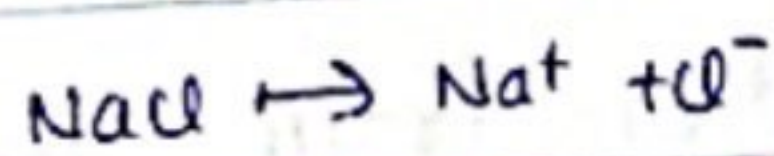
$$E_T = E_{298} + \left[\frac{(T - 298) \Delta S}{nF} \right]$$

31.10.2023

Conductivity (K)



conductivity



* Conductivity: A ability of the ion of the dissolved or molten compound to transport electric charge.

→ Molten - not dissolve in water; liquid form of solid.

Conductance (G)

• It directly measure the ability of ion species to carried to change-charge.
more ion ↑, low Resistant ↓

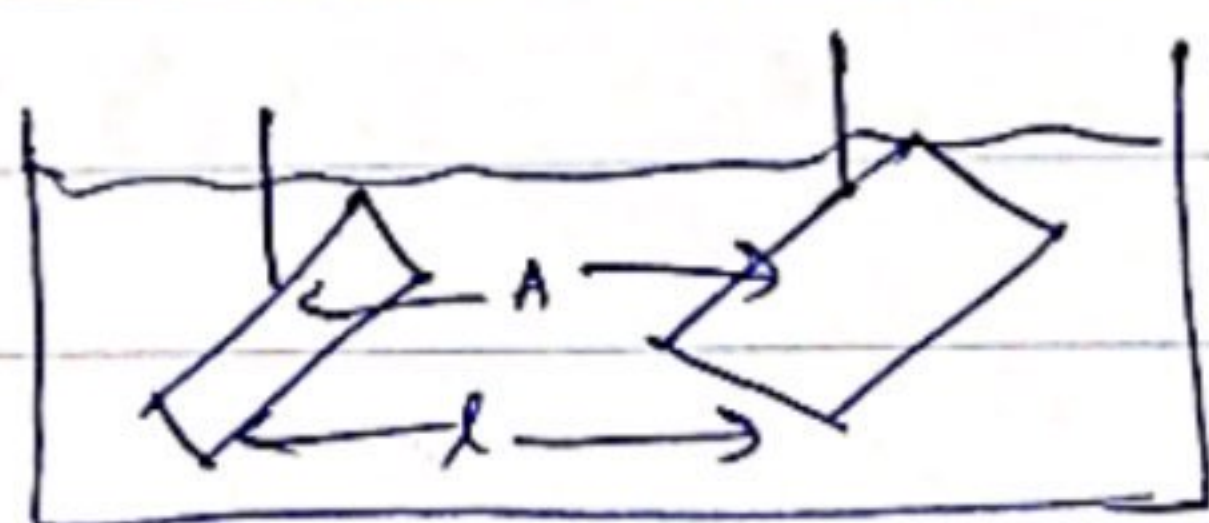
• conductance of a solution is the inverse of the its resistance i.

$$G = \frac{1}{R} = \Omega^{-1}$$

Simens

$$R = \frac{\rho l}{A}$$

Resistivity



$$k = \frac{l}{A}$$

$$k = \frac{G l}{A}$$

$$\frac{l}{A} = \text{Cell constant}$$

$$\frac{\text{m}}{\text{m}^2} = \text{m}^{-1}$$

• Conductivity is an additive property,

The conductivity of the solution is equal to the sum of the individual conductivity of all the ions present in the solution.

$$k = \sum k_i$$

Ex: $\text{NaCl}_{(aq)}$; Ions : Na^+ , Cl^- , H^+ , OH^-

$$k_{\text{NaCl}} = k_{\text{Na}^+} + k_{\text{Cl}^-} + k_{\text{H}^+} + k_{\text{OH}^-}$$

Wheatstone Bridge Circuit

Cell constant normally calculated (l/A) by using 0.100 M KCl solution of known conductivity.

$$k_{0.1\text{M KCl}} = G_{0.1\text{M KCl}} (l/A) \quad \text{--- (1)}$$

$$k_{\text{solution}} = G_{\text{solution}} (l/A) \quad \text{--- (2)}$$

$$k_{\text{H}_2\text{O}} = G_{\text{H}_2\text{O}} (l/A) \quad \text{--- (3)}$$

$$\frac{k_{\text{H}_2\text{O}}}{k_{\text{KCl}}} = \frac{G_{\text{H}_2\text{O}}}{G_{\text{KCl}}}$$

$$k = k_{\text{solution}} - k_{\text{water}}$$

Molar Conductivity (Λ)

Conductivity depends on the

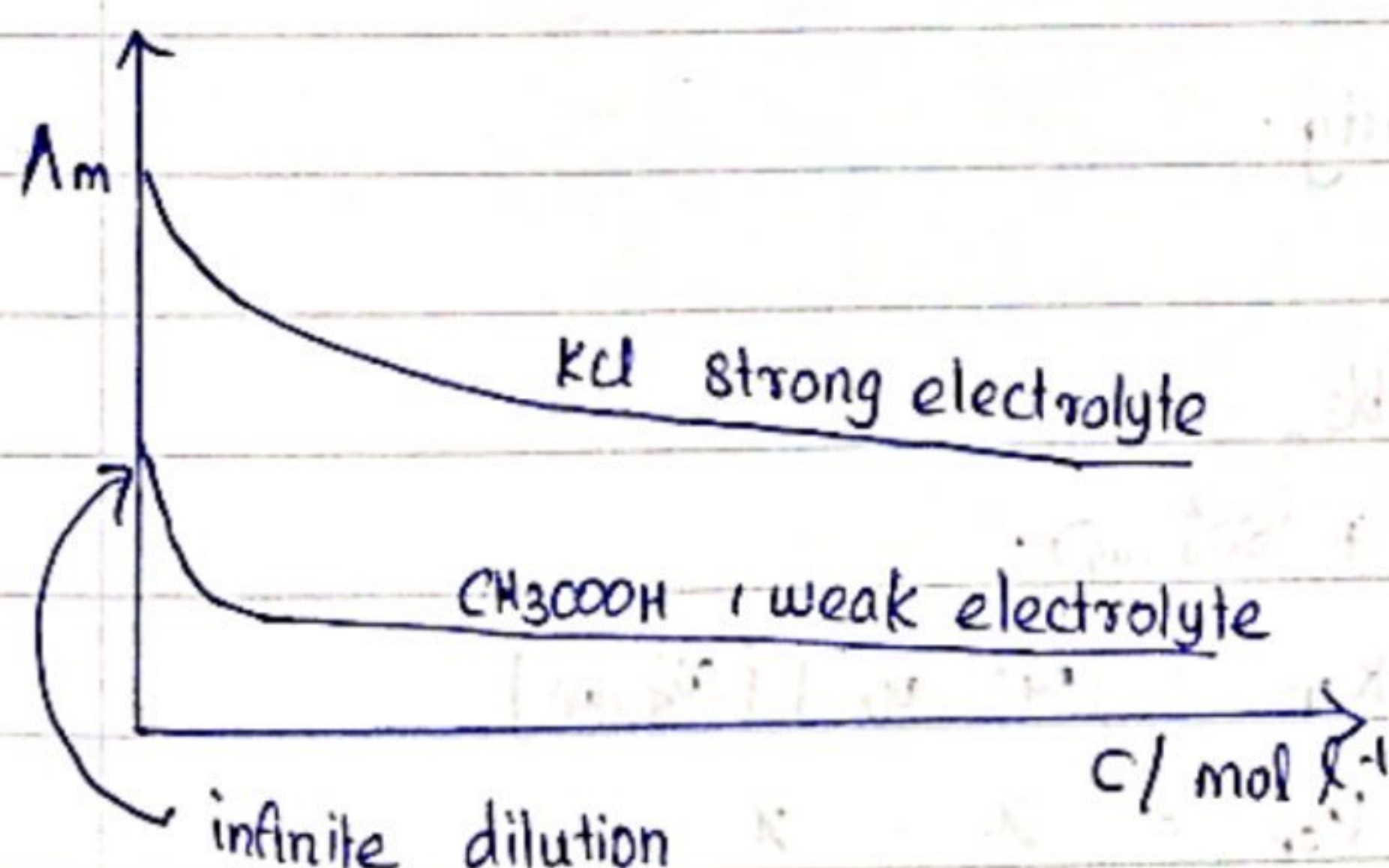
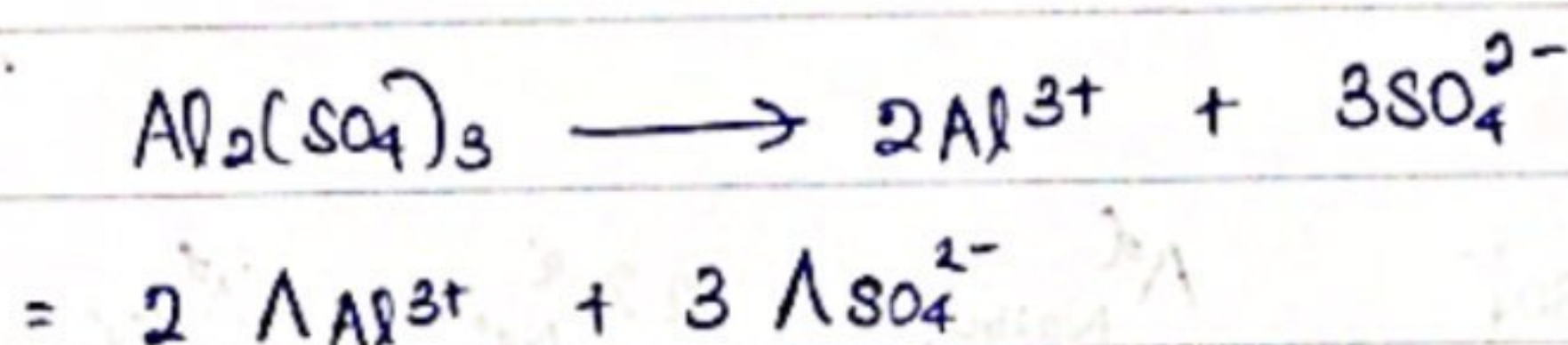
1. Number of ions.
2. Charge of the ions.
3. Mobility of the ion.

$$\Lambda = \frac{k}{c} \quad \begin{array}{l} \text{conductivity} \\ \text{concentration} \end{array} \quad \frac{\text{Scm}^{-1}}{\text{mol dm}^{-3}} \Rightarrow \frac{\text{Scm}^{-1}}{\text{mol m}^3 \times 10^{-3}} = \text{Sm}^2 \text{mol}^{-1} \times 10^{-3}$$

The molar conductivity of the electrolyte is predicted to the molar conductivity of the ion to the electrolyte.

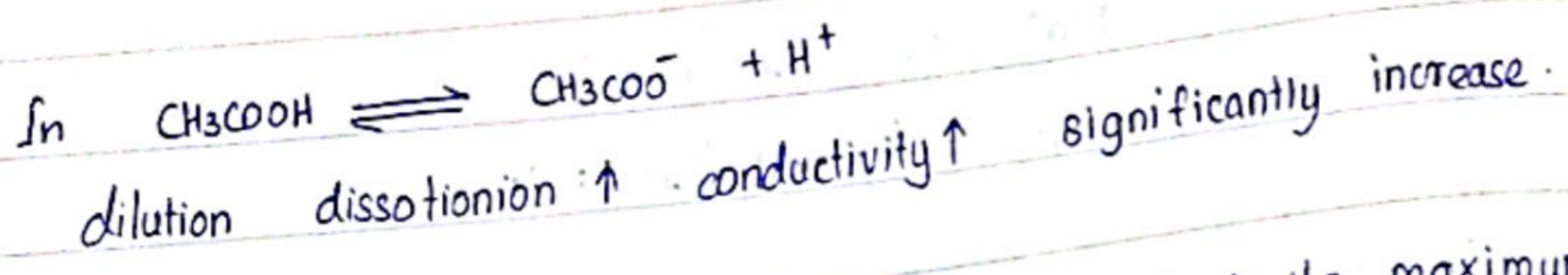
$$\Lambda = \sum \nu_i \Lambda_i$$

↑
Number of the type i formed from one molecule of the electrolyte.



* Strong electrolyte completely dissociate in water. But with dilution, conductivity is constant. The number of ion present in the solution doesn't change. But the ions are very separated.

Infinite dilution, ions speed is increase.



For both electrolytes molar conductivity reached to its maximum value, when the electrolyte concentration approaches to zero (infinite dilution).

This value is known as limiting molar conductivity. OR conductivity at infinite dilution. (Λ^∞)

- Molar conductivity of electrolyte depends on the num of ions
1. Present in the solution.
 2. The speed of ion
 3. The magnitude of the charge of the ion.

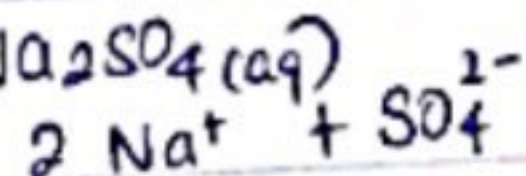
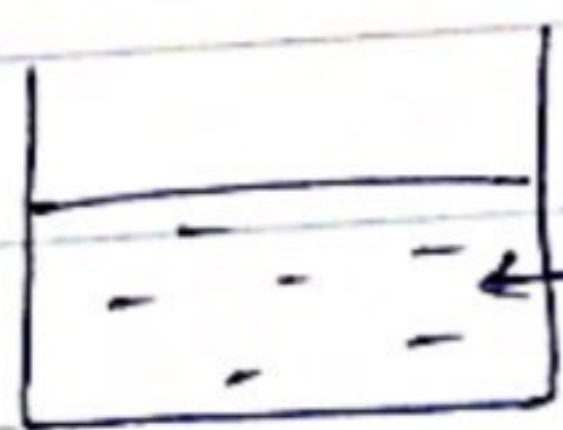
** 0.1 M (LiCl , NaCl , KCl) which have higher conductor?

- Charge density $\text{Li}^+ > \text{K}^+, \text{Na}^+$ Bcs more water molecules are attracted
- Li^+ has higher surface charge density.
- Li^+ more hydrolysis from H_2O molecules. Bcs Li^+ conductivity $>$ K^+ conductivity.

Kohlrausch conform that limiting molar conductivity (Λ_m^∞) its due to the limiting molar conductivity of individual ion denoted by λ .

$$\Lambda_m^\infty = \nu_+ \lambda_+ + \nu_- \lambda_-$$

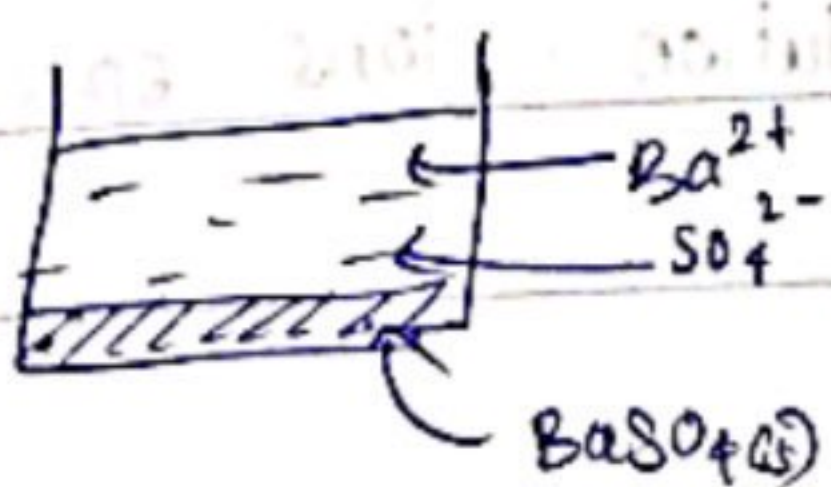
number of anions, cations per formula with the electrolyte.



$$\Lambda_{\text{Na}_2\text{SO}_4}^\infty = 2\lambda_{\text{Na}^+}^\infty + \lambda_{\text{SO}_4^{2-}}^\infty$$

Application of Conductivity.

(oi) Solubility product of a sparingly soluble salt.



$$K_{sp} = [\text{Ba}^{2+}(\text{aq})][\text{SO}_4^{2-}(\text{aq})]$$

$$K_{sp} = x \cdot x$$

$$K_{sp} = x^2$$

This is at infinite dilution $\Lambda = k_{\text{BaSO}_4} / c$

$$k_{\text{solution}} = k_{\text{BaSO}_4} + k_{\text{H}_2\text{O}}$$

- 1) Measure the conductivity of H_2O
- 2) Measure the conductivity of solution (aq)
- 3) Calculate conductivity of given

$$\Lambda_{\text{BaSO}_4}^{\infty} = \underbrace{\Lambda_{\text{Ba}^{2+}}^{\infty} + \Lambda_{\text{SO}_4^{2-}}^{\infty}}$$

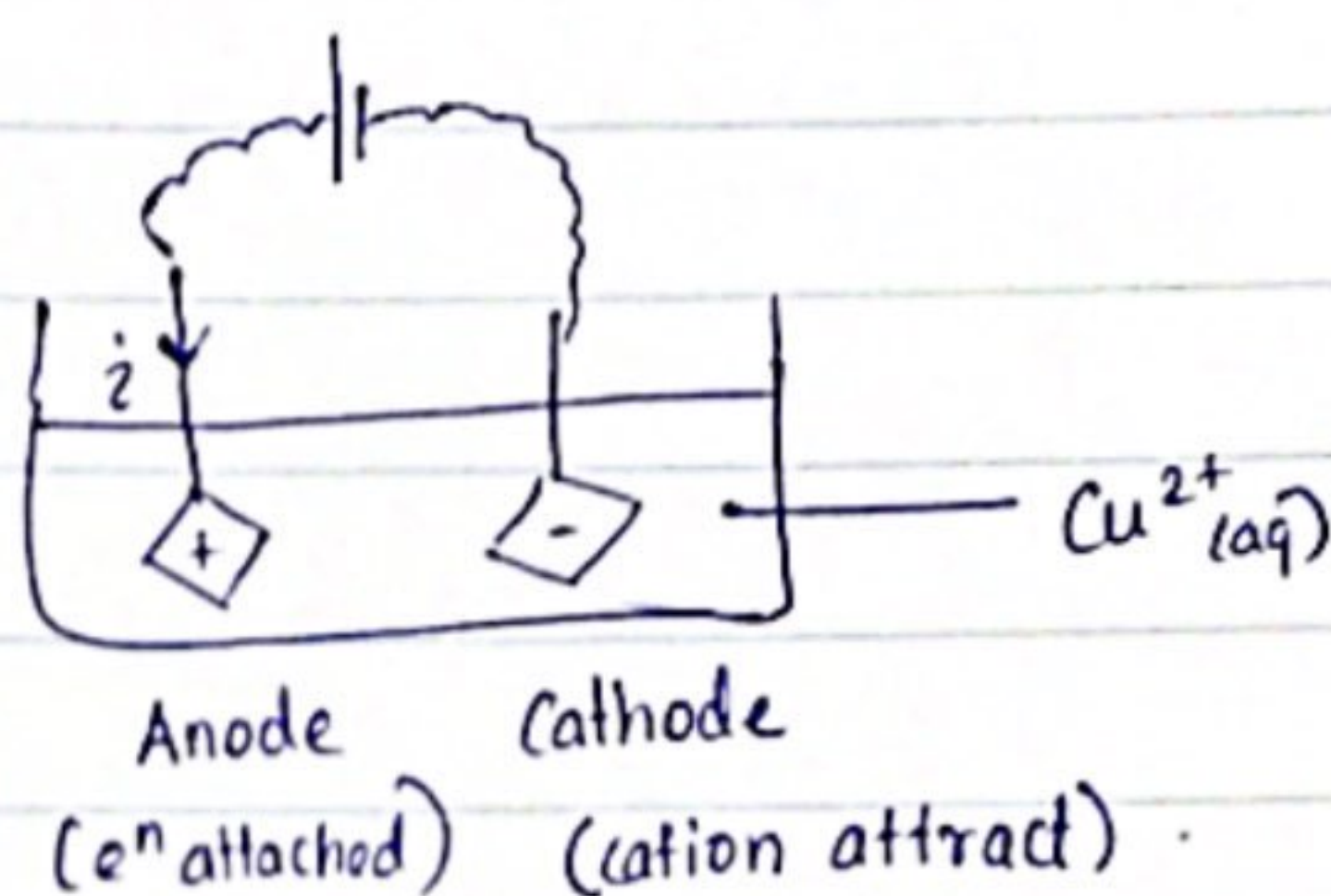
$$\Lambda_{\text{BaSO}_4} = \Lambda_{\text{BaSO}_4}^{\infty}$$

Faraday's Law

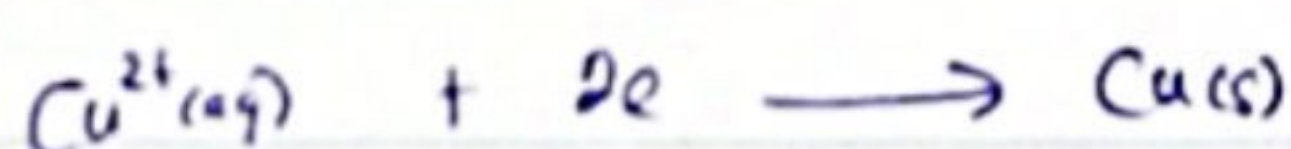
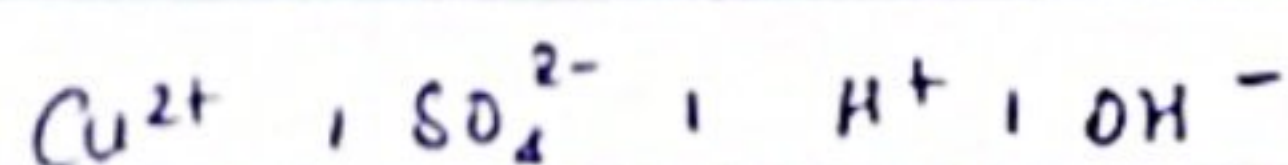


$$Q = It$$

Ex :

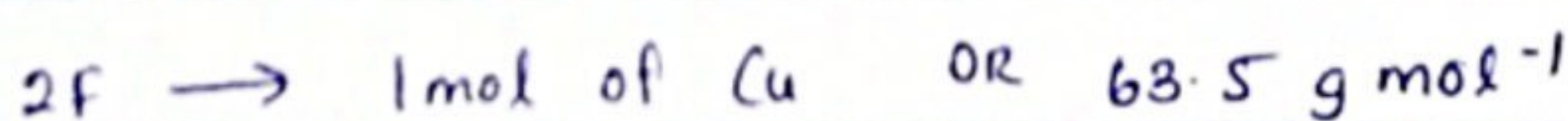


• (But Galvan cell it is differ) -

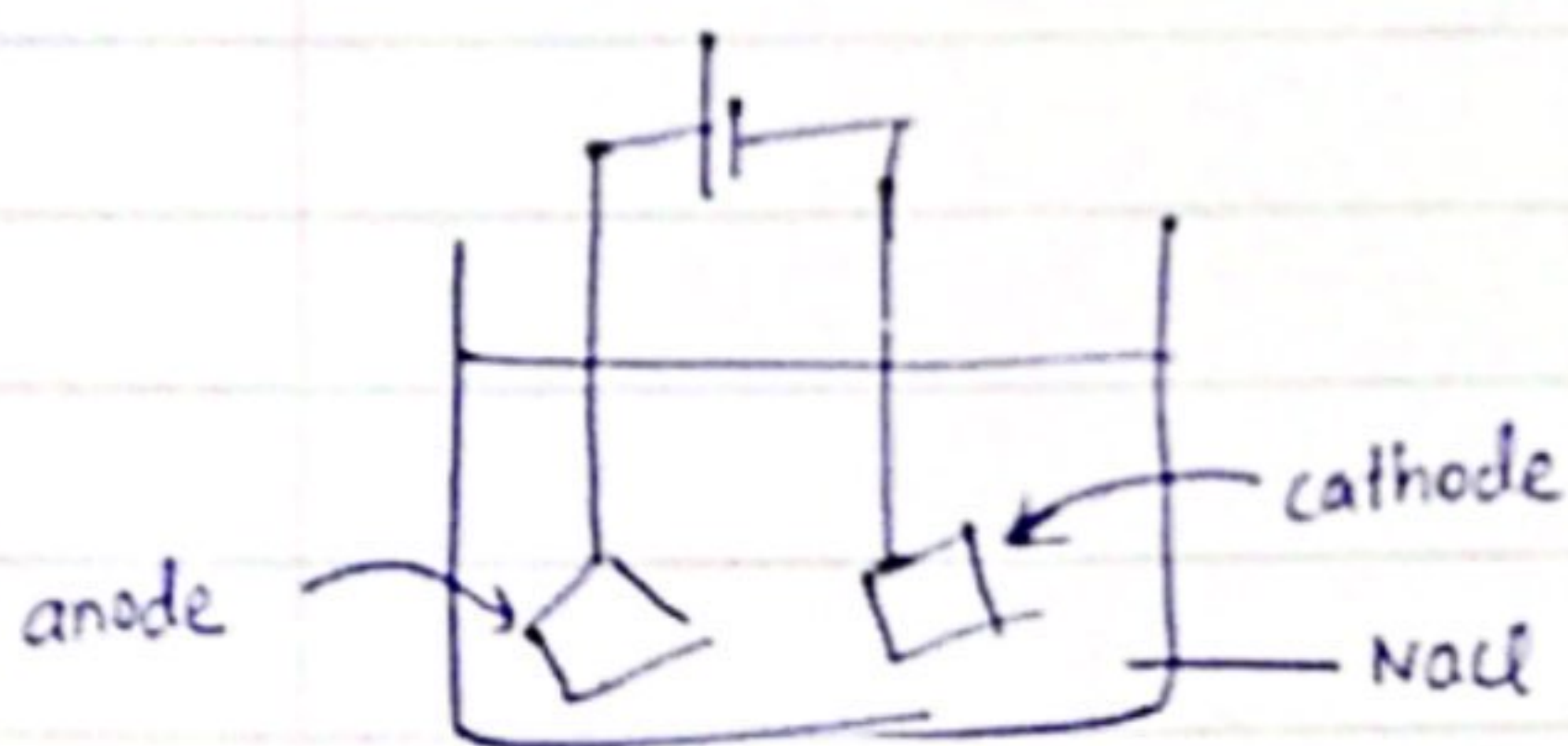
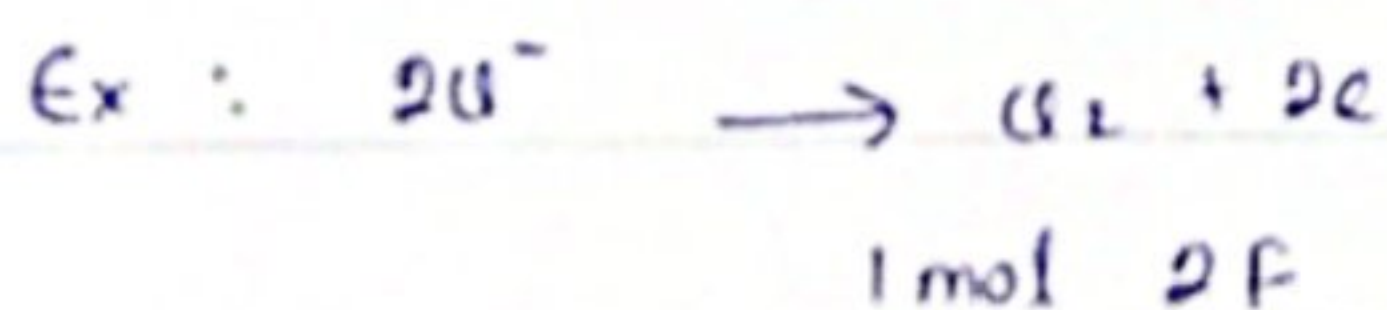


$$1.602 \times 10^{-19} \times 6.02 \times 10^{23}$$

$$96480 \text{ C} = F$$



$$20F = 10 \text{ mol}$$



$$i = 0.5 \text{ A}$$

$$t = ? \longrightarrow 22.4 \text{ l of Cl}_2 \text{ at STP}$$



$$1 \text{ mol } 2F$$

$$\uparrow 2 \times 96480 \text{ C}$$

$$Q = It$$

$$2 \times 96480 \text{ C} = 0.5 \text{ A} \times t$$

$$t = \frac{2 \times 96480}{0.5}$$